

tramdag documentation

Contents

tramdag: Interpretable Neural Causal Models (TRAM-DAGs) in PyTorch	3
Start here	3
Install	3
Documentation	4
Experiments	4
Layout	4
Use of AI tools	5
Citation	5
The model	5
The transformation function	5
The components	6
The components in code	6
Ordinal nodes	7
What the model cannot do	7
Notation	8
Symbols	8
The model's symbols	8
Reading a fitted TRAM-DAG	8
Which term owns which edge	9
A linear shift is a log-odds ratio	9
Ordinal nodes: cutpoints and the interventional PMF	9
Continuous nodes: density and one fitted shift	9
The price of a complex intercept	10
An effect that varies with a covariate	10
Fitting a TRAM-DAG	10
One module, one sub-model per node	10
The likelihood	11
One node alone	11
Path A: stochastic optimization with <code>fit</code>	11
Path B: classical optimization with <code>fit_classical</code>	14
Memory and disk	14
Per-observation scores and the effect-modifier scan	15
The idea	15
Reading the running sum	15
The statistic	15
Read it as screening	16
Using it	16
How the scores are computed	17

Varying-coefficient treatment effects: VC(*modifiers, t=, penalty=, propensity=)	17
Why not a joint CS over treatment and modifiers	17
Semantics	18
Propensity centering: propensity="col"	18
Validation	19
Paper replication: protocol, hyperparameters and results, per experiment	19
What is common to all eight	19
Triangle, continuous (triangle.py): paper Sec. 6.1, App. C.3	21
Triangle, mixed (triangle_mixed.py): paper Sec. 6.2, App. C.4, App. B	22
VACA / CNF benchmark (vaca.py): paper Sec. 5.1 and 5.2, App. C.1	23
CAREFL benchmark (carefl.py): paper Sec. 5.3, App. C.2	25
D4: the marginal start, measured per variant	26
D5: the plateau rule per node	27
Runtime and the CI deviations	28
Repository choices the paper does not state	28
Code map: every module of src/tramdag/, the public names and the private plumbing	29
spec.py: declare the model	29
transforms.py: the monotone map h and the ordinal transform	30
flow.py: the model	30
modules.py: the term modules	31
nodes.py: the node model	32
fitting.py: NodeFitMixin and FitMixin	32
readouts.py: ReadoutsMixin	33
scores.py: effect-modifier detection	33
callbacks.py: the shipped fit callbacks	33
plots.py: the figures (matplotlib optional: tramdag[plots])	34
What is <i>not</i> in the package	34
Where every training hyperparameter lives	34
Upstream candidates for zuko	35
1. Analytic call_and_ladj for BernsteinTransform	35
2. Learnable/linear tails for MonotonicRQSTransform	36
3. Public inverse of _constrain_theta	36
4. Docstring fix: the Bernstein $\bar{\theta}$ -shape off-by-one	36
5. Logistic in zuko.distributions	36
Anti-candidates (look upstreamable, are not)	36
Additive vs joint complex intercept: reading per-parent effects	37
The data	37
Fit both models	38
intercept_contributions: the centered decomposition	39
The structural difference: separable curves vs an entangled cloud	39
Takeaway	40
Classical fitting of all-ls TRAM-DAGs (fit_classical)	40
0. The zeroth example: logistic regression on MASS::birthwt	41
1. Continuous case	45
1b. A bimodal DAG: the demo data	49
2. Ordinal case	52
3. Warm start: classical fit, then further training	54
TRAM-DAG: one causal model, all three rungs	54
1. The benchmark process	55
2. The spec is the DAG, and one fit	56

3. Rung 1: the observational fit	58
4. Rung 2: interventions	59
5. Rung 3: individual counterfactuals	60
What this notebook checked	61
Training strategies: driving fit from the API side	62
1. What fit records without any callback	63
2. Plain Adam	64
3. Two phases	64
4. EarlyStopping: keep the best weights	64
5. EarlyStopping(patience=): stop each node on its own	65
6. Writing your own	66
7. One node alone	68
8. The scoreboard: a runtime comparison	68
Which one, when	69
Heterogeneous treatment effects: the VC term	69
1. A DGP with a known effect function	70
2. Which covariates modify the effect? (effect_modifier_scan)	71
3. The spec: prognostic part and effect head are separate	73
4. Reading the effect out	74
5. The read-out is an identity, not a summary	75
6. Confounding: why propensity= exists	75
What to take away	76
References	77

tramdag: Interpretable Neural Causal Models (TRAM-DAGs) in PyTorch

TRAM-DAGs model each variable of a structural causal model with a monotone normalizing flow, the statistician’s *transformation model*. The whole model is one triangular normalizing flow from iid standard-logistic latents to the observed variables, and its triangular structure is exactly your causal DAG. One fit on observational data answers all three rungs of Pearl’s causal hierarchy: observational, interventional (the do-operator) and counterfactual (abduction). The model keeps **interpretable effects**: every linear-shift coefficient is a log-odds ratio, as in a classical proportional-odds model.

Beate Sick & Oliver Dürr, *Interpretable Neural Causal Models with TRAM-DAGs*, CLeaR 2025 (arXiv:2503.16206). This repository is the reference implementation, in PyTorch on zuko. It replicates the paper’s experiments under continuous integration.

Start here

The demo notebook is the introduction. It fits the paper’s bimodal benchmark, walks the three rungs, and checks every answer against analytic ground truth. It runs on CPU in a few minutes, in Colab or locally.

Install

The package is on PyPI as tramdag. `uv add tramdag` or `pip install tramdag` installs it, and the plots extra (`tramdag[plots]`) adds matplotlib for the figures. The PyPI release lags main. For the current state, install from git (`tramdag @ git+https://github.com/tensorchiefs/tramdag.git@main`, pinned to a commit for reproducibility). A clone with `uv sync` gives the development setup with tests and experiments.

Documentation

Guides, executed notebooks and the API reference are published at the documentation site. Each subject has at most one guide and one example notebook.

Subject	Guide	Example
The model, its notation, what it cannot do	docs/model.md, docs/notation.md	demo
Reading a fitted model: coefficients, curves, interventional distributions	docs/interpretation.md	classical fitting
Fitting: the likelihood, one node alone, the Adam loop and its callbacks, the classical fit, a runtime comparison of the recipes	docs/fitting.md	training strategies
Complex intercepts, joint versus additive		additive vs joint intercepts
Treatment effects that vary with covariates	docs/varying-coefficients.md	varying coefficients
Scores and the effect-modifier scan	docs/scores.md	varying coefficients
Agreement with statsmodels and R, standard errors		classical fitting
What the tests guarantee	tests/README.md	
The API, every default and where it lives	docs/code-map.md	
How the package is built inside	docs/architecture.md	
zuko, and what tramdag would upstream	docs/zuko-upstream.md	
The cited literature	docs/references.md	

CONTRIBUTING.md covers setup, tests, linting and releases; notebooks/README.md how the notebooks are written and run.

Experiments

experiments/ replicates the paper's experiments against its own R code and checks them against committed ground truth in continuous integration. experiments/README.md explains the layout, the YAML variants and the frozen data; docs/paper-replication.md gives the protocol, the hyperparameters and the numbers, experiment by experiment.

Layout

- src/tramdag/ is the framework, and nothing else.
- tests/ measures it against three inline data-generating processes; it never imports research code.
- experiments/ holds the paper replications and their frozen data sets.
- notebooks/ are jupyter .py files, executed by the docs workflow.
- docs/ are the guides; the API pages render from the docstrings.

Use of AI tools

This project was developed with the help of LLM coding assistants. The authors reviewed and take responsibility for all content.

Citation

If you use tramdag, please cite the method paper:

```
@inproceedings{sick2025tramdag,
  title = {Interpretable Neural Causal Models with {TRAM-DAGs}},
  author = {Sick, Beate and D{\'u}rr, Oliver},
  booktitle = {Proceedings of the 4th Conference on Causal Learning and Reasoning (CLeaR)},
  series = {Proceedings of Machine Learning Research},
  volume = {275},
  pages = {606-630},
  year = {2025},
  eprint = {2503.16206},
  url = {https://arxiv.org/abs/2503.16206},
}
```

The model

A TRAM-DAG is one triangular normalizing flow whose sparsity is a causal DAG. This page states the model in the notation of the paper and maps each part of it onto the constructors in tramdag.spec. For the symbols see notation.md; to read a model after it is fitted see interpretation.md; for a worked fit see the demo notebook.

The transformation function

The variables have a causal ordering. For each variable a TRAM-DAG fits a monotone transformation function h . The function maps the observed value to a latent scale, conditional on the parents of that variable:

$$\begin{aligned} u_1 &= h(x_1) \\ u_2 &= h(x_2 \mid x_1) \\ u_3 &= h(x_3 \mid x_1, x_2) \\ u_p &= h(x_p \mid x_1, x_2, \dots, x_{p-1}) \end{aligned}$$

This is the observed-to-latent direction. It is Eq. 2 of the paper, $F_{X|pa}(x) = F_U(h(x \mid pa))$, and in code it is `u = h(x) + shift`.

The direction matters for cost. Training evaluates h directly to score the likelihood, which is cheap. Sampling runs the inverse $x_i = h^{-1}(u_i \mid pa)$. That inverse has no closed form, so bracketed bisection solves it, and sampling is the more expensive direction.

Together the h 's form one triangular flow. Each variable depends only on its causal parents $pa(x_i)$, which is a subset of its predecessors. The Jacobian sparsity of the flow is therefore the DAG itself.

Each $h(x_i \mid pa)$ is one transformation model of x_i given its parents. In code it is a `Node`, which also fits on its own, without the DAG (fitting.md).

The latents $u_1, \dots, u_p$ are standard logistic. That choice is what makes the fitted parameters interpretable, because a shift on the latent scale is then a log-odds ratio.

The components

The transformation decomposes additively on the latent scale, which is what keeps the interpretation valid. Each node's h is

$$u_i = h(x_i | \text{pa}(x_i)) = \underbrace{h_{\vartheta}(x_i)}_{\text{intercept}} + \underbrace{\sum_j \beta_{ij} x_j}_{\text{linear shifts}} + \underbrace{\sum_k g_{ik}(x_k)}_{\text{complex shifts}} + \underbrace{(\beta_0 + b_{\Theta}(x_{\text{mod}})) x_t}_{\text{varying coefficient}}$$

and every causal parent enters through exactly one term. Take x_5 with parents $\text{pa}(x_5) = \{x_1, x_2, x_4\}$ as the example.

- **Simple intercept, SI.** $h_{\vartheta}(x_5)$ has constant parameters. It is a flexible monotone baseline, a Bernstein polynomial by default, and it is the same for every observation.
- **Complex intercept, CI.** The parameters ϑ are themselves a function of some parents. The whole transformation bends with the parent, which permits interactions that an additive shift cannot express.
- **Linear shift, LS.** $\beta_{51}x_1 + \beta_{52}x_2$. One interpretable number per parent.
- **Complex shift, CS.** $g(x_4)$, an unrestricted network of the parent. It stays additive on the latent scale.
- **Varying coefficient, VC.** $(\beta_0 + b_{\Theta}(x_{\text{mod}})) x_t$: a treatment x_t whose effect β varies with modifier covariates through a small penalized network. `varying-coefficients.md` is its guide.

Sampling has to invert only the intercept, because the shifts move to the other side:

$$x_5 = h^{-1}(u_5 | x_1, x_2, x_4) = h_{\vartheta}^{-1}\left(u_5 - \underbrace{\beta_{51}x_1 + \beta_{52}x_2}_{\text{LS}} - \underbrace{g(x_4)}_{\text{CS}}\right).$$

The components in code

Each node declares its transformation as an additive formula of terms. The formula is the node's first argument, a + sum of terms. Each constructor names the parents its term depends on. The paper's symbols I, LS, CS, VC are the classes Intercept, LinearShift, ComplexShift, VaryingCoefficient.

formula for a continuous node x_3	$u_3 = h(x_3 \text{pa})$
nothing, or <code>I()</code>	$h_{\vartheta}(x_3)$, the simple intercept
<code>LS("X1")</code>	$h_{\vartheta}(x_3) + \beta x_1$
<code>I("X1")</code>	$h_{\vartheta(x_1)}(x_3)$, the complex intercept
<code>CS("X1")</code>	$h_{\vartheta}(x_3) + g_1(x_1)$
<code>LS("X1") + CS("X2")</code>	$h_{\vartheta}(x_3) + \beta x_1 + g_2(x_2)$
<code>CS("X1", "X2")</code>	$h_{\vartheta}(x_3) + g_{12}(x_1, x_2)$, one joint network
<code>CS("X1") + CS("X2")</code>	$h_{\vartheta}(x_3) + g_1(x_1) + g_2(x_2)$, two additive networks
<code>I("X1", "X2")</code>	$h_{\vartheta(x_1, x_2)}(x_3)$, one joint network
<code>I("X1", "X2", allow_interaction=False)</code>	$h_{\vartheta(x_1) + \vartheta(x_2)}(x_3)$, one network per parent, summed in coefficient space
<code>CS("X1") + VC("X2", t="T")</code>	$h_{\vartheta}(x_3) + g_1(x_1) + \beta(x_2) x_T$

Three rules follow from the model.

- **Exactly one intercept, first.** A formula written without one gets `I()` prepended. `I()` without parents is the simple intercept, `I(...)` with parents the complex one; `SI()` and `CI(...)` spell that out with the arity checked.

- **Joint versus additive is argument grouping.** Several parents in one term form one network over all of them, an interaction. The same parents in separate terms act additively. A node takes at most one intercept with parents, so for the intercept `allow_interaction` sets the grouping.
- **Every parent enters through exactly one edge-owning term.** The VC modifiers are the one exception: `CS("X2") + VC("X2", t="T")` is the intended pattern. There x_2 acts prognostically through the shift and also modifies the treatment effect.

The three knobs on a term

transform= on I picks the class of h_ϑ for a continuous node:

- "bernstein", the default: a Bernstein polynomial with `n_coeffs=20` unconstrained coefficients whose tails extrapolate along the boundary slope;
- "spline": a monotone rational-quadratic spline with `bins=8`, whose tail slope is fixed (`zuko-upstream.md` explains the consequence);
- "affine": location and scale only, so the node-conditional is a logistic GLM.

Extra keyword arguments of **I** pass straight to the transform class, `I(transform="spline", bins=16)`. Each transform pre-scales the data from the training `range_q` and `$1 - $range_q` quantiles onto a fixed domain. `range_q` is an intercept option with default 0.05, and `range_q=0` uses the minimum and maximum. Ordinal nodes have no transform to pick.

input_transform= on I, CS, VC transforms that term's continuous network inputs, with statistics frozen at the first fit: "minmax", "standardize", or a callable `fn(x, train)`. LS and the VC treatment stay raw, so their coefficients keep their units.

units=, activation=, batch_norm= on I, CS, VC size the term's network. `code-map.md` lists every default.

Ordinal nodes

An ordinal node has no monotone transform to choose. Its intercept is the cutpoint vector of an ordered logit,

$$P(Y \leq k \mid \text{pa}) = \sigma(\vartheta_k - s(\text{pa})),$$

with increasing cutpoints ϑ and s the total shift. The shift is subtracted where a continuous node adds it. `notation.md` states the convention. `interpretation.md` shows what it does to the sign of a coefficient.

The log-probability of an observed level is the difference of two sigmoids. The code computes it in log space with `logsigmoid` and a stable `log(1 - exp(x))`, and per element it takes the better-conditioned side. The direct difference of two sigmoids has exactly zero gradient once they saturate in float32, and a node that starts there never recovers.

Parents enter every network as features: a continuous parent raw, in one column; an ordinal parent one-hot, in one column per level. Abduction is exact for continuous nodes and truncated-logistic for ordinal ones, so `flow.sample(u=flow.abduct(df))` reproduces `df` exactly, level-exactly for ordinal nodes. A counterfactual takes three steps: abduction, action and prediction (Pearl, 2009). `abduct` gives the latents, `do=` sets the intervention, and `sample(u=..., do=...)` predicts.

What the model cannot do

- The latent distribution is fixed to standard logistic. It is not learned.
- A VC treatment is continuous or a binary ordinal node; a multi-level ordinal treatment is not supported. Propensity centering needs a binary ordinal treatment.

- The DAG is an input. `tramdag` fits the mechanisms of a graph you supply, and does not discover the graph.
- Identification is the usual causal one. No hidden confounding, and the graph has to be right.

Notation

This page defines the notation. Every guide, docstring and notebook in this repository follows it.

Symbols

Random variables are capital Latin letters, for example Y . Their realizations, observed or sampled values, are lowercase Latin letters, for example y . The cumulative distribution function of Y is F_Y and the density is f_Y . A hat marks an estimate: $\hat{F}$ is the fitted distribution, F the true and often unknown one.

Sets and vectors are bold, for example $\mathbf{X} = [X_1, \dots, X_J]$.

A single optimized parameter is a lowercase Greek letter, for example ϑ or β . A vector of parameters is bold lowercase, for example $\vartheta = (\vartheta_1, \dots, \vartheta_M)^\top$. A tuple that collects several parameter groups, for example all weight matrices of a network, is bold capital, for example Θ .

A function that a parameter vector defines carries it as a subscript: h_{ϑ} is the monotone transformation with coefficients ϑ .

The model's symbols

Symbol	Meaning	In code
X_j, Y	the variables of the DAG	node names in the spec
$\text{pa}(X_j)$	the causal parents of X_j	<code>node_parents</code>
U_j	the standard-logistic latent of node j	<code>u = flow.abduct(df)</code>
h_{ϑ}	the monotone transformation of a node	the I term; <code>theta</code> in docstrings
ϑ_k	the ordinal cutpoints, elements of ϑ	<code>ordinal_cutpoints</code>
β	a linear-shift coefficient; e^β is an odds ratio	<code>LS; flow.ls_coefficients()</code>
g_j	a complex shift, an NN of one term's parents	<code>CS</code>
β_0	the constant treatment effect of a VC term	<code>beta0</code>
b_{Θ}	the penalized effect-modification network	the VC net; <code>units=</code> sets its layers
λ	the L2 weight on b_{Θ}	<code>penalty=</code>
σ	the logistic function	<code>torch.sigmoid</code>

Every node's model is one additive formula on the latent scale. `model.md` writes it out. For a continuous node the shifts are added. For an ordinal node the shift is subtracted inside the sigmoid: $P(Y \leq k \mid \text{pa}) = \sigma(\vartheta_k - \text{shift})$.

Reading a fitted TRAM-DAG

A TRAM-DAG is a normalizing flow, but it is built so that parts of it stay readable. This page collects the read-outs and says what each number means. For the model itself see `model.md`, and for the symbols see `notation.md`.

Every claim on this page is checked somewhere that runs. The links say where, so nothing here is a number typed into prose.

Which term owns which edge

`flow.to_matrix()` gives the labeled adjacency matrix of the model. Rows are parents and columns are children. A cell holds the term tag: LS, CS, CI, VC for a treatment edge, or VCm for a varying-coefficient modifier. An empty cell means no edge. A multi-parent term carries its parent group as a suffix, and several terms in one cell join with +.

This is the meta-adjacency view of the paper. It is also the fastest way to confirm that the model you wrote is the model you meant. `plot_dag(spec)` draws the same thing, with one edge style per term.

Exercised in the varying-coefficients notebook.

A linear shift is a log-odds ratio

`flow.ls_coefficients()` gives `{node: {parent: weights}}` for the LS terms only. A CS or VC shift is a network and has no single weight, so those are skipped.

Two things decide how to read a weight.

The sign follows the latent-scale convention of notation.md. A continuous node adds its shift, so a parent that raises the child gets a *negative* weight. An ordinal node subtracts its shift, so the sign is the one you expect there.

The scale is the latent scale, not the data scale. For a continuous node the weights carry the transform's scaling. A single weight is therefore not comparable to a regression coefficient. Their *ratio* is comparable, because the scaling cancels. An ordinal node has no such scaling, so $\exp(\beta)$ is an odds ratio directly.

An ordinal parent gives one weight per level, from its one-hot encoding. Those weights are identified only up to a common constant. The one-hot columns sum to one in every row. Adding a constant to all of them, and subtracting it from the intercept, leaves the likelihood unchanged. Read them as differences. $w[k] - w[0]$ is the level-k-against-level-0 log-odds ratio. It matches the coefficient of column `{parent}[k]` in `flow.design_matrix(df, node, drop_first=True)`, which drops the level-0 column.

Checked in `notebooks/classical_fit_tram_dag.py`, which compares the weights against statsmodels and R, and pinned in continuous integration by `experiments/triangle_mixed.py`, whose odds-ratio check is in `paper-replication.md`.

Ordinal nodes: cutpoints and the interventional PMF

An ordinal node's intercept is its cutpoint vector, not a transform (model.md). `tramdag.transforms.ordinal_cut` turns the unconstrained parameters into the increasing cutpoints, which is how you read a fitted baseline.

`flow.pmf(df, node, do=...)` gives the class probabilities per row, in closed form. It needs no sampling, and `do=` makes it interventional by overriding columns before the parents are read. That combination is what lets a treatment-effect question be answered exactly rather than by Monte Carlo.

Exercised in the classical-fitting notebook, and under `do=` in `experiments/triangle_mixed.py`, which is checked against committed ground truth on every run.

Continuous nodes: density and one fitted shift

`flow.density(df, node, grid, do=...)` is the continuous counterpart of `pmf`. For every row it evaluates $p(\text{node} = g \mid \text{pa})$ at each grid value, in closed form from the transform. Use it to look at a fitted conditional without sampling it.

`flow.shift_curve(node, parent, grid)` evaluates one fitted shift term along a grid of parent values. Use it to plot a CS term against the function it was supposed to learn. It goes through the term's own evaluation, so a custom term works too.

A caution on both: a CS term is identified only up to a constant, because that constant can move into the intercept. Compare a fitted curve to the truth after mean-centering both, not point by point.

`density` runs in the classical-fitting notebook, section 1b. `experiments/triangle.py` compares the `shift_curve` of each CS variant with the true function against committed ground truth.

The price of a complex intercept

`I(...)` with parents is the flexible end of the family. The parents control the transform parameters, so the node can bend in ways no additive shift can express. The cost is that there is no single coefficient to read.

`paper-replication.md` measures that trade-off on the paper's triangle process, fitted once with an LS edge and once with a CS edge.

A middle option exists. `I("a", "b", allow_interaction=False)` builds one network per parent and sums their parameter vectors, so the parents act additively on the transform. `flow.intercept_contributions(df, node)` then decomposes that sum into mean-centered per-term parts, using the usual additive model convention, so the parts become comparable. This is a post-hoc read-out and changes nothing about the fitted model.

Worked through in the intercept notebook.

An effect that varies with a covariate

`VC(*modifiers, t=...)` fits $\beta(\text{modifiers}) \cdot x_t$, and `flow.varying_coef(df, node)` reads the fitted β back in closed form.

`flow.scores(df, node)` and `flow.effect_modifier_scan(df, node, t=...)` come before that decision rather than after it. They rank candidate modifiers by how much the treatment coefficient's per-observation scores drift when the rows are ordered by each candidate. A cheap all-LS fit is enough, so the shortlist is measured rather than guessed.

See `varying-coefficients.md` and `scores.md`, and `notebooks/varying_coefficients.py`.

Fitting a TRAM-DAG

This page is the reference for training a `CausalFlowDAG` and its nodes. It covers the likelihood, the two fitting paths `fit` and `fit_classical`, and the hooks of the Adam path. `notebooks/training_strategies.py` runs the Adam recipes end to end. It closes with a runtime comparison of the recipes on one workload. `n_jobs`, `marginal_init`, `validation_split` and per-node epochs or `learning_rate` functions have no notebook example.

One module, one sub-model per node

A `CausalFlowDAG` is a single `torch.nn.Module` with one independent sub-model per variable. A sub-model has an intercept that produces the transform parameters θ and one shift module per shift term. It also has the monotone transform h , which holds range buffers but no weights. The nodes share no parameters. The DAG lives entirely in which parents each node reads. There is no edge weight matrix and no shared trunk. `code-map.md` says which class implements which term.

The likelihood

The flow maps iid standard-logistic latents U to the observed X in causal order, and node i reads its parents as data. The Jacobian is therefore triangular, and the joint log-likelihood decomposes per node, $\log p(x) = \sum_i \log p(x_i \mid \text{pa}(x_i))$. `Node.row_log_prob` computes one term and `CausalFlowDAG.log_prob` sums them.

- **Continuous node.** Change of variables through the monotone transform: $u = h(x; \theta) + s$ and $\log p(x \mid \text{pa}) = \log f_{\text{logistic}}(u) + \log |h'(x)|$. The transform returns the log-derivative alongside u .
- **Ordinal node.** The ordered-logit head of `model.md`, evaluated as the log of the cutpoint-interval probability in log space.

Because parents enter as data, the per-node terms have disjoint parameters, and maximizing each term maximizes the sum. So every node fits on its own: `Node.fit` minimizes the node's mean negative log-likelihood over the batch, plus the node's VC penalty. `CausalFlowDAG.fit` runs `Node.fit` for each node. `log_prob` returns the per-row joint.

One node alone

A `Node` is a TRAM regression of one variable on its parents, and it needs no DAG. It takes its name, its node spec and a parent schema, `{parent: "continuous" | n_levels}`. `notebooks/training_strategies.py` fits one with Adam, and `notebooks/classical_fit_tram_dag.py` fits one classically against R glm. The frame needs the node's column, its parents' columns and its side columns; the fit ignores other columns. `fit`, `fit_classical`, `nll`, `save` and `load` work as on the flow, for one node. `nll` gives one float, and the `fit_classical` report is the per-node report. `fit` takes an optimizer instance or a factory `f(node)`, but no `n_jobs`. A centered VC needs its propensity column in every frame, because only the flow can compute it live from the treatment node. A flow derives each node's shuffling seed from `fit(seed=)` and the node's index. So `Node.fit(seed=)` alone does not repeat a node of a seeded flow fit. A node built alone draws its initial weights from torch's global RNG; call `torch.manual_seed` first for a reproducible start. It always uses torch's default init; `init=` exists on `CausalFlowDAG` only.

Path A: stochastic optimization with fit

`fit` is the general-purpose trainer, and the only one for a model with a CS, CI or VC term. Per node, it is one minibatch Adam loop that keeps the final weights.

- **Nodes.** `CausalFlowDAG.fit` fits the nodes in topological order. With validation data, the nodes with a centered VC fit last. `n_jobs=N` forks N worker processes (Linux only), one task per node. A worker runs torch on one thread. With the same `seed=`, it gives the weights of a serial fit on one thread. Without `seed=`, `n_jobs > 1` draws a base seed from torch's global RNG. `epochs=`, `learning_rate=` and `callbacks=` take one value for all nodes or a function of the node name.
- **Optimizer.** One optimizer per node, Adam(`lr=learning_rate`) by default. `optimizer=` takes a factory `f(node)`, for example to give the networks weight decay in their own parameter group. `Node.fit` also takes an optimizer instance.
- **Minibatches.** A fresh shuffle each epoch. `seed=` seeds the shuffle only, not the weight init; each node gets its own seed derived from `(seed, node index)`.
- **Epochs.** `epochs` has no default: a fixed budget under-spends on one workload and wastes on the next, so every caller states its own.
- **Calibration.** The first fit or `fit_classical` calibrates each node on `train_df`. Every term freezes its data-dependent state there, the intercept its `range_q` quantiles and each `input_transform=` its statistics. A loaded checkpoint is never recalibrated.
- **Marginal start.** Off by default: every simple intercept starts with all its parameters at zero, the zero start. `fit(marginal_init=True)` starts each Bernstein or ordinal simple intercept at the empirical distribution of its column instead. At shift zero the untrained model then

already fits each marginal, and the fit needs fewer epochs. Only the fit that calibrates the node applies it, so a node that `fit_classical` calibrated ignores the flag. The marginal start says how, and when the endpoint changes.

- **Validation.** `validation_data=` takes a frame. `validation_split=` takes a float and uses the last fraction of `train_df` unshuffled. So shuffle the frame first if its row order means anything. Only the head calibrates. The flow splits once, so all nodes see the same rows. With either, `fit` writes the node's validation NLL after every epoch into `node.history["val"]`.
- **History.** `node.history` holds "train", "val", "val_epoch" and "lr", the optimizer's rates per epoch as {group index: lr}. `flow.history` is a view with one {node: value} dict per epoch and key, plus the `val_epoch` list. A node that stopped earlier repeats its last entry. `verbose=N` prints every Nth epoch and the last one, per node; the default 0 is silent.
- **Callbacks.** `callbacks=` takes one Callback or a list. The hooks get the node: `on_fit_begin(node, optimizer)`, `on_epoch_end(node, epoch, optimizer)` and `on_fit_end(node, optimizer)`. In `Node.fit` a bare callable is an `on_epoch_end` hook; any True return stops that node's fit. `on_fit_end` runs before the VC re-centering. On the flow, a bare callable is a function of the node name that gives that node its callbacks. So on the flow, a bare hook goes into a list. The shipped callback is `EarlyStopping`: it restores the node's best-validation weights and takes an optional `patience` and `min_delta`. It reads `node.history["val"]`. `fit` refuses two callbacks that restore weights.
- **Centered VC propensities** ride the training frame as the column that `VC(propensity=)` names, and `split` and `minibatch` with it. On the flow, the validation NLL uses the live propensity of the fitted treatment node, as every query does. `Node.fit` alone reads the column from the validation frame. `varying-coefficients.md` is the guide.

The marginal start

The marginal start puts the optimizer near the data before the first step. It changes the parameters the optimizer starts from, not the model or the loss.

`marginal_init=True` sets the ϑ of each Bernstein or ordinal simple intercept. At shift $s = 0$, the node's CDF then equals the empirical marginal $\hat{F}$ of its training column. The match is exact for an ordinal node and a smooth approximation for a continuous node. An intercept with parents (`I("x1")`) has no single ϑ . The flag leaves it at the seeded initialization of its network.

Why logit. The latent U is standard logistic, so its CDF is the sigmoid σ and its quantile function is logit. At $s = 0$ the model says $P(X \leq x) = \sigma(h(x))$. For this CDF to equal $\hat{F}$, the transform must be

$$h(x) = \text{logit } \hat{F}(x) = \log \hat{F}(x) - \log(1 - \hat{F}(x)).$$

The code evaluates $\log(1 - p)$ as `log1p(-p)`, which stays accurate for p near 0.

Ordinal node. `ordinal_marginal_init_theta` takes the counts of the K classes $0, \dots, K-1$. The model is $P(X \leq k) = \sigma(\vartheta_k - s)$, so the targets are the cumulative class log-odds $c_k = \text{logit } \hat{F}(k)$ for $k = 0, \dots, K-2$. `ordinal_cutpoints` builds increasing cutpoints from unconstrained parameters: $\vartheta_0 = \tilde{\vartheta}_0$ and $\vartheta_k = \vartheta_{k-1} + e^{\tilde{\vartheta}_k}$. The start inverts this map. It sets $\tilde{\vartheta}_0 = c_0$ and $\tilde{\vartheta}_k = \log(c_k - c_{k-1})$. At $s = 0$ the class probabilities are then the empirical frequencies. For counts [50, 30, 15, 5] the start gives the PMF [0.50, 0.30, 0.15, 0.05].

Continuous node. `BernsteinUT.marginal_init_theta` takes the column. Take a Bernstein polynomial of order M with control points ϑ_k . It is close to the function whose values at k/M are ϑ_k . The transform pre-scales the calibrated range onto the domain of the polynomial (`model.md`). The point k/M of the domain then belongs to a data value x_k . `zuko` turns `n_coeffs` parameters into $M + 1$ control points, with $M = \text{n_coeffs} + 1$. Step 3 explains the two extra points. The start uses the closeness property in three steps:

1. It places $M + 1$ equally spaced points x_k on the calibrated range, from the `range_q` quantile to the $1 - \text{range_q}$ quantile of the column. With `n_coeffs = 20`, $M = 21$ and there are 22 points.
2. It sets each control point to $\text{logit } \hat{F}(x_k)$, with $\hat{F}(x_k)$ the fraction of rows at or below x_k .
3. It inverts zuko's parameterization. zuko builds the control points as a cumulative sum of softplus steps. It ties the first two and the last two steps. The start averages each tied pair and inverts the softplus with $\log(e^d - 1)$. It adds zuko's centering offset $n \log 2/2$, with $n = \text{n_coeffs}$, to the first parameter. zuko-upstream.md lists the zuko internals this relies on.

The polynomial then starts as the Bernstein approximation of $\text{logit } \hat{F}$. The approximation smooths $\hat{F}$, and a sharp feature such as the dip between two modes smooths most.

The ends of the range carry the empirical quantiles `range_q` and $1 - \text{range_q}$. The end control points are therefore close to $\text{logit } q$ and $\text{logit}(1 - q)$, with $q = \text{range_q}$. At the default $q = 0.05$ these are -2.944 and 2.944 . With fewer than 3 coefficients, the tied steps use all the parameters and leave only a straight line. The start then takes the line between these two end values.

Guards. Two limits keep the start finite and increasing:

- The start clips $\hat{F}$ to $[10^{-3}, 1 - 10^{-3}]$. A control point or a cutpoint then stays within about ± 6.9 .
- Each step between adjacent control points or cutpoints is at least 10^{-3} , because the softplus and log inverses need strictly increasing points. An empty class or a gap in the data therefore gets a small but nonzero probability. An empty middle class starts at $2.5 \cdot 10^{-4}$ or less.

Where it does not apply. The spline and affine transforms have no marginal start, and the flag skips them. A `range_q=0` Bernstein transform maps the data minimum and maximum onto the range ends. The start ties the range ends to $\text{logit } q$ and $\text{logit}(1 - q)$, which are infinite for $q = 0$. So fit raises before it reads the column. It raises for such a transform also in an intercept with parents, which does not use the start. Fit such a model with `marginal_init=False`.

What it changes. At `n_coeffs = 20` the zero start is close to a straight line too. Its control points span a wider range than the line from -2.944 to 2.944 . A steeper h gives a narrower distribution than the data, so the zero start has the higher NLL at shift zero.

The likelihood of an all-LS model has one optimum, and the fit reaches it with or without the start. So the start only shortens the way, and `fit_classical` takes no such flag.

A model with a network shift or a learning-rate anneal can reach a different local optimum. For such a model the start can improve or worsen the causal estimate. D4 in paper-replication.md measures this per variant.

Which recipe

All-LS models train to the MLE and keep the final weights. Flexible models with a CI, CS or VC term overfit observational confounding at the MLE and need the best-validation weights to recover the causal effect.

recipe	when
<code>fit_classical</code> plain Adam	all-LS spec: exact, deterministic, seconds an all-LS spec whose <code>fit_classical</code> report is not converged, quick looks
multi-phase Adam, one fit call per rate <code>EarlyStopping()</code>	a tighter MLE without a scheduler any CI, CS or VC model; register it, fit has no default
<code>EarlyStopping(patience=)</code>	also stop each node once its best epoch is that old

recipe	when
a plateau Callback around torch's ReduceLROnPlateau	a decaying rate per node

Three details are easy to get wrong.

- A second fit call continues training and history accumulates. That is what makes a multi-phase schedule a loop of fit calls.
- A callback instance passed to the flow serves every node in turn. A shipped callback resets at each node's fit begin, so its attributes describe the last node only. Use a function of the node name to keep one instance per node. Under `n_jobs` the state of every callback stays in the workers; read `node.history` instead.
- A post-fit `load_state_dict` skips the VC re-centering. Restore weights from `on_fit_end` instead, as `EarlyStopping` does.

Path B: classical optimization with `fit_classical`

`fit_classical` is the optimizer for all-LS models, where every node-conditional is an ordered logit or a transformation model with linear shifts. It raises on any CS, CI or VC term.

- **Full-batch, float64, L-BFGS** with a strong-Wolfe line search; no minibatches, schedule or early stopping, so the same init gives bit-identical results. A converged fit matches statsmodels and R to about four decimals, where a converged Adam fit gets to about $1e-3$.
- **Budget.** `max_iter=5000` per node is a default, not a promise. Torch ends the run when the NLL or the parameters move by less than $1e-9$. A node with many parameters can need thousands of iterations to stop on that rule. A report that is not converged issues a `UserWarning`.
- **Per node.** The flow's `fit_classical` runs `Node.fit_classical` for each node. Its report holds converged (every node converged), the summed `final_nll` and seconds. It also holds the coefficients as `{node: {parent: array}}` and nodes, the per-node reports.
- **The report.** Per node, `stop_reason` is "tolerance", "max_iter" or "max_eval" (torch's budget of closure calls, `max_iter * 5 // 4`). converged needs both: the run stopped on its own AND `grad_norm` is at most `tramdag.fitting.GRAD_TOL` ($1e-2$). Both conditions are necessary, because the same tolerance fires when the line search stalls far from the optimum. Such a run stops on "tolerance" with a large gradient norm and is not converged.
- **float64 is transient.** The fit restores float32 afterwards, and checkpoints stay float32.
- **Weakly identified directions never settle.** A Bernstein intercept, rare one-hot levels or a flat treatment-effect ridge drift along zero-curvature valleys after the likelihood is at its optimum. This is why `tolerance_grad` is 0 and the gradient bound is loose. Correctness comes from the comparison with classical software in `notebooks/classical_fit_tram_dag.py`.

A converged `fit_classical` leaves the model at the MLE, ready for any operation. A fit call from there stays put. This checks that the classical solution is the optimum, and it is a warm start. The VC guide uses it for `beta0`.

Memory and disk

Neither path writes to disk. The parameters, the optimizer state and the history dicts live in RAM, and whatever a callback records is yours. The only disk I/O is the explicit `save()` and `load()`. `fit(n_jobs=N)` forks N copy-on-write copies of the flow and the training frame.

Per-observation scores and the effect-modifier scan

A treatment effect is often not the same for every patient. A covariate that changes the effect is an effect modifier. A VC term models an effect $\beta(x)$ that changes with the modifiers, but you must tell it which covariates they are. The effect-modifier scan finds candidates for that list. It uses a simple model that takes seconds to fit, before you fit the flexible one. `varying-coefficients.md` covers the VC term itself.

The idea

1. **Fit a simple model.** It assumes one constant effect β for everyone. Every parent enters as a linear shift (LS), and `fit_classical` fits this all-LS model in seconds.
2. **Ask each row which way β should move.** The score of row i is $\psi_i = \partial \ell_i / \partial \beta$, the slope of that row's log-likelihood ℓ_i . A positive score means the row fits better with a larger β . A negative score means it fits better with a smaller one.
3. **Check that the answers cancel in every group.** At the fitted β the scores sum to about zero, because β is the best compromise. If the effect is really constant, the scores of any group of rows also sum to about zero, apart from noise.
4. **Sort and add up.** Sort the rows by a candidate covariate, for example age. The sum of the first j scores is the score sum of the group "age up to the j -th value". So the running sum checks all these groups at once.

Plotted over the sorted rows, the running sum forms a path. Suppose young patients all ask for a larger β and old patients for a smaller one. The path then climbs first and falls back later. A large swing of the path along a covariate means that one constant β hides a pattern along that covariate.

Reading the running sum

Section 2 of `notebooks/varying_coefficients.py` draws the running sum for three covariates. The true effect changes with X2 and X3 and not with X1. The figure has two panels with the same effect. In the left panel X1 has a quadratic effect on the outcome, and in the right panel a linear one.

The figure plots the running sum after a scaling, and the grey band spans ± 1.358 on that scale (The statistic explains both). If a path leaves the band, the scan flags its covariate.

- The paths of X2 and X3 swing far out of the band and come back. They flag in both panels.
- In the right panel the path of X1 stays in the band. The simple model describes the linear effect of X1 correctly, and X1 does not flag.
- In the left panel the path of X1 swings out as well. The simple model cannot describe the quadratic effect, and X1 flags although it does not change the treatment effect. The section Read it as screening explains why.

The statistic

The scan is the structural-change test (Zeileis & Hornik, 2007) behind model-based recursive partitioning (Dandl et al., 2024; Zeileis et al., 2008). That method is a tree that splits the data where the coefficients of a model change. The scan applies its test to the treatment coefficient of a TRAM-DAG.

For one candidate, the scan sorts the n rows by that candidate. It then forms the scaled running sum, also called a CUSUM (cumulative sum):

$$B_j = \frac{1}{\text{sd}(\psi)\sqrt{n}} \sum_{i \leq j} \psi_{(i)}, \quad j = 1, \dots, n,$$

with $\psi_{(i)}$ the score of the i -th row in the sorted order. The statistic `stat` is $\max_j |B_j|$, the largest distance of the path from zero.

- The path ends near 0, because the scores sum to about zero. It starts near 0, because one score divided by $\sqrt{n}$ is small.
- The scaling by $\text{sd}(\psi)\sqrt{n}$ makes the distribution of `stat` approximately independent of the sample size and of the size of the scores.
- With a constant effect, B behaves approximately like a Brownian bridge: a random path that starts and ends at 0.
- The largest distance of a Brownian bridge from zero has a known distribution, the Kolmogorov distribution. It gives the `p_value` without simulation.
- A `stat` above 1.358, the 5 % critical value, sets `flag`.

The p-value is approximate. The scan uses the raw treatment score. The full test (Zeileis & Hornik, 2007) removes its correlation with the scores of the other fitted coefficients, and the scan does not.

The error has a direction. For a candidate that is also an LS parent of the node, the fitted LS weight already removes part of the pattern. The scan then flags too rarely for that candidate. Read the p-value as a guide, not as an exact error rate.

Read it as screening

A flag means that one constant β does not fit all rows along the covariate. Two causes give this:

- The effect really changes with the covariate.
- The simple model describes the covariate's own effect on the outcome, the prognostic part, wrongly.

The second cause works through the score. The treatment score of row i is $\psi_i = r_i$ times the term's regressor: t_i , or $t_i - \hat{e}_i$ for a centered VC. Here t_i is the treatment of the row. The factor $r_i = \partial \ell_i / \partial s_i$ is the derivative of ℓ_i with respect to the row's total shift s_i , and it acts like a residual. If the simple model gets the prognostic effect of X1 wrong, r_i follows a pattern in X1. The treatment score takes on that pattern, so the path of X1 swings.

So a flag means "look here". What you find is a modifier or a misspecification, and both are worth knowing.

Two cases make the p-value less reliable:

- **Few distinct values.** A binary or three-level candidate gives many tied rows. Inside a tie the order is arbitrary, and the path depends on it. Read the p-value as a ranking, not as an exact test.
- **No converged fit.** The scores sum to zero only at the optimum. Scan an all-LS model whose `fit_classical` report says converged. The scan also takes the β_0 of a VC term, but a VC model stopped at its best-validation weights is not at the optimum. Read such a scan with more care.

Using it

`flow.effect_modifier_scan(df, node, t=)` runs the scan for the treatment `t` of `node`. It returns one row per candidate with `stat`, `p_value`, `crit_5pct` and `flag`, sorted by `stat`.

- `candidates=` lists the covariates to scan. By default it is every node except `node`, `t` and the descendants of `t`: a modifier is fixed before the treatment, and a descendant would pick up the effect itself. A candidate does not have to be a parent.
- For a binary ordinal LS treatment, `t` resolves to the identified contrast of level 1 against level 0. For a VC treatment it resolves to β_0 , the constant part of $\beta(x)$.

- `column=` picks one score column by name, for example one level contrast "t[2]" of a treatment with more than two levels.

`flow.scores(df, node)` returns the scores themselves: one row for each row of `df`, and one column per interpretable shift coefficient of the node.

- A continuous LS parent gives one column, named after the parent.
- An ordinal LS parent gives one column per level of its one-hot encoding, named "{parent}[0]", "{parent}[1]" and so on. A row's score goes to the column of the level that the row has.
- A VC term gives one column for its β_0 , named after the treatment.
- A CS term is a network with no single coefficient, so it has no column.

The scores also serve influence analyses and robust standard errors.

How the scores are computed

Every shift coefficient γ adds γz to the node's total shift s . Here z is the value of a parent, or one column of its one-hot encoding. The chain rule therefore gives

$$\frac{\partial \ell_i}{\partial \gamma} = \frac{\partial \ell_i}{\partial s_i} z_i = r_i z_i,$$

so one derivative r_i per row serves every coefficient. The derivative has a closed form for both node kinds, and the computation uses no autograd. Below, h is the fitted monotone transformation and σ the logistic function.

- **Continuous node.** For the node value y the latent value is $u = h(y) + s$ and $\ell = \log f(u) + \log h'(y)$, with f the standard-logistic density. So $\partial \ell / \partial s = 1 - 2\sigma(u)$.
- **Ordinal node.** $P(Y = y) = \sigma(b) - \sigma(a)$ with the shifted cutpoints $a = \vartheta_{y-1} - s$ and $b = \vartheta_y - s$, as in `model.md`. So $\partial \ell / \partial s = (\sigma'(a) - \sigma'(b)) / (\sigma(b) - \sigma(a))$, with $\sigma' = \sigma(1 - \sigma)$. For the first and last class the outer cutpoint is $\mp\infty$ and its σ' is zero.

The function only reads the fitted model. It changes neither fitting nor sampling. The tests check that each column sums to about zero at a fitted optimum, and that the scores agree with float64 finite differences.

Varying-coefficient treatment effects: `VC(*modifiers, t=, penalty=, propensity=)`

A VC term gives a node a treatment-effect head with its own bias-variance budget. It contributes $\beta(x) x_t$ with $\beta(x) = \beta_0 + b_\Theta(x)$ to the node's shift, where b_Θ is a deliberately small, penalized network of the modifiers. The point is not expressiveness. The point is that the flow estimates the effect function with care, not as a by-product. Every claim here runs in `notebooks/varying_coefficients.py`, which is the one place for the code, or in the tests that the text names.

Modifiers may appear twice in a node, as prognostic parents through CS or LS and as effect modifiers through VC. That pattern is intended. Only the treatment named by `t=` owns its edge, and a second term that declares it raises an error.

Why not a joint CS over treatment and modifiers

For a binary treatment the multi-parent CS is equivalent in expressiveness. Any shift decomposes as $s(x, t) = s(x, 0) + [s(x, 1) - s(x, 0)] t$. But the CS form has no effect-specific regularization.

The likelihood rewards a good average fit of $s(x, t)$, and nothing rewards a smooth difference between the arms. So the read-out is the difference of two jointly fitted, unregularized networks, and it amplifies noise. On the `vc_hetero` process the VC term reaches a correlation of about 0.99 with the true effect function. The CS reduced form reaches a clearly lower correlation on the same protocol, although the model is in-class. Causal forests and R-learners do not work because they target the effect. They work because they regularize it (Athey et al., 2019; Nie & Wager, 2021). VC brings that ingredient into the TRAM framework.

Semantics

- **Scale.** $\beta(x)$ lives on the node's latent scale, like an LS weight. A continuous node adds it, and an ordinal node subtracts it from the cutpoints. With no modifiers, `VC(t="T")` is `LS("T")` bit-exactly, so VC against LS is a nested question.
- **Penalty.** The objective is $\sum_i \text{NLL}_i + \lambda \|b_\Theta\|^2$ on the total-NLL scale, with `penalty=` as λ . It is a fixed Gaussian prior whose shrinkage vanishes as n grows, it never applies to β_0 , and $\lambda \rightarrow \infty$ recovers the LS fit. Raise it when the modifiers are many or n is small.
- **Identification.** A constant moves freely between β_0 and b_Θ . The head's output layer is zero-initialized, so $\beta(x) = \beta_0$ at step 0. After fit each node re-centers its head to mean zero over the training data, which preserves the function. β_0 is therefore the main effect in the training population, the Colr reading when β is constant.
- **Warm start.** Fit the all-LS version classically and copy the treatment weight into `beta0` before fit. Training then starts at the classical answer and learns only deviations.
- **Treatments.** Continuous or binary ordinal, entering as its 0/1 level so that β is the identified level-1-against-0 contrast. The term is linear in the treatment.
- **Read-out.** `flow.varying_coef(df, node)` evaluates $\beta(x)$ in closed form, deterministic and free of the outcome. `tramdag.plots.plot_varying_coef(flow, df, node, by=)` draws it along one modifier. For a continuous outcome, $\beta(x) = \text{logit } P(Y \leq y \mid x, \text{do}(T = 1)) - \text{logit } P(Y \leq y \mid x, \text{do}(T = 0))$ at every y . This is an interventional quantity and needs no counterfactual. Because `abduct` inverts a continuous node exactly, $\beta(x)$ also equals the abduction difference $u(x, 1, y) - u(x, 0, y)$, which a test pins. For an ordinal outcome `abduct` samples the latent, so this identity does not apply.

Propensity centering: `propensity="col"`

With a `propensity=` column, the term contributes $\beta(x)(t - \hat{e}(x))$ instead of $\beta(x)t$. This is Robinson's R-learner orthogonalization (Nie & Wager, 2021; Robinson, 1988) inside the likelihood. In model-based forests, this ingredient is decisive for effect estimation under confounding (Dandl et al., 2024). That finding reproduces here on the confounded process, where the model deliberately under-specifies its prognostic part. The uncentered $\hat{\beta}$ absorbs the misfit, and centering brings it back near the truth. `tests/test_vc_centered.py` requires at least a factor of two. If the prognostic part is correctly specified, centering changes little.

Centering guards against a misspecified prognostic part. This also applies to randomized data: with $\hat{e} = 0.5$ the regressor is $\pm \frac{1}{2}$, and a misfit that both arms share cannot follow it. Centering does not guard against unmeasured confounding. The propensity uses only the observed parents of the treatment, so a hidden common cause of T and Y biases $\beta(x)$ with or without centering.

The design is two-stage and frozen, because the naive versions are wrong.

- **Training** uses out-of-fold propensities, one value per row. Compute them and pass them as the training-frame column that `propensity=` names. Any propensity model works if a fit that never saw a fold predicts that fold. This is the cross-fitting requirement of double machine learning (Chernozhukov et al., 2018). In-sample propensities reintroduce the own-observation bias and can be worse than no centering. The values enter the loss as frozen data, so no gradient reaches the treatment node and the per-node factorization stays intact.

fit refuses a centered spec whose training frame lacks the column. The flow’s validation NLL uses the live propensity of the fitted treatment node.

- **Inference**, in `log_prob`, `sample`, `abduct`, `pmf` and `scores`, recomputes $\hat{e}$ from the flow’s own fitted treatment node on the current parent values, detached. Under `do(T=t)` the regressor becomes $t - \hat{e}(x)$. Nothing is cached.
- **Interpretation**. β_0 is the mean of $\beta(x)$ over the training rows, with or without centering, because fit re-centers the head in both cases. `varying_coef` is unchanged, because centering moves the regressor and not β . Centering requires a binary ordinal treatment.

Validation

The `vc_hetero` process in `tests/conftest.py` is a logistic-shift SCM with a known $\beta(x) = -1 + 0.8 X_2 - 0.6 X_3$. It has a nonlinear prognostic part and confounded assignment, and X_2 is confounder and modifier at once. That is the configuration in which the CS reduced form fails hard-est. `tests/test_vc_term.py` requires three things: a recovery correlation of at least 0.9 with 4500 training rows, a fitted β_0 that matches `fit_classical` under a large penalty, and the read-out identities. `tests/test_vc_centered.py` tests the centering claims on the confounded process. `scores.md` covers the scan that shortlists modifiers before a VC term is declared.

Paper replication: protocol, hyperparameters and results, per experiment

The eight variants under `experiments/` replicate the TRAM-DAG paper (Sick & Dürr, 2025) against its own R code (`tensorchiefs/tram-dag`). For each experiment this page lists the DGP, the model, every hyperparameter with its source, what deviates from the paper and why, and the numbers. The paper states five training numbers: $n = 40000$, 500 epochs, Adam, lr 0.001 and Bernstein `len_theta` 20 (order 19). It shows its results as figures. Where it gives no number, the “paper” column names the figure and what it shows. The pinned ground truth is `experiments/ground_truth/*.json`; `experiments/README.md` explains the YAML variants, the frozen data and the check.

Seeds are DGP 42, init 7 and shuffle 0. The triangle scripts of the reference run unseeded and the comparison scripts seed R’s RNG, which torch cannot replay, so every seed here is a repository choice.

What is common to all eight

item	reference	here
latent	standard logistic, shifts added on the continuous scale, subtracted for ordinal $P(Y \leq k) = \sigma(\vartheta_k - s)$	same (<code>notation.md</code>)

item	reference	here
Bernstein basis	len_theta unconstrained coefficients, to_theta softplus-cumsum, domain $\rightarrow [0, 1]$ with tangent-linear extrapolation outside; the domain comes from the train 5 %/95 % quantiles in the triangle scripts (quantile(..., c(0.05, 0.95))) and from min/max in the comparison scripts (scale_df)	zuko Bernstein, n_coeffs unconstrained (zuko ties two control points on, so n_coeffs=20 is order 21 against the reference's order 19; the free-parameter count is what matches), domain = train range_q/1-range_q quantiles $\rightarrow [-5, 5]$, linear extrapolation. Triangle: range_q: 0.05, a match up to the reparametrization. CAREFL: range_q: 0, the reference's min/max. VACA: quantiles kept, deviation D1, measured below
networks	triangle scripts create_param_net with hidden_features = c(2, 25, 25, 2) continuous and c(2, 2, 2, 2) mixed, sigmoid (the ReLU line is commented out); the vector reads as in/out dims around the hidden stack, so the hidden layers are (25, 25) and (2, 2). Comparison scripts make_model: dense(10, tanh) $\rightarrow$ dense(100, tanh) $\rightarrow$ dense(len_theta), one net per node	units and activation set per variant to exactly those stacks. The package defaults in code-map.md replicate the PyTorch reference buehlpa/TramDag instead and are not the paper's nets
init	triangle scripts: LinearMasked layers with Keras random_normal (N(0, 0.05 ²)) on weights and biases, the LS beta layer included; comparison scripts: layer_dense default, gloriot-uniform weights and zero biases	init: normal (triangle) and init: gloriot (VACA/CAREFL) through CausalFlowDAG(init=); torch's default init remains the framework default, and under the full-batch protocol the init decides the fit (see VACA)
optimizer	Keras Adam, eps 1e-7	torch Adam, eps 1e-8; measured: no effect (VACA identical to four digits)
calibrated start	none	marginal_init: true for the two triangle LS models, atan-cs and exp-cs, deviation D4 below; off where it moves the endpoint (linear-cs, sin-cs, VACA) and impossible for CAREFL's range_q: 0 domain. Calibration without the flag never touches the weights

item	reference	here
intercept output layer	Keras dense with bias	bias-free, deviation D3: the same function class, because the bias adds a constant to all unconstrained coefficients
plateau rule (VACA/CAREFL)	update_learning_rate: one optimizer, reduce when the summed validation NLL has not improved for 50 epochs (strict <), factor 0.1, min 1e-7	torch ReduceLR0n-Plateau(patience=49, threshold=0, threshold_mode="abs", factor=0.1, min_lr=1e-7), the same rule verified against torch's source, but per node on the node's own validation NLL, deviation D5 below; experiments/helpers.py::fit_paper drives it

Triangle, continuous (triangle.py): paper Sec. 6.1, App. C.3

DGP (summerof24/triangle_structured_continous.R):

- $x_1 \sim 0.5 N(0.25, 0.1^2) + 0.5 N(0.73, 0.05^2)$
- $h(x_2 | x_1) = 5 x_2 + 2 x_1 = u_2$
- $h(x_3 | x_1, x_2) = 0.63 x_3 - 0.2 x_1 - f(x_2) = u_3$, $u \sim \text{logistic}$

f has three forms: linear $-0.3 x$, atan $0.75 \text{atan}(5(x + 0.12))$, $\sin 2 \sin(3x) + x$. True weights in the flow's convention: $\beta_{12} = +2$, $\beta_{13} = -0.2$, $\beta_{23} = +0.3$ (linear only). A fitted CS learns $-f(x_2) + \text{const}$.

Model: x_1 : SI, x_2 : SI + LS(x_1), x_3 : SI + LS(x_1) + {LS(x_2) | CS(x_2)}, Bernstein, the CS net hidden (25, 25) with sigmoid.

hyperparameter	paper / R code	here
train / validation	dgp(40000) / dgp(40000), two draws	40000 / 40000, two draws
epochs	500	300 (linear-cs: 500), one continuous run; the CI deviation
lr	0.001 (optimizer_adam() default)	0.004, the CI deviation
batch	32 (Keras fit() default)	256, the CI deviation
len_theta / n_coefs	20	20
schedule / early stop / init	none / none, final weights / random_normal	none / none / init: normal
coefficient read-out	after every epoch (Keras loop)	fit(callbacks=), every epoch

Results

variant	metric	paper	paper protocol, init: normal (batch 32 / lr 0.001 / 500 epochs)	CI config, the pinned ground truth
linear-ls	$\beta_{12} / \beta_{13} / \beta_{23}$	Fig. 14 trajectories; Sec. 6.1 text: 1.98 / -0.21 / 0.26	1.987 / -0.170 / 0.282	1.981 / -0.161 / 0.281
linear-cs	β_{13} ; max $ \hat{g} - (-f) $ on $[-1, 1]$	Fig. 17: fitted CS is a straight line	-0.173; 0.122	-0.181; 0.025
atan-cs	β_{13} ; cs max err	Fig. 7 right / 15 / 16: CS on -f; text: $\beta_{12} = 2.07$, $\beta_{13} = -0.203$	-0.168; 0.059	-0.168; 0.104
sin-cs	β_{13} ; cs max err	Fig. 18: CS follows the non-monotone f on $x_2 \in [-1, 0]$	-0.195; 1.10	-0.168; 0.240 on the paper's $[-1, 0]$ window
all	$ E[x_3 \mid \text{do}(x_1 = -1)] \text{ flow} - \text{DGP} $	Figs. 16/17: histograms overlap	0.09 to 0.14	CS variants 0.075 to 0.100; linear-ls bound 0.267
all	val NLL x3	(none)	2.4607 to 2.4764	2.4606 to 2.4614

The pinned column gives the value of each entry in experiments/ground_truth/*.json. An error entry has only a max bound. Where the file's _note says that the bounds are 2.5x the measurement, the column gives the bound divided by 2.5. Otherwise it gives the bound, labeled as one. The triangle pins predate the marginal start (D4) and the per-node shuffle seeds. A CI run at the current config does not reproduce them exactly.

β_{13} measures -0.16 to -0.18 instead of -0.2. β_{13} multiplies x_1 , whose two mixture components sit at 0.25 and 0.73. So x_1 has sd 0.254 against 0.375 for x_2 and 2.918 for x_3 , and $\text{SE}(\beta_{13}) \approx 0.036$ at $n = 40000$. The value stays within about one SE. At $n = 5000$, β_{13} is too weakly identified for the frozen 5000-row CSVs to serve as fit checks. So the fit checks train on the paper's $n = 40000$ protocol.

The (25, 25) hidden stack is what lands sin-cs on Fig. 18, including its small -1-endpoint deviation. A literal [2, 25, 25, 2] reading puts a 2-sigmoid bottleneck on the input. It cannot reproduce the figure at any protocol (cs max err 1.22).

Triangle, mixed (triangle_mixed.py): paper Sec. 6.2, App. C.4, App. B

DGP (triangle_structured_mixed.R): x_1 and x_2 as above. x_3 is ordinal with four levels and cutpoints $\theta = (-2, 0.42, 1.02)$ from the R code, which the paper does not state. The level is $\#\{k : u_3 > \theta_k + 0.2 x_1 + f(x_2)\}$, with f either linear $-0.3 x$ or $\exp 0.5 \exp(x)$. The paper adds the ordinal shift and the flow subtracts it, so the fitted weights are $\beta_{13} = -0.2$, $\beta_{23} = +0.3$.

Model: x_1 : SI, x_2 : SI + LS(x_1), x_3 : OrdinalNode(4, LS(x_1) + {LS(x_2) | CS(x_2)}), the CS net hidden (2, 2) with sigmoid.

hyperparameter	paper / R code	here
train / validation	dgp(40000) / dgp(10000)	40000 / 10000, two draws

hyperparameter	paper / R code	here
epochs, lr, batch, schedule, init	500, 0.001, 32, none, random_normal	350 (exp-cs) / 200 (linear-ls) continuous, none, init: normal; lr 0.004 (linear-ls: 0.002) / batch 256, the CI deviations
n_coeffs (x1, x2)	20	20
odds-ratio check (App. C.4)	odds(x2 $\leq$ -1) under do(x1 += 1), theory $e^2 \approx 7.39$	40000 rows, seed 99
counterfactual PMF (App. B)	Fig. 10 explains why point counterfactuals fail for the ordinal node	2000 rows $\times$ 200 draws

Results

variant	metric	paper	paper protocol, init: normal (500 epochs)	CI config, pinned
linear-ls	$\beta_{12} / \beta_{13} / \beta_{23}$	Fig. 19: 2 / 0.2 / -0.3 (paper convention; -0.2 / +0.3 in the flow's)	1.987 / -0.246 / 0.319	1.978 / -0.258 / 0.333
linear-ls	odds ratio predicted / DGP	$e^2 \approx 7.39$; C.4 text: 7.74, CI [7.16, 8.38]	7.29 / 7.19	7.23 / 7.19
linear-ls	CF PMF TV vs analytic; P(true level) flow / analytic / mode bound	(App. B qualitative)	0.044; 0.718 / 0.728 / 0.806	TV bound 0.124; 0.716 / 0.728 / 0.806
exp-cs	β_{13} ; cs max err	Fig. 20: distributions match (none)	-0.207; 0.142	-0.205; 0.143
exp-cs	CF PMF TV; P(true level)		0.023; 0.917 / 0.921	0.023; 0.923 / 0.921

The pinned column follows the rule of the continuous table. The triangle-mixed-linear-ls note does not state the 2.5x rule, so its TV entry is the bound.

VACA / CNF benchmark (vaca.py): paper Sec. 5.1 and 5.2, App. C.1

DGP (Sánchez-Martín et al., 2022, App. E.1): $x_1 \sim 0.5 N(-2, 1.5) + 0.5 N(1.5, 1)$, $x_2 = -x_1 + N(0, 1)$, $x_3 = x_1 + 0.25 x_2 + N(0, 1)$. The noise is Gaussian, outside the logistic-latent family, and the all-CI flow must fit it. The analytic target is $E[x_3 \mid \text{do}(x_2 = a)] = -0.25 + 0.25 a$. The paper's Sec. 5.2 text says $a \in \{-3, -2, 0\}$. Its Fig. 5 panels and the R code (Figure_Triangle_Linear_Bimodal.R) intervene at $a \in \{-3, -1, 0\}$, and this repository follows the code. $\text{do}(x_2 = -3)$ is off-manifold extrapolation and takes a looser tolerance.

Model: x_1 : SI, x_2 : CI(x_1), x_3 : CI(x_1, x_2), Bernstein n_coeffs = 31 (reference M = 30, len_theta = 31), the make_model nets.

hyperparameter	R code	here
train / validation epochs	nTrain 2500 / dgp(5000) 10000 (comparison/Figure_Triangle_Linear_Bimodal.R: EPOCHS = 10000, M = 30, nTrain = 2500)	2500 / 5000, two draws 10000, one run, 1:1
lr	0.001	0.001
batch	full batch (one apply_gradients per epoch)	2500 = n_train
schedule	the plateau rule above	the same rule per node, deviation D5
input scaling	scale_df: everything min-max to [0, 1]	input_transform: minmax on the CI terms
Bernstein domain	train min/max (scale_df)	5 %/95 % quantiles (range_q: 0.05), deviation D1, chosen under the global plateau. At seed 7 the reference's min/max domain scored 0.289 / 0.040 / 0.067 against the quantiles' 0.096 / 0.080 / 0.018. Under the per-node plateau (D5) the order turns. Min/max scores 0.072 / 0.001 / 0.049 (val NLL 1.4342) against the quantiles' 0.080 / 0.068 / 0.092 (1.4347), so D1 is open for revision. CAREFL, same nets, uses range_q: 0
n_compare	(none)	50000

Results: the check is the flow's error against the analytic mean, not a pinned flow value.

metric	paper	here, the pinned ground truth
$ E[x_3 \mid \text{do}(x_2 = -3)] - (-1.0) $	Fig. 5: densities overlap	0.080
$ E[x_3 \mid \text{do}(x_2 = -1)] - (-0.5) $	Fig. 5	0.068
$ E[x_3 \mid \text{do}(x_2 = 0)] - (-0.25) $	Fig. 5	0.092
sd(x1) flow vs analytic 2.0767	Fig. 4: bimodal x1 fitted; the default CNF (Javaloy et al., 2023) fails	2.069, error 0.008
val NLL x3	(none)	1.4347

The do(x2) errors are seed-sensitive; D5 below gives the spread over four init draws. The paper shows one run's densities. The committed bounds are $2.5\times$ the seed-7 measurement.

The table below traces what each protocol ingredient is worth under the reference protocol with its global plateau, one change at a time. It uses the paper text's grid $-3 / -2 / 0$ and min-max inputs unless the row says otherwise.

variant	error	reading
raw parents into the nets	0.731 / 0.426 / 0.017	the tanh nets saturate: 40 % of the rows have $ x_1 > 2$, 43 % $ x_2 > 2$
global plateau, torch init	0.523 / 0.334 / 0.129	the summed NLL keeps improving through x_1 while x_3 overfits
... without any plateau	0.562 / 0.354 / 0.142	the rule is what stops the full-batch overfit
... Adam eps 1e-7	0.523 / 0.334 / 0.129	no effect
... Bernstein on train min/max (D1 off)	0.154 / 0.012 / 0.062	helps, not the cause
... glorot init	0.035 / 0.006 / 0.007	the cause (a different random draw than the config's seed 7)
... glorot + min/max	0.035 / 0.056 / 0.057	

A probe of the reference trajectory shows that the $\text{do}(x_2)$ errors descend until epoch 6000 to 8500 and then creep back up. The plateau anneal fires at about 9050 and freezes them. The reference protocol's quality is its anneal landing inside that window.

Rejected alternatives, each measured:

- relu with raw parents wanders 0.17 to 0.45 on $\text{do}(x_2 = -3)$. It holds the bound only at isolated epochs.
- tanh on raw parents saturates outright at 0.731.
- Minibatch stepping (batch 256, even with minmax + tanh) converges with a systematic -0.11 common-mode offset of all three do -means. The observational fit stays perfect.

CAREFL benchmark (carefl.py): paper Sec. 5.3, App. C.2

This benchmark is the reference run 1:1 on the reference's own data, except D5 (the plateau per node). `carefl_fig5.r` sets `USE_EXTERNAL_DATA = TRUE` and trains on CAREFL's own committed 2500 rows (`X.csv`, x_3/x_4 sd-standardized by 6.0104/1.9114) with `val = train`. CAREFL's repository also commits the observation `x0bs.csv`, the analytic truth curves and its own predictions on the grid `seq(-3, 2.9, 0.1)`. This repository freezes all of it under `experiments/data/carefl-cf`, external input with no generator, so the Fig. 6 curves are comparable point by point and every metric is in the reference's standardized units.

DGP (Khemakhem et al., 2021): $x_1, x_2 \sim \text{Laplace}(0, 1/\sqrt{2})$, $x_3 = x_1 + 0.5 x_2^3 + \varepsilon$, $x_4 = -x_2 + 0.5 x_1^2 + \varepsilon$ with $\varepsilon \sim \text{Laplace}(0, 1/\sqrt{2})$, x_3/x_4 divided by their sample sds. Counterfactuals are analytic by noise abduction. The observation is `x0bs.csv` = (2, 1.5, 0.8465, -0.2616); the paper prints (2, 1.5, 0.81, -0.28), slightly off it. The queries are $x_3^{\text{cf}} \mid \text{do}(x_2 = \alpha)$ and $x_4^{\text{cf}} \mid \text{do}(x_1 = \alpha)$ on the committed grid.

Model: x_1, x_2 : SI, x_3, x_4 : CI(x_1, x_2), Bernstein `n_coeffs` = 31, the same `make_model` nets as VACA, `range_q`: 0 for the reference's `scale_df` min-max Bernstein domain.

hyperparameter	R code (carefl_fig5.r)	here
train / validation	CAREFL's own <code>X.csv</code> , sd-standardized, <code>val = train</code>	the same committed <code>X.csv</code> , <code>val = train</code> ; a fresh standardized draw is scored, never trained or annealed on
epochs, lr	7000 @ 0.001	7000 @ 0.001
batch, schedule, input scaling, init	full batch, plateau 0.1/50/1e-7, <code>scale_df</code> , glorot	same, the plateau per node (D5)

hyperparameter	R code (carefl_fig5.r)	here
Bernstein domain scoring	train min/max (scale_df) the single x_{obs} , curves over α (Fig. 6)	train min/max (range_q: 0) the same, plus 300 held-out rows at $\alpha \in \{-1.5, 0, 1.5\}$, all in standardized units

Results, in standardized units.

metric	paper / CAREFL	here, the pinned ground truth
Fig. 6 max $ x_3^{\text{cf}} \text{ error} $	Fig. 6: flow tracks the DGP curve; CAREFL's committed x_3 preds err up to 1.21 at $\alpha = -3$	0.063
Fig. 6 max $ x_4^{\text{cf}} \text{ error} $	CAREFL's committed x_4 preds: max 0.174	0.204, at the $\alpha = -3$ grid edge; 0.074 at $\alpha = 0$
held-out CF MAE x_3 , $\alpha = -1.5$ / 0 / 1.5	(none)	0.014 / 0.008 / 0.013
held-out CF MAE x_4 , $\alpha = -1.5$ / 0 / 1.5	(none)	0.080 / 0.027 / 0.048

Three ingredients carry the agreement. Training on the committed rows instead of a fresh 2500-row draw lowers the Fig. 6 x_4 error. The reference optimization, 7000 @ 0.001 with `val = train`, fits the parabola bottom at $\alpha = 0$. The min-max domain lowers the errors at the grid edges.

On fresh draws the parabola bottom sits below the truth near $\alpha = 0$. The reason is neither protocol nor init. It is the finite-sample variance of the Laplace 12 % quantile at which the observed noise sits, in the sparse region near $x_1 = 2$. CAREFL's own committed draw is a mild one, which is one more reason to train on it.

Rejected alternatives:

- Raw parents saturate the sigmoid on the Laplace parents, as tanh does.
- relu underfits x_3 and grows a fragile Bernstein tail. One held-out row inverts to $x_4 \approx 580$.
- Minibatch underweights the sparse x_2 tail where the Fig. 6 grid ends. The held-out x_3 error at $\text{do}(x_2 = 1.5)$ and the Fig. 6 x_3 error both grow.

D4: the marginal start, measured per variant

`marginal_init` starts every simple intercept at its column's empirical marginal (fitting.md). The reference starts `bernp$beta` at zero. Measured on this machine at the CI protocol, one run per variant with the start off and on, seeds unchanged:

variant	epochs to within 0.01 of the final train NLL, off $\rightarrow$ on	endpoint, off $\rightarrow$ on	verdict
triangle linear-ls	28 $\rightarrow$ 5	identical (β to 4 digits, val NLL 2.4606)	on
triangle linear-cs	28 $\rightarrow$ 5	cs max err 0.025 $\rightarrow$ 0.056; $\text{do}(x_1)$ err 0.132 $\rightarrow$ 0.080	off: the misspecified line is what this variant measures, and the start bends it

variant	epochs to within 0.01 of the final train NLL, off → on	endpoint, off → on	verdict
triangle atan-cs	28 → 6	cs max err 0.108 → 0.074; do(x1) err 0.095 → 0.072	on
triangle sin-cs	34 → 10	cs max err 0.240 → 0.088; do(x1) err 0.075 → 0.201, past the 0.188 bound	off: a different optimum, better curve, worse L2
mixed linear-ls	42 → 9	identical	on
mixed exp-cs	28 → 6	cs max err 0.143 → 0.060; do(x1) err 0.018 → 0.010	on
VACA	within 0.5: 2245 → 168 (global plateau)	per-node plateau (D5): val NLL 1.4347 → 1.4347; do(x2) err 0.080 / 0.068 / 0.092 → 0.094 / 0.079 / 0.102	off: no gain, slightly worse at every do point
CAREFL	(none)	(none)	impossible: range_q: 0 has no marginal start

The start is only an initialization. Where the optimizer reaches the same basin, the endpoint is identical. Only the epoch count changes. Where the model has a network shift or the anneal decides the endpoint, the basin changes. Of the five such cases, two switch the start on and three leave it off. The triangle rows were measured before the per-node fit and its per-node shuffle seeds. The VACA endpoint was measured under the per-node plateau.

The pinned ground truth of the four variants that switched on is unchanged. Every metric stays within its tolerance. `check.py` prints advisory notes where a bound is far from the current measurement, and these notes are kept for the next deliberate re-pin. They cover the triangle-linear-cs do(x1) bound and the triangle-mixed-exp-cs cs max err and do(x1) bounds.

D5: the plateau rule per node

The reference reduces the learning rate of one optimizer when the summed validation NLL stops improving. Here every node fits on its own, so no summed NLL exists during a fit. Each node runs the same rule on its own validation NLL, with its own optimizer. The rule and its constants are unchanged.

The table gives the VACA do(x2) errors at $a = -3 / -1 / 0$ and the x3 validation NLL, one run per init seed on the same data.

init seed	per node (here)	global (reference)
1	0.038 / 0.047 / 0.074, 1.4344	0.331 / 0.051 / 0.037, 1.4514
2	0.038 / 0.032 / 0.076, 1.4351	0.213 / 0.044 / 0.041, 1.4476
3	0.070 / 0.005 / 0.039, 1.4345	0.221 / 0.066 / 0.048, 1.4481
4	0.012 / 0.037 / 0.072, 1.4345	0.317 / 0.053 / 0.098, 1.4541
7 (config)	0.080 / 0.068 / 0.092, 1.4347	0.096 / 0.080 / 0.018, 1.4495

The per-node rule gives a lower x3 validation NLL on every seed. Its error at the off-manifold point do(x2 = -3) is smaller on every seed, and much smaller on seeds 1 to 4. At do(x2 = 0) it is

worse on three of the five seeds, and seed 7 shows the largest gap. CAREFL moves by less than 0.003 in validation NLL, and all its bounds hold.

Runtime and the CI deviations

The reference protocol’s cost is the motivation for every deviation. The triangle runs 500 epochs $\times$ 1250 steps of batch 32. One epoch takes 3.2 s single-process at 2 to 4 torch threads and 5.5 s at 32 threads, because the step is overhead-bound. So one variant takes 42 to 69 min on the 2-core CI runners. The triangle scripts therefore use batch 256 at lr 0.004, 8 $\times$ fewer optimizer steps per epoch. This grid at the paper’s 500 epochs with glorot init chose them. Every entry is the cs max err unless the column says otherwise.

batch / lr / epochs	linear-ls β_{13}	linear-cs	atan-cs	mixed exp-cs (TV)	steps vs paper
32 / 0.001 / 500 (paper)	−0.170	0.110	0.080	0.071 (0.019)	1 $\times$
128 / 0.001 / 500	−0.170	0.170	0.041	0.126 (0.023)	1/4
128 / 0.004 / 500	−0.178	0.160	0.094	0.070 (0.014)	1/4
128 / 0.004 / 250	−0.170	0.147	0.070	0.131 (0.035)	1/8
256 / 0.004 / 500 (CI batch and lr)	−0.171	0.132	0.073	0.072 (0.017)	1/8
256 / 0.008 / 500	−0.181	0.163	0.113	0.073 (0.017)	1/8
512 / 0.010 / 500	−0.169	0.183	0.104	0.119 (0.015)	1/16

Batch 256 / lr 0.004 is the only row that keeps every cs error within about 0.02 of the paper protocol. Larger batches or rates bend the misspecified linear-cs curve (0.16 to 0.18), and lr 0.008 hurts atan-cs. At 500 epochs a triangle job takes 7 to 11 min instead of 42 to 69.

The epoch floors, measured at that batch and rate:

- **Triangle** holds every bound at 300 epochs. At 150 the run fails atan’s do(x1) bound (0.235 > 0.200). linear-cs keeps 500, because at 300 its fitted cs curve flattens past $|x_2| > 0.5$ (max err 0.140 against the pinned 0.025 at 500). The DGP line spans only ± 0.3 , and the sigmoid net’s slow tail convergence is half of that. relu is out, because the 2-unit input layer dies with it (atan: $\beta_{13} + 0.53$, cs err 1.42).
- **Triangle-mixed** exp-cs has two slow parts, the 2-unit sigmoid CS net and the do(x1) mean. At 100 / 250 / 350 epochs the cs max err is 0.344 / 0.177 / 0.156 and the do err 0.033 / 0.060 / 0.022, so it trains 350. linear-ls has no CS net to wait for and trains 200 epochs at lr 0.002. At that rate it reads the weakly identified β_{13}/β_{23} slightly closer to the 500-epoch values than at lr 0.004. relu dies at the sd-0.05 normal init (cs err 0.859, $\beta_{13} - 0.055$).
- **VACA and CAREFL** run their references 1:1 except D5 (and D1 for VACA); the rejected shortcuts are in their sections.

Repository choices the paper does not state

- the seeds,
- the validation draw sizes for the triangle scripts (the R code’s test),

- `n_compare`,
- `n_heldout = 300` and the α set $\{-1.5, 0, 1.5\}$, scored next to the single `x_obs`,
- the mixed cutpoints from the R code,
- the odds-ratio sample (40000 rows),
- the counterfactual-PMF sample (2000 rows $\times$ 200 draws).

Code map: every module of `src/tramdag/`, the public names and the private plumbing

This document has one entry per name, with its role and its place in the pipeline. Names in parentheses are private machinery. They are useful to know, and they are not part of the API. The last section lists every training hyperparameter and where it lives.

`spec.py`: declare the model

Every term is a `Term` subclass under its pythonic name. The paper's symbol is the same object, so `LS` is `LinearShift`.

Name	Role
<code>Term</code>	One additive term of a node's transformation, plain data: a subclass per term with name and module as class attributes and its options as the keyword arguments of <code>__init__</code> assigned to <code>self</code> , so <code>options()</code> is <code>dict(vars(term))</code> . <code>+</code> on terms builds plain lists. Carries the spec-level rules check, <code>edge_parents</code> , <code>cells</code> , <code>classical</code> and <code>from_serialized</code> . The contract is in <code>architecture.md</code> .
<code>SI()</code>	The parentless intercept, the paper's SI. Free transform parameters, the same for every row. Carries the transform choice (<code>transform=</code> , default "bernstein"); extra keyword arguments pass straight to the transform class.
<code>CI()</code>	The parent-conditioned intercept, the paper's CI: the parents reshape the monotone transform. Needs at least one parent. Also carries <code>units=</code> and <code>allow_interaction=</code> (joint vs. additive multi-parent intercept).
<code>Intercept / I</code>	The intercept term class: without parents the paper's SI, with parents the CI; <code>SI()/CI()</code> are the two spellings with their arity checked.
<code>LinearShift / LS</code>	Linear shift βx , the interpretable log-odds coefficient. Exactly one parent.
<code>ComplexShift / CS</code>	Complex shift: an NN $g(x)$, additive on the latent scale. Several parents form one joint network.
<code>VaryingCoefficient / VC</code>	Varying-coefficient shift $(\beta_0 + b_{\Theta}(\text{mod})) x_t$, the penalized treatment-effect head. <code>propensity=</code> names the propensity column for centering.
<code>ContinuousNode</code>	Continuous variable: monotone 1-D transform plus shifts. <code>terms</code> is the first positional argument.
<code>OrdinalNode</code>	Ordinal variable with levels classes: ordered logit (cutpoints) plus shifts.
<code>node_parents()</code>	Ordered de-duplicated parent names of a node (the canonical term list is <code>node.terms</code>).
<code>validate_and_sort()</code>	Edge-ownership validation plus Kahn topological sort. The returned order makes the flow triangular.

Name	Role
<code>spec_to_dict()</code> / <code>spec_from_dict()</code>	Checkpoint (de)serialization. A term serializes as {term, parents, options} with every option it carries; a hand-written spec may name only some, and the constructor fills the rest. A built-in term's term key is its name; a custom term writes its <code>module.ClassName</code> import path, which <code>_term_class</code> imports. <code>spec_from_dict</code> rejects a term without options, the node constructors normalize the formula, and <code>validate_and_sort</code> checks the DAG.
<code>(_normalize_terms,</code> <code>_check_term, _term_class,</code> <code>_checked_input_transform,</code> <code>_serialized)</code> <code>(_check_node, _kahn_sort)</code>	Formula flattening and per-entry validation (a + sum nested in a list is rejected), the one-parented-I rule plus transform hoisting in one pass. An option another term takes is refused by Python's own argument binding, as a <code>TypeError</code> . The stages behind <code>validate_and_sort</code> : parents exist, then each term's check (the VC treatment and centering rules), then edge ownership through <code>edge_parents</code> ; a term's own shape (arity, option values) and a node's own (ordinal levels, the transform) are checked when they are built. Kahn's sort emits ready nodes in sorted batches, so the order is deterministic.

transforms.py: the monotone map h and the ordinal transform

Name	Role
<code>StandardLogistic</code>	The TRAM base distribution: <code>log_prob</code> , <code>sample</code> (generator-aware), <code>icdf</code> .
<code>BernsteinUT</code>	Bernstein-polynomial transform, the default. Linear tail extrapolation follows the boundary derivative. <code>marginal_init_theta(column)</code> gives the start <code>marginal_init=True</code> applies.
<code>SplineUT</code>	Monotone rational-quadratic spline. Tails extrapolate with a fixed slope (zuko-upstream.md).
<code>AffineUT</code>	Monotone affine transform: the node-conditional is a logistic GLM.
<code>make_univariate_transform()</code>	Transform registry: name $\rightarrow$ transform instance.
<code>ordinal_cutpoints()</code>	Unconstrained $(n, K-1) \rightarrow$ increasing cutpoints with $\pm\text{inf}$ ends.
<code>ordinal_log_prob()</code>	$\log P(Y = y)$, computed in log space (model.md).
<code>ordinal_pmf()</code> / <code>ordinal_sample()</code> / <code>ordinal_abduct()</code>	Class probabilities / latent $\rightarrow$ level / truncated-logistic latent recovery (abduction) for ordinal nodes.
<code>ordinal_marginal_init_theta</code>	Cutpoint start that matches the empirical class frequencies (<code>marginal_init=True</code>).
<code>ordinal_bounds()</code>	The shifted cutpoint interval of each observed level. <code>scores.py</code> reads it for the latent-scale derivative.
<code>(_ScaledUT, _loglmexp)</code>	Quantile pre-scaling base class, whose inverse is zuko's with its closed-form tail. Stable $\log(1 - e^x)$.

flow.py: the model

Name	Role
CausalFlowDAG	The flow: one Node per variable in topological order. Construction seeds the weights.
fit()	Maximum likelihood node by node, serially or in <code>n_jobs</code> forked processes. The mechanics are in <code>fitting.md</code> .
fit_classical()	Float64 full-batch L-BFGS for all-ls specs, node by node; refuses flexible specs.
sample()	Observational, interventional (<code>do=</code> , graph mutilation) and counterfactual (<code>u=</code>) sampling.
abduct()	Abduction: recover the latents.
pmf()	Analytic class probabilities of an ordinal node, with <code>do=</code> overrides.
density()	Analytic conditional density of a continuous node on a grid, with <code>do=</code> overrides, the continuous counterpart of <code>pmf</code> .
log_prob() / nll()	Joint per-row log-likelihood, or a <code>nodes=</code> subset (exact, and the log-space way to get one node's conditional likelihood per row) / mean per-node NLL diagnostic.
varying_coef()	Closed-form read-out $\beta(x)$ of a fitted VC term. Deterministic, y-free.
scores() / effect_modifier_scan()	Analytic per-observation scores and the CUSUM modifier scan (delegate to <code>scores.py</code>).
intercept_contributions()	Post-hoc GAM-style decomposition of a complex intercept into mean-centered per-term parts.
ls_coefficients()	The per-node linear-shift weights, the interpretable coefficients.
design_matrix()	Parent encoding as a DataFrame (<code>drop_first=</code> gives the classical statsmodels/polr design).
to_matrix()	The labeled meta-adjacency matrix of term tags.
save() / load()	Checkpoints with the nodes' histories and provenance (version, time, device). <code>load</code> requires a complete checkpoint and fails loudly otherwise.
(<code>_node</code> , <code>_schema</code> , <code>_features</code> , <code>_tensorize</code> , <code>_generator</code> , <code>_dtype</code> , <code>_init_linear</code>)	Node lookup with one shared error; each node's parent schema; parent encoding through <code>Node.encode</code> (continuous raw, ordinal one-hot); <code>_tensorize(df, cols=None)</code> for any column subset, checking every ordinal column against its levels on the way in; seeded-generator and dtype plumbing.
(<code>_calibrate</code> , <code>_propensity</code> , <code>_side_feats</code> , <code>_query_side_columns</code>)	The flow-wide calibration (every node before the first fit, flags rolled back when a node refuses the frame), the generic side-column plumbing (each term names/validates/recomputes its own columns via the <code>ShiftModule</code> hooks) plus the binary propensity fit.

modules.py: the term modules

There is one module class per term, and the term class holds it as module (`ComplexShift.module` is `ComplexShiftModule`); the module holds the term's network and owns the runtime hooks (`architecture.md`).

The default architectures replicate the PyTorch reference `buehlpa/TramDag`, file `tram_models.py`, so a fitted model stays comparable to it. They are not the paper's nets, which `paper-replication.md` lists per experiment.

Name	Role
<code>intercept_module()</code>	The one factory: an I term becomes the free theta, one joint net, or one net per parent (<code>allow_interaction=False</code>).

Name	Role
feat_width()	Total encoded width of a parent set: levels per ordinal parent, 1 per continuous one.
ShiftModule / InterceptModule	The behavior hooks a term module owns: <code>__init__(term, schema)</code> (builds the net from the parent schema, sets key and parents), <code>shift_value/theta_value</code> , <code>post_init</code> , <code>regularizer</code> , <code>post-fit finalize</code> , <code>score_columns</code> , the side-input contract.
LinearShiftModule	LS: $\text{Linear}(n, 1, \text{bias}=\text{False})$. <code>.weight</code> is the interpretable coefficient; no bias because the intercept slot owns the constant.
ComplexShiftModule	CS: an NN to one shift value.
VaryingCoefficientModule	VC: $\beta_0 + b_{\Theta}(\text{mod})$ with the L2 hook <code>l2()</code> ; <code>beta()</code> evaluates the effect and <code>recenter()</code> re-splits <code>beta0/b_theta</code> after training (<code>varying-coefficients.md</code>). <code>regressor</code> is both the forward regressor and the <code>beta0</code> score.
SimpleInterceptModule / ComplexInterceptModule / AdditiveInterceptModule	The intercept slot: free <code>theta</code> (<code>I()</code>), one joint net from the parent features to the transform parameters, or one net per parent summed in coefficient space. The additive one holds its nets in <code>nets</code> .
(<code>_nn</code>)	The one NN builder: a stack of the given units with the term's activation (optional <code>batch_norm</code> before it), then a bias-free output layer.
(<code>_InputTransform</code>)	One term's frozen network-input transform (<code>minmax</code> / <code>standardize</code> / callable over frozen train columns).

nodes.py: the node model

Name	Role
Node	One sub-model per variable, a TRAM regression on its parents, exported as <code>tramdag.Node</code> : built from its name, node spec and parent schema <code>{parent: "continuous" \ n_levels}</code> ; builds its intercept and shift terms through each term's module; <code>theta_shift()</code> sums the terms' <code>shift_values</code> (plain shifts first, then VC); <code>net_input()</code> feeds every term network, <code>input_transform</code> applied.
Node <code>.log_prob</code> / <code>.sample</code> / <code>.abduct</code> / <code>.marginal_theta</code> / <code>.encode</code>	The continuous-vs-ordinal branches and the parent encoding (<code>architecture.md</code>).
Node <code>.features</code> / <code>.tensorize</code> / <code>.calibrate</code> / <code>.ls_coefficients</code> / <code>.save</code> / <code>.load</code>	The node's parent encoding from its schema, its columns as tensors (ordinal levels checked), its own calibration and calibrated flag, its LS coefficients, its checkpoint.
(<code>encode</code> , <code>schema_entry</code> , <code>tensorize</code> , <code>check_columns</code> , <code>check_level_values</code> , <code>write_checkpoint</code> , <code>load_weights</code>)	Shared by the node and the flow: one column's encoding, a node's schema entry, columns as tensors, the column and level checks, the checkpoint writer and loader.

fitting.py: NodeFitMixin and FitMixin

Name	Role
Node.fit() / Node.fit_classical() / Node.nll() fit() / fit_classical() (_split_validation, _normalize_callbacks, _check_fit_sizes, _learning_rates, _log_epoch, _fit_epoch, _FnCallback, _node_seed, _per_node, NodeFit- Mixin._check_classical, _fit_in_child, _pad_last, FitMixin._check_frames, FitMixin._fit_nodes, FitMixin._with_live_side)	One node's Adam loop and L-BFGS fit, mixed into Node via NodeFitMixin; row_log_prob, _recenter (the VC re-centering) and the side-column check live here too. The flow's loops over the nodes, methods of the flow via FitMixin; history is the flow's view over the nodes' histories. The loop plumbing: Keras-shaped validation split, callback normalization, the rate record, verbose printing, the per-node seeds and arguments, the frame checks before the first fit, the forked worker and the serial loop, the padded history view, the live propensity for the validation NLL.

readouts.py: ReadoutsMixin

Name	Role
shift_curve() the read-out methods	One fitted shift term on a 1-D grid, through the term's own shift_value. varying_coef, ls_coefficients, to_matrix, intercept_contributions, design_matrix, defined here once, methods of the flow via the mixin.

scores.py: effect-modifier detection

Name	Role
node_scores() effect_modifier_scan() sup_bb_pvalue() (_dl_ds, CRIT_5PCT)	Per-observation scores of the interpretable shift coefficients (scores.md). The fluctuation scan over candidate modifiers (scores.md). $P(\sup B > \text{stat})$ for a Brownian bridge B , the Kolmogorov series. Closed-form latent-scale derivative and the 5 % critical value. The per-term columns come from each term's score_columns hook.

callbacks.py: the shipped fit callbacks

Name	Role
EarlyStopping	The node's best-validation weights, restored at fit end (restore_best=False keeps the final ones); patience= also stops the node's fit, and min_delta= sets the smallest gain that counts.

plots.py: the figures (matplotlib optional: tramdag[plots])

Name	Role
plot_dag()	The labelled DAG of a spec or flow: layered left to right, ellipses for continuous and rounded boxes for ordinal nodes, every edge drawn by the term that owns it (LS / CS / CI / VC + modifiers, joint for a multi-parent net). Exported as tramdag.plot_dag. style= ("light", "dark" or a dict), node_kind= and modifiers= ("edge" points a VC modifier at its treatment edge).
plot_marginals()	Observed vs sampled marginal per node, one panel each; colors=, legend= and title=. Every plot function takes ax= to draw into a given layout.
plot_training()	Summed train/val NLL per epoch, with a dashed mark per stops= entry.
plot_varying_coef()	A VC term's $\beta(x)$ along one modifier, the other node columns at an observed median.
(_layout, _term_edges, _dag_edges, _bulges, _clear)	Longest-path layers with one barycenter sweep; the edge list with the VC treatment/modifier split; each edge's bend, the nearest one that misses every node. matplotlib is imported on the first call, never at package import.

What is *not* in the package

Three groups of files are research code:

- the SCM generators
- the frozen datasets
- the replication scripts

They live in experiments/, outside the installed package; experiments/README.md describes them and tests/README.md what the framework tests measure instead.

Where every training hyperparameter lives

Everything that shapes a fit is either a keyword you pass or a documented default you can read at the call site. Nothing numeric is buried.

Knob	Where	Default
learning rate, batch size validation, progress	fit() fit(validation_data=, validation_split=, verbose=)	1e-2 / 512 (in-repo callers state them explicitly anyway) validation off, verbose=0
schedules, early stopping	fit(optimizer=, callbacks=)	tramdag.callbacks ships EarlyStopping, per node; anything else is torch's lr_scheduler and a few lines of callback (fitting.md)
marginal start	fit(marginal_init=)	False: the zero start; applied only by the fit that calibrates a node, so a node that fit_classical calibrated ignores it
VC stage-1 propensities	the training-frame column VC(propensity=) names	required for a centered VC term (varying-coefficients.md)
VC penalty and centering	VC(penalty=, propensity=)	1.0 / None (propensity="col" names the propensity column)

Knob	Where	Default
L-BFGS budget	<code>fit_classical(max_iter=, history_size=)</code>	5000 / 50; torch's <code>tolerance_change</code> is 1e-9 and <code>tolerance_grad</code> is off. One full-batch run, no chunks; converged also needs <code>GRAD_TOL</code> (1e-2)
training budget	<code>fit(epochs=)</code>	required (fitting.md)
network widths	<code>units=</code> on I/CS/VC	(8, 8) / (64, 128, 64), parity with the PyTorch reference's default classes; VC's (16,) has no counterpart there and comes from the recovery measurement
activation	<code>activation=</code> on I/CS/VC	"relu" (the reference default classes); "sigmoid" and "tanh" are the paper's
batch norm	<code>batch_norm=</code> on I/CS/VC	False: neither reference uses it. True puts a <code>BatchNorm1d</code> between each hidden layer and its activation, so the fit needs more than one row per batch and inference needs <code>eval()</code> mode (fit and load leave the flow there)
transform class	<code>I(transform=, **kwargs)</code> (extra kwargs go to the transform class)	"bernstein", <code>n_coeffs=20</code> unconstrained coefficients (zuko-upstream.md on the order they give); <code>spline bins=8</code> , zuko's NSF default; the domain is fixed at [-5, 5], <code>transforms.BOUND</code>
shuffling / weight init	<code>fit(seed=)</code> / <code>CausalFlowDAG(seed=)</code>	init happens at construction, so the constructor seed is the reproducibility knob; a lone <code>Node</code> takes <code>torch.manual_seed</code> before <code>Node(...)</code>
weight init	<code>CausalFlowDAG(init=)</code>	"torch" (<code>nn.Linear</code> Kaiming-uniform); "glorot" = Keras Dense default, glorot-uniform weights and zero biases, the paper's reference; decisive under its full-batch protocol
network inputs	CI/CS/VC(<code>input_transform=</code>)	None, raw parents; "minmax", "standardize" or a callable <code>fn(x, train)</code> (model.md)

Upstream candidates for zuko

This document lists what tramdag works around in zuko (1.6.0). The list is ranked by value against effort, as candidate upstream PRs or issues. tramdag uses zuko in one place only. It uses these three transforms from `zuko.transforms`:

- `BernsteinTransform`
- `MonotonicRQSTransform`
- `MonotonicAffineTransform`

The three wrappers in `src/tramdag/transforms.py` hold them. The flow machinery, the base distribution and the DAG structure are tramdag's own. The suggested order of attack is $4 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 5$.

1. Analytic `call_and_ladj` for `BernsteinTransform`

`BernsteinTransform` inherits `MonotonicTransform.call_and_ladj`. That method gets the Jacobian by `torch.autograd.grad` in the forward pass. It therefore roughly doubles the training graph.

It also makes the per-node loss un-torch.compile-able. Autograd in the forward pass gives a double backward, and `aot_autograd` does not support one. Therefore the op-count axis stays closed in eager mode.

The derivative is closed form: $f'(x) = \text{order} \cdot \Delta\theta$ against the order-1 basis. zuko already computes it in `_offset_and_slope` for the tail slopes. An override is therefore ~25 lines and needs no

API change. `MonotonicRQSTransform` already has such an override. `tramdag` gets a free speedup and the `torch.compile` axis back.

2. Learnable/linear tails for `MonotonicRQSTransform`

`zuko` pads the knot derivatives with $\exp(0)=1$. It extrapolates as the identity outside $[-B, B]$. Therefore the tail slope stays fixed at 1 for every θ . This is the structural reason why spline consistently trails bernstein here, because $\sim 10\%$ of the data sits beyond the 5%/95% pre-scaling range.

Upstream, `zuko` can accept boundary derivatives of shape $(*, K+1)$, or it can take an opt-in `tails="linear"`. Identity tails are deliberate in the NSF design. The framing must therefore stay back-compatible. This change will delete the caveats and make spline a first-class transform choice.

3. Public inverse of `_constrain_theta`

`BernsteinUT.marginal_init_theta` inverts `zuko`'s `private _constrain_theta` (cumsum-of-softplus) in closed form. It hard-codes two internal details of that function:

- the $\log(2) \cdot n/2$ centering
- the tying of the first two and the last two diffs

Any upstream reparametrization therefore breaks the marginal start silently. A public `unconstrain_theta(theta)` that takes the target control points is ~ 15 lines upstream. It removes the coupling to these private details. The framing is “identity/linear initialization support”, a standard flow trick.

4. Docstring fix: the Bernstein θ -shape off-by-one

`zuko`'s docstring says that θ has shape $(*, M-2)$ for a degree- M polynomial. n unconstrained coefficients become $n+2$ control points, which is degree $n+1$. The correct claim is therefore $(*, M-1)$. This off-by-one is a real replication trap: order 21 against the paper's `len_theta=20`, which is order 19. The PR is trivial and an accept is near-certain.

5. Logistic in `zuko.distributions`

Neither `zuko` nor `torch` ships a Logistic distribution. `tramdag` therefore hand-rolls `StandardLogistic` in ~ 50 lines, with `log_prob`, `sample` and `icdf`. Logistic latents are standard in TRAM-DAGs and in discrete flows, and `zuko` has a precedent in `GeneralizedNormal`. But `zuko` can point at `TransformedUniform(SigmoidTransform().inv)` instead. `tramdag` keeps its `_U_EPS` clamp locally in both cases. This candidate has the lowest value of the five.

Anti-candidates (look upstreamable, are not)

- **Quantile pre-scaling and `_ScaledUT`**: this is a modeling choice. `zuko`'s `ComposedTransform` with `MonotonicAffineTransform` already composes it.
- **The ordinal ordered-logit transform and log-space `ordinal_log_prob`**: this transform is not a bijection. It stays outside `zuko`'s flow scope. It is `torch.distributions` material, if anywhere.
- **LS, CS, CI and VC terms, the marginal-start policy, propensity centering and scores**: these are TRAM-DAG semantics. `zuko`'s `MaskedMLP` already supports arbitrary adjacency. `tramdag` does not use it, because it needs per-term interpretability.
- **The spline slope knob and the Bernstein eps knob**: `zuko` already exposes both upstream.

Additive vs joint complex intercept: reading per-parent effects

A complex intercept groups its parents jointly, `CI("x1", "x2")`, or additively, `CI("x1", "x2", allow_interaction=False)`. `docs/model.md` states the rule, and `docs/interpretation.md` why the additive form needs `flow.intercept_contributions(data, node)` to be read.

This notebook fits both models and shows the difference in interpretation. The data have an interaction, so the joint model fits them better. Only the additive model splits into per-parent parts, and `intercept_contributions` gives these parts.

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

```
from tramdag import CI, CausalFlowDAG, ContinuousNode
from tramdag.callbacks import EarlyStopping
```

The data

$x_1, x_2 \sim U(-2, 2)$ and $x_3 = x_1 + x_2 + \gamma x_1 x_2 + \frac{1}{2} \varepsilon$, $\varepsilon \sim \text{Logistic}(0, 1)$. The term $\gamma x_1 x_2$ is an interaction of the two parents.

GAMMA = 0.6

```
def simulate(n, seed):
    rng = np.random.default_rng(seed)
    x1 = rng.uniform(-2, 2, n)
    x2 = rng.uniform(-2, 2, n)
    eps = rng.logistic(0, 1, n)
    x3 = x1 + x2 + GAMMA * x1 * x2 + 0.5 * eps
    return pd.DataFrame({"x1": x1, "x2": x2, "x3": x3})
```

```
train = simulate(4000, seed=1)
val = simulate(1000, seed=2)
train.head()
```

```
x1
x2
x3
0
0.047286
0.755745
1.239427
1
1.801855
0.126365
0.853817
2
```

```

-1.423362
0.970406
-0.294179
3
1.794598
1.590401
4.880817
4
-0.752674
-0.168717
-0.689205

```

Fit both models

The data and the node are the same. Only the grouping of the two intercept parents is different.

```

def make_flow(joint: bool):
    spec = {
        "x1": ContinuousNode(),
        "x2": ContinuousNode(),
        "x3": ContinuousNode(
            CI("x1", "x2") if joint else CI("x1", "x2", allow_interaction=False)
        ),
    }
    return CausalFlowDAG(spec, seed=0)

flow_joint = make_flow(joint=True)
flow_add = make_flow(joint=False)

for f in (flow_joint, flow_add):
    f.fit(
        train,
        epochs=1200,
        learning_rate=1e-2,
        validation_data=val,
        callbacks=EarlyStopping(), # keep the best-validation weights
    )

# The validation NLL is per row. The joint model can represent the interaction,
# so it has the lower validation NLL.
nll_joint = sum(flow_joint.nll(val).values())
nll_add = sum(flow_add.nll(val).values())
print(f"val NLL joint : {nll_joint:.4f}")
print(f"val NLL additive: {nll_add:.4f}")
print(
    f"gap over the {len(val)} validation rows: {len(val) * (nll_add - nll_joint):.1f}"
)
assert nll_joint < nll_add, "the joint model should fit the interaction better"
val NLL joint : 4.1693

```

```
val NLL additive: 4.2020
gap over the 1000 validation rows: 32.7
```

intercept_contributions: the centered decomposition

It returns the mean-centered (sum-to-zero) contribution of each parent's network to the transform parameters theta. The removed means collect in baseline. The centering is over the rows of the df that you pass.

```
res_add = flow_add.intercept_contributions(train, "x3")
res_joint = flow_joint.intercept_contributions(train, "x3")

print("additive terms :", list(res_add["contributions"])) # 'x1', 'x2': separable
print("joint terms    :", list(res_joint["contributions"])) # 'x1+x2': inseparable

# the centering: every column mean is about 0
col_means = {
    k: float(np.abs(v.mean(0)).max()) for k, v in res_add["contributions"].items()
}
print("per-term column means (≈0) :", col_means)
assert max(col_means.values()) < 1e-5

additive terms : ['x1', 'x2']
joint terms     : ['x1+x2']
per-term column means (≈0) : {'x1': 1.0779500314583856e-07, 'x2': 1.411438006471144e-07}
```

The structural difference: separable curves vs an entangled cloud

Pick the transform coefficient that varies most across the data and plot each term's centered contribution to it against a parent value.

- **Additive** (net(x1) depends only on x1): its contribution to any coefficient is a deterministic 1-D function of x1. The points collapse to a **clean curve**, and the same applies to x2. These curves *are* the per-parent partial effects, because the structure is separable.
- **Joint** (one network over both parents): the single x1+x2 component depends on *both*. Plotted against x1 alone it is a **cloud**, and the hidden x2 (color) sets its height. There is nothing to read off per parent. The joint panel picks its own most-varying coefficient and has its own y-axis.

```
c1 = res_add["contributions"]
c2 = res_add["contributions"]
cj = res_joint["contributions"]
k = int((c1.var(0) + c2.var(0)).argmax()) # most-varying coefficient, additive
kj = int(cj.var(0).argmax()) # most-varying coefficient, joint
x1v, x2v = train["x1"].values, train["x2"].values

fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
o1, o2 = np.argsort(x1v), np.argsort(x2v)
axes[0].plot(x1v[o1], c1[o1, k], color="#1b9e77", lw=2.5, label="net(x1) vs x1")
axes[0].plot(x2v[o2], c2[o2, k], color="#d95f02", lw=2.5, label="net(x2) vs x2")
axes[0].axhline(0, color="0.7", lw=0.8)
axes[0].set_title("additive: separable per-parent curves")
axes[0].set_xlabel("parent value")
axes[0].set_ylabel(f"centered contribution to theta[{k}]")
axes[0].legend()
```

```

sc = axes[1].scatter(x1v, cj[:, kj], c=x2v, s=8, alpha=0.6, cmap="coolwarm")
axes[1].axhline(0, color="0.7", lw=0.8)
axes[1].set_title("joint: against x1 a cloud (height set by x2)")
axes[1].set_xlabel("x1")
axes[1].set_ylabel(f"centered contribution to theta[{kj}]")
fig.colorbar(sc, ax=axes[1], label="x2")
fig.tight_layout()
plt.show()

```

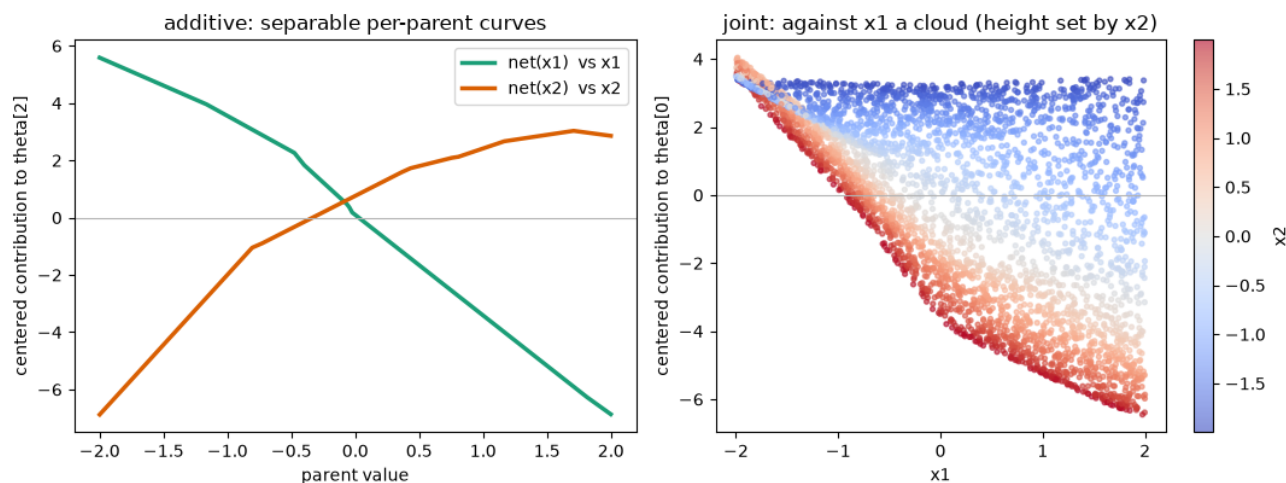


Figure 1: png

Takeaway

you want...	use	what you get
a per-parent partial-effect plot ("what does x1 do?")	additive CI("x1", "x2", allow_interaction=False)	intercept_contributions → mean-centered, separable components
interactions between parents in the transform	joint CI("x1", "x2")	one entangled network: flexible, not separable

The choice is between fit and interpretation. Here the joint model has the lower validation NLL, and only the additive model is readable per parent.

Caveat: the contributions live in the **unconstrained** parameter space of the transform. The additive terms are summed there, before the monotonicity constraint. Thus they are partial effects on the parameters. In general, they are not an additive split of the curve h itself.

Classical fitting of all-ls TRAM-DAGs (fit_classical)

When **every edge of a TRAM-DAG is a linear shift (LS)**, each node-conditional is a classical model

- an ordered-logit / proportional-odds model for ordinal nodes
- a Colr model (R's `tram::Colr`) for continuous ones.

For such a model `flow.fit_classical()` is the optimizer. It is a deterministic full-batch L-BFGS, and its report says when it reached the maximum-likelihood estimate (docs/fitting.md). This notebook checks the result against statsmodels and R.

Every R reference printed below comes from a real fit, typed in as a constant. notebooks/classical_fit_tram_dag.R holds all of these fits. To check them again, run Rscript notebooks/classical_fit_tram_dag.R from the repo root. It needs MASS (ships with R) and tram.

```
import time
from pathlib import Path

import numpy as np
import pandas as pd
import torch
from statsmodels.miscmodels.ordinal_model import OrderedModel

from tramdag import LS, SI, CausalFlowDAG, ContinuousNode, Node, OrdinalNode
from tramdag.callbacks import EarlyStopping

# repo-relative data, whether the notebook runs from the repo root or notebooks/
HERE = [Path.cwd(), *Path.cwd().parents]
REPO = next(p for p in HERE if (p / "pyproject.toml").exists())
DATA = REPO / "experiments" / "data"
NB_DATA = REPO / "notebooks" / "data"

def claim(text, value, bound, fmt="{:.2e}"):
    """Assert a computed quantity stays inside its bound, and print both.

    Every agreement this notebook asserts goes through here, so no number is
    typed into a markdown cell where it can drift away from the code. The
    prose says what is checked. The bound is the only number in the source,
    and it leaves room for another machine.
    """
    ok = "ok " if value <= bound else "FAIL"
    print(f"{ok}{text}: {fmt.format(value)} <= {fmt.format(bound)}")
    assert value <= bound, f"{text}: {fmt.format(value)} exceeds {fmt.format(bound)}"
```

0. The zeroth example: logistic regression on MASS::birthwt

Before any DAG, take a well-known model on a dataset that every R installation has. A **two-level** OrdinalNode whose terms are all LS is logistic regression. It is the same model, not an analogue. The ordinal transform is

$$P(Y \leq 0 \mid x) = \sigma(\theta_0 - \sum_p w_p x_p),$$

and with two levels there is a single cutpoint θ_0 , so

$$\text{logit } P(Y = 1 \mid x) = -\theta_0 + \sum_p w_p x_p.$$

The shift weights **are** the logistic-regression coefficients. The intercept is **minus** the cutpoint, because an ordinal node *subtracts* its shift (docs/notation.md). Section 2 uses the same model with $K - 1$ cutpoints instead of one.

The data is birthwt from **MASS**, the low-birth-weight study of Hosmer & Lemeshow. It has 189 births and the outcome low (birth weight under 2.5 kg). The predictors are age (mother's

age), lwt (mother's weight at last period, lbs) and smoke (smoking during pregnancy). `notebooks/data/birthwt.csv` holds these columns, exported verbatim from MASS. This is not a **causal** model.

The R you can copy-paste

`birthwt` ships with MASS, so nothing needs a download. Paste this into any R session to get the reference fit:

```
library(MASS)
m <- glm(low ~ age + lwt + smoke, data = birthwt, family = binomial)
coef(m)
logLik(m)
```

which prints

```
(Intercept)          age          lwt          smoke
 1.36822527 -0.03899458 -0.01213854  0.67076374
'log Lik.' -111.4396765 (df=4)
```

(R 4.2.3, MASS 7.3-58.2.) The cell below holds these numbers as `R_GLM`, so the notebook shows the three-way comparison without an R installation.

```
R_GLM = { # coef(glm(low ~ age + lwt + smoke, birthwt, family = binomial))
  "intercept": 1.3682252685,
  "age": -0.0389945827,
  "lwt": -0.0121385423,
  "smoke": 0.6707637407,
}
R_LOGLIK = -111.43967649
```

```
bw = pd.read_csv(NB_DATA / "birthwt.csv")
print(
  bw.head(3).to_string(index=False),
  f"\n\n{len(bw)} births, {bw['low'].sum()} of them low birth weight",
)
```

low	age	lwt	smoke	bwt
0	19	182	0	2523
0	33	155	0	2551
0	20	105	1	2557

189 births, 59 of them low birth weight

The spec. `low` is the outcome. `smoke` is a binary **parent**, so it is an ordinal node too. `age` and `lwt` are source nodes. The flow also models their marginals, which a plain `glm` does not. Here the marginals are a nuisance, so they get the cheapest transform (`SI(transform="affine")`). They cannot change the outcome node: the joint NLL decomposes per node, so the gradient of the `low` node never sees the covariate marginals.

```
spec_bw = {
  "age": ContinuousNode(SI(transform="affine")),
  "lwt": ContinuousNode(SI(transform="affine")),
  "smoke": OrdinalNode(2),
  "low": OrdinalNode(2, LS("age") + LS("lwt") + LS("smoke")),
}
```

```
flow_bw = CausalFlowDAG(spec_bw, seed=0)
flow_bw.fit_classical(bw)
```

```
{'converged': True,
'final_nll': 9.1376129294278,
'seconds': 0.8537037770000211,
'coefficients': {'low': {'age': array([-0.03899491]),
'lw': array([-0.01213854]),
'smoke': array([-0.13497349, 0.5358056 ])}},
'nodes': {'age': {'converged': True,
'stop_reason': 'tolerance',
'n_iter': 7,
'final_nll': 3.089470463926682,
'grad_norm': 1.0146105993900928e-06,
'seconds': 0.8174441849999994,
'coefficients': {}},
'lw': {'converged': True,
'stop_reason': 'tolerance',
'n_iter': 8,
'final_nll': 4.789085054515023,
'grad_norm': 1.1217799113443504e-05,
'seconds': 0.002510951000033401,
'coefficients': {}},
'smoke': {'converged': True,
'stop_reason': 'tolerance',
'n_iter': 5,
'final_nll': 0.6694294930549758,
'grad_norm': 2.9695561377940383e-08,
'seconds': 0.002773366999974769,
'coefficients': {}},
'low': {'converged': True,
'stop_reason': 'tolerance',
'n_iter': 41,
'final_nll': 0.5896279179311191,
'grad_norm': 4.893858897949371e-05,
'seconds': 0.030975274000013542,
'coefficients': {'age': array([-0.03899491]),
'lw': array([-0.01213854]),
'smoke': array([-0.13497349, 0.5358056 ])}}}
```

Now read the coefficients back out. age and lw are continuous parents, so they enter raw and their weights compare directly. smoke is ordinal, so it enters **one-hot over both levels**.

The coding is different from R. R gives a binary predictor one column. The flow gives it two, w_0 and w_1 , and only their difference is identified (docs/interpretation.md). This difference is the single smoke coefficient of R.

The second column also moves the intercept. With $s \in \{0, 1\}$, $\mathbf{1}[s = 0] = 1 - s$, so the one-hot pair collapses to

$$w_0 \mathbf{1}[s = 0] + w_1 \mathbf{1}[s = 1] = w_0 + (w_1 - w_0) s,$$

and the node reads

$$\text{logit } P(\text{low} = 1) = (-\theta_0 + w_0) + w_{\text{age}} \text{age} + w_{\text{lw}} \text{lw} + (w_1 - w_0) s.$$

The first bracket is the (Intercept) of R, which is $-\theta_0 + w_0$ and not $-\theta_0$. The last term is the smoke of R.

```

import statsmodels.formula.api as smf # noqa: E402

from tramdag.transforms import ordinal_cutpoints # noqa: E402

# the same formula as the R call above, and the same names in the output
res_bw = smf.logit("low ~ age + lwt + smoke", data=bw).fit(dispatch=False)

w = flow_bw.ls_coefficients()["low"]
with torch.no_grad(): # cutpoints are [-inf, theta_0, +inf]; take the finite one
    theta_0 = float(ordinal_cutpoints(flow_bw.nodes["low"].intercept(1))[0, 1])

rows_bw = [
    # -theta_0 + w_smoke[0]: the one-hot level-0 column is part of the intercept
    (
        "intercept",
        -theta_0 + float(w["smoke"]),
        res_bw.params["Intercept"],
        R_GLM["intercept"],
    ),
    ("age", float(w["age"]), res_bw.params["age"], R_GLM["age"]),
    ("lwt", float(w["lwt"]), res_bw.params["lwt"], R_GLM["lwt"]),
    (
        "smoke",
        float(w["smoke"] - w["smoke"]),
        res_bw.params["smoke"],
        R_GLM["smoke"],
    ),
]

print(f"{'':<11}{'fit_classical':>14}{'sm.Logit':>10}{'R glm':>10}{'max|diff|':>11}")
spreads = []
for name, flow_value, sm_value, r_value in rows_bw:
    spreads.append(max(abs(flow_value - sm_value), abs(flow_value - r_value)))
    print(
        f"{name:<11}{flow_value:>14.4f}{sm_value:>10.4f}{r_value:>10.4f}"
        f"{spreads[-1]:>11.2e}"
    )

claim("max coefficient spread of the three fits", max(spreads), 6e-5)


```

	fit_classical	sm.Logit	R glm	max diff
intercept	1.3682	1.3682	1.3682	1.54e-06
age	-0.0390	-0.0390	-0.0390	3.26e-07
lwt	-0.0121	-0.0121	-0.0121	7.07e-09
smoke	0.6708	0.6708	0.6708	1.53e-05

```

ok max coefficient spread of the three fits: 1.53e-05 <= 6.00e-05

Three optimizers land on the same maximum likelihood, because there is only one. They are L-
BFGS on the flow, the Newton solver of statsmodels and the IRLS of R. The log-likelihood of the
low node agrees with logLik(m) of R too. flow.pmf gives the fitted probabilities of glm row by
row.

# nll() is the *mean* NLL per node; times n and negated it is the log-likelihood
ll_flow = -flow_bw.nll(bw)["low"] * len(bw)
p_flow = flow_bw.pmf(bw, node="low")[:, 1]
p_classical = res_bw.predict(bw).values
print(f"log-likelihood flow {ll_flow:.6f} R glm {R_LOGLIK:.6f}")
claim("|logLik_flow - logLik_glm|", abs(ll_flow - R_LOGLIK), 3e-6)
claim(

```

```

    f"max |P_flow(low=1) - P_glm(low=1)| over {len(bw)} rows",
    float(np.abs(p_flow - p_classical).max()),
    1e-5,
)
log-likelihood    flow -111.439677    R glm -111.439676
ok |logLik_flow - logLik_glm|: 6.89e-07 <= 3.00e-06
ok max |P_flow(low=1) - P_glm(low=1)| over 189 rows: 3.20e-06 <= 1.00e-05

```

The same regression without the DAG. The marginals of age and lwt are a nuisance here, and a Node drops them. It is the low node alone. It takes its name, its node spec and the parent schema, {parent: "continuous" | n_levels}. The frame needs only low and its parents. Node.fit_classical finds the same maximum likelihood as the flow, R and statsmodels.

```

low = Node(
    "low", spec_bw["low"], {"age": "continuous", "lwt": "continuous", "smoke": 2}
)
low.fit_classical(bw[["age", "lwt", "smoke", "low"]])
w_low = low.ls_coefficients()
ll_low = -low.nll(bw) * len(bw)
print(f"log-likelihood    node alone {ll_low:.6f}    R glm {R_LOGLIK:.6f}")
print(
    f"smoke    node alone {float(w_low['smoke'] - w_low['smoke']):.4f}"
    f"    R glm {R_GLM['smoke']:.4f}"
)
assert abs(ll_low - R_LOGLIK) < 1e-4, "the lone node misses the glm likelihood"
log-likelihood    node alone -111.439677    R glm -111.439676
smoke    node alone 0.6708    R glm 0.6708

```

1. Continuous case

Section 0 modelled low, the *dichotomized* birth weight. birthwt also holds bwt, the weight in grams that low was cut from. Thus the same three predictors can be fitted against a continuous outcome. With a flexible monotone transform and a logistic latent, this is the Colr model of R:

```

tram::Colr(bwt ~ age + lwt + smoke, order = 21)

```

glm cannot fit this model. glm(family = gaussian) is lm, with a linear transform and a normal latent.

```

R_COLR_BWT = { # Colr(bwt ~ age + lwt + smoke, order = 21); R 4.2.3 / tram 1.0.4
    "age": -0.021971,
    "lwt": -0.010233,
    "smoke": 0.668698,
    "loglik": -1499.6678,
}

```

```

spec_bwt = {
    "age": ContinuousNode(SI(transform="affine")), # nuisance marginals
    "lwt": ContinuousNode(SI(transform="affine")),
    "smoke": OrdinalNode(2),
    "bwt": ContinuousNode(
        SI(n_coeffs=20) + LS("age") + LS("lwt") + LS("smoke") # degree 21
    ),
}

```

```

flow_bwt = CausalFlowDAG(spec_bwt, seed=0)
flow_bwt.fit_classical(bw)

```

```

w = flow_bwt.ls_coefficients()["bwt"]
fitted = {
    "age": w"age",
    "lwt": w"lwt",
    "smoke": w"smoke" - w"smoke", # one-hot: read the difference
    "loglik": -flow_bwt.nll(bw)["bwt"] * len(bw),
}

print(f"{'':<9}{'age':>9}{'lwt':>9}{'smoke':>9}{'logLik':>11}")
for name, r in [("flow", fitted), ("R Colr", R_COLR_BWT)]:
    print(
        f"{name:<9}{r['age']:>9.4f}{r['lwt']:>9.4f}{r['smoke']:>9.4f}"
        f"{r['loglik']:>11.2f}"
    )
diff = {k: abs(fitted[k] - R_COLR_BWT[k]) for k in fitted}
print(
    f"{'|diff|':<9}{diff['age']:>9.4f}{diff['lwt']:>9.4f}{diff['smoke']:>9.4f}"
    f"{diff['loglik']:>11.2f}"
)

```

	age	lwt	smoke	logLik
flow	-0.0223	-0.0103	0.6641	-1500.55
R Colr	-0.0220	-0.0102	0.6687	-1499.67
diff	0.0004	0.0000	0.0046	0.88

This agreement is not exact, and neither fit is the more correct one. The two libraries place the Bernstein basis differently. The flow puts it on the range_q quantiles with linear tails (docs/model.md). Colr uses the full range of the data. Thus the two fits maximize slightly different likelihoods.

A coefficient difference matters when it is large relative to the **sampling uncertainty** of the estimate. The next cells compute the standard errors and compare the differences with them.

Standard errors and confidence intervals

The sampling distribution of the maximum likelihood estimator is

$$\hat{\theta} \sim N(\theta, I^{-1})$$

where I is the Fisher information matrix. θ is unknown, so I is evaluated at $\hat{\theta}$. The standard errors are on the diagonal of I^{-1} . For a contrast c , the standard error is $\sqrt{c^\top I^{-1} c}$.

The **observed** Fisher information is the Hessian of the *negative* log-likelihood at the MLE,

$$I_{ab} = \frac{\partial^2(-\ell)}{\partial \theta_a \partial \theta_b} \bigg|_{\hat{\theta}},$$

which autograd gives directly. One backward pass gives the gradient with the graph retained. Then one more backward pass per component gives a row of I .

Here I is singular, so I^{-1} does not exist. The one-hot smoke pair and the Bernstein intercept have directions without curvature (see “weakly identified directions” in docs/fitting.md). I is symmetric, so it has an orthonormal eigendecomposition. The **pseudo-inverse** inverts only the directions that have curvature:

$$I = \sum_{k=1}^P \lambda_k e_k e_k^\top \quad \Rightarrow \quad I^+ = \sum_{k: \lambda_k > \tau} \frac{1}{\lambda_k} e_k e_k^\top, \quad \tau = 10^{-7} \lambda_{\max}.$$

Terms with $\lambda_k \leq \tau$ are dropped and not inverted. This is correct for a contrast that avoids the flat subspace. The leak column measures how far a contrast enters it:

$$\text{leak}(c) = \max_{k: \lambda_k \leq \tau} |c^\top e_k|.$$

Only interpret rows whose leak is close to zero. The choice of τ is not critical. The spectrum splits into a flat block and a curved block, with orders of magnitude between them. Any threshold inside that gap gives the same answer. The cell below measures the gap.

```
from scipy.stats import norm # noqa: E402
```

```
def observed_information(flow, node, data):
    """Give one node's NLL Hessian, the gradient norm, and parameter offsets."""
    flow.double() # second derivatives need float64
    names, params = zip(*flow.nodes[node].named_parameters(), strict=True)
    nd = flow.nodes[node]
    nll = -nd.row_log_prob(nd.tensorize(data)).sum()
    grad = torch.cat(
        [g.reshape(-1) for g in torch.autograd.grad(nll, params, create_graph=True)]
    )
    hess = torch.stack(
        [
            torch.cat(
                [
                    h.reshape(-1)
                    for h in torch.autograd.grad(grad[i], params, retain_graph=True)
                ]
            )
            for i in range(grad.numel())
        ]
    )
    offsets, at = {}, 0
    for name, p in zip(names, params, strict=True):
        offsets[name] = at
        at += p.numel()
    result = hess.detach().numpy(), float(grad.detach().norm()), offsets
    flow.float() # back to the stored precision
    return result
```

```
def spectrum_split(evals, tau):
    """Describe the curvature spectrum either side of ``tau``.
```

A wide gap between the flat block and the curved block is what makes the threshold uncritical, so this reports both edges of the gap and lets the caller check it instead of describing it in prose.

```
    """
    absolute = np.abs(evals)
    flat, curved = absolute[absolute <= tau], absolute[absolute > tau]
```

```

return {
    "n_flat": int(flat.size),
    "flat_max": float(flat.max()) if flat.size else 0.0,
    "curved_min": float(curved.min()) if curved.size else 0.0,
}

def ls_conf_int(flow, node, data, level=0.95):
    """Give Wald intervals for the linear-shift coefficients of one node."""
    hess, grad_norm, offsets = observed_information(flow, node, data)
    evals, evecs = np.linalg.eigh(hess)
    rcond = 1e-7
    tau = rcond * np.abs(evals).max() # one threshold, used for both
    flat = evecs[:, np.abs(evals) <= tau] # the directions without curvature
    cov = np.linalg.pinv(hess, rcond=rcond) # invert only the rest
    z = norm.ppf(0.5 + level / 2)

    rows = []
    for parent, w in flow.ls_coefficients()[node].items():
        c = np.zeros(hess.shape[0])
        key = next(n for n in offsets if n.startswith(f"shifts.{parent}"))
        if len(w) == 1: # continuous parent: one weight
            c[offsets[key]], estimate, term = 1.0, w[0], parent
        else: # ordinal parent: the contrast w[1] - w[0]
            c[offsets[key]], c[offsets[key] + 1] = -1.0, 1.0
            estimate, term = w[1] - w[0], f"{parent} (1 vs 0)"
        se = float(np.sqrt(c @ cov @ c))
        rows.append(
            {
                "term": term,
                "estimate": estimate,
                "se": se,
                "lower": estimate - z * se,
                "upper": estimate + z * se,
                # 0 = the contrast avoids the flat subspace, so the SE is real
                "leak": float(np.abs(c @ flat).max()) if flat.size else 0.0,
            }
        )
    diagnostics = {
        "n_params": hess.shape[0],
        "grad_norm": grad_norm,
        **spectrum_split(evals, tau),
    }
    return pd.DataFrame(rows).set_index("term"), diagnostics

table, diag = ls_conf_int(flow_bwt, "bwt", bw)
print(table.round(6).to_string())
print(
    f"\n{diag['n_params']} parameters, {diag['n_flat']} of them without curvature"
    f" | grad | = {diag['grad_norm']:.1e}"
)
# Colr(bwt ~ age + lwt + smoke, order = 21) on birthwt; R 4.2.3 / tram 1.0.4,
# produced by classical_fit_tram_dag.R. This is the oracle, not the claim.
R_COLR_BWT_SE = {"age": 0.025305, "lwt": 0.004116, "smoke": 0.260131}

print()

```

```

claim(
  "flat_max / curved_min, so smaller is a wider gap",
  diag["flat_max"] / diag["curved_min"],
  1e-4,
)
rows_se = [("age", "age"), ("lwt", "lwt"), ("smoke (1 vs 0)", "smoke")]
se_gap = max(abs(float(table.loc[t, "se"]) - R_COLR_BWT_SE[r]) for t, r in rows_se)
claim("max |SE_flow - SE_Colr|", se_gap, 1e-3)
claim(
  "max |coef_flow - coef_Colr| / SE",
  max(diff[r] / float(table.loc[t, "se"]) for t, r in rows_se),
  0.07,
)

```

term	estimate	se	lower	upper	leak
age	-0.022340	0.025265	-0.071858	0.027177	0.0
lwt	-0.010276	0.004113	-0.018337	-0.002216	0.0
smoke (1 vs 0)	0.664066	0.259577	0.155305	1.172828	0.0

24 parameters, 13 of them without curvature |grad| = 2.1e-02

ok flat_max / curved_min, so smaller is a wider gap: 4.98e-07 <= 1.00e-04

ok max |SE_flow - SE_Colr|: 5.54e-04 <= 1.00e-03

ok max |coef_flow - coef_Colr| / SE: 1.78e-02 <= 7.00e-02

The standard errors of all three parents agree with those of Colr. The confidence intervals of `confint()` on a Colr fit are Wald intervals too. The coefficient differences above are a small fraction of one standard error.

1b. A bimodal DAG: the demo data

This is the VACA benchmark triangle $x_1 \rightarrow x_2 \rightarrow x_3 \leftarrow x_1$, the bimodal SCM of the demo notebook. The model is **all-ls**. Each node is a Colr model, a Bernstein baseline with linear shifts.

The model is misspecified on purpose. The noise of the DGP is Gaussian, and the TRAM latent is logistic. Thus the all-ls model is not the true generator. `fit_classical` still finds its MLE.

if False:

```

def sample_vaca(n, seed):
    """Draw n rows from the VACA bimodal triangle SCM."""
    rng = np.random.default_rng(seed)
    mix, a, b = rng.uniform(size=n), rng.normal(size=n), rng.normal(size=n)
    x1 = np.where(mix < 0.5, -2.0 + np.sqrt(1.5) * a, 1.5 + b)
    x2 = -x1 + rng.normal(size=n)
    x3 = x1 + 0.25 * x2 + rng.normal(size=n)
    return pd.DataFrame({"x1": x1, "x2": x2, "x3": x3})

df = sample_vaca(1000, seed=42)
df.to_csv(NB_DATA / "vaca.csv", index=False)

```

The R you can copy-paste

`tram::Colr` fits the same model, a Bernstein baseline with linear shifts. The cell above wrote `notebooks/data/vaca.csv`. The file is tracked, so R reads the same rows that the flow is fitted on.

The two libraries **count the basis differently**. The comparison is only meaningful at the same polynomial degree. `N_COEFFS = 20` below is order = 21 in `tram`, because `n_coeffs` counts unconstrained coefficients and `zuko` adds two control points (`docs/zuko-upstream.md`).

```
library(tram)
d <- read.csv("notebooks/data/vaca.csv")

m2 <- Colr(x2 ~ x1, data = d, order = 21)
coef(m2)      # -> x1 1.800042
logLik(m2)    # -> -1401.213542

m3 <- Colr(x3 ~ x1 + x2, data = d, order = 21)
coef(m3)      # -> x1 -1.777289 x2 -0.455282
logLik(m3)    # -> -1411.901958
```

The coefficients need **no sign flip**: `Colr` reports the shift on the same side as the flow.

```
R_COLR = { # Colr(..., order = 21) on vaca.csv; R 4.2.3 / tram 1.0.4
  "x2": {"coef": {"x1": 1.800042}, "loglik": -1401.213542},
  "x3": {"coef": {"x1": -1.777289, "x2": -0.455282}, "loglik": -1411.901958},
}

df = pd.read_csv(NB_DATA / "vaca.csv")
# stated rather than left to the default, for the reason the markdown cell
# above the R block gives
N_COEFFS = 20 # -> Bernstein degree 21, which is tram's `order = 21`

spec_vaca = {
  "x1": ContinuousNode(SI(n_coeffs=N_COEFFS)),
  "x2": ContinuousNode(SI(n_coeffs=N_COEFFS) + LS("x1")),
  "x3": ContinuousNode(SI(n_coeffs=N_COEFFS) + LS("x1") + LS("x2")),
}

flow_c = CausalFlowDAG(spec_vaca, seed=0)
rep = flow_c.fit_classical(df)
print(f"\n{'':<12}{'fit_classical':>14}{'R Colr':>12}{'|diff|':>10}")
loglik = {k: -v * len(df) for k, v in flow_c.nll(df).items()} # nll() is a mean
for node, parents in flow_c.ls_coefficients().items():
  for p, w in parents.items():
    c, r = w[0], R_COLRnode[p]
    print(f"{p + ' -> ' + node:<12}{c:>14.4f}{r:>12.4f}{abs(c - r):>10.4f}")
  c, r = loglik[node], R_COLRnode
  print(f"{'logLik ' + node:<12}{c:>14.4f}{r:>12.4f}{abs(c - r):>10.4f}")
```

	fit_classical	R Colr	diff
x1 -> x2	1.7986	1.8000	0.0014
logLik x2	-1401.8707	-1401.2135	0.6572
x1 -> x3	-1.7796	-1.7773	0.0024
x2 -> x3	-0.4549	-0.4553	0.0004
logLik x3	-1410.9657	-1411.9020	0.9363

Why this one is not an exact-MLE check

The coefficients here agree far less closely than in Sections 0 and 2. The cause is the basis placement of Section 1. The shift coefficients are robust to it: their relative gap stays small.

```
gaps = [
  (abs(float(w[0]) - R_COLRnode[parent]), abs(R_COLRnode[parent]))
```

```

    for node in R_COLR
    for parent, w in flow_c.ls_coefficients()[node].items()
]
claim("max |coef_flow - coef_Colr|", max(g for g, _ in gaps), 0.01)
claim("max relative coefficient gap", max(g / r for g, r in gaps), 0.02)
ok max |coef_flow - coef_Colr|: 2.36e-03 <= 1.00e-02
ok max relative coefficient gap: 1.33e-03 <= 2.00e-02

```

The fitted conditional density

`flow.density(df, node, grid)` evaluates $p(\text{node} = g \mid \text{pa})$ at each grid value, in closed form from the transform and with no sampling. It is the continuous counterpart of `flow.pmf`.

Two checks are worth making on it, and neither needs an external reference. A density must integrate to one over a wide enough grid, and its mode must track the conditional mean that the linear shifts imply. The first is the stronger check, because it exercises the log-determinant of the transform rather than only its location.

```

grid = np.linspace(df["x3"].min() - 2, df["x3"].max() + 2, 601)
rows = df.iloc[:200]
dens = flow_c.density(rows, "x3", grid) # (200, 601)
mass = np.trapezoid(dens, grid, axis=1)
claim("worst |integral of the density - 1|", float(np.abs(mass - 1.0).max()), 5e-3)

# the mode should move with the parents in the direction the shifts say: a
# continuous node adds the shift to h(x), so a positive weight lowers the value
modes = grid[dens.argmax(axis=1)]
shift = -(
    float(flow_c.ls_coefficients()["x3"][0]) * rows["x1"]
    + float(flow_c.ls_coefficients()["x3"][0]) * rows["x2"]
)
corr = float(np.corrcoef(modes, shift)[0, 1])
print(f"corr(mode of the fitted density, the linear predictor) = {corr:.4f}")
assert corr > 0.95, f"the density's mode does not track the shifts: r = {corr:.4f}"

ok worst |integral of the density - 1|: 1.29e-03 <= 5.00e-03
corr(mode of the fitted density, the linear predictor) = 0.9951

```

Classical against Adam: the same optimum. An Adam fit with `EarlyStopping` reaches the same coefficients. Here the training rows also serve as the validation rows. The guarantees of the classical fit are determinism and the exact maximum-likelihood estimate, not speed. Which fit is faster depends on the model. The cell prints both timings, because a wall clock is a property of the machine.

```

flow_a = CausalFlowDAG(spec_vaca, seed=0)
t0 = time.perf_counter()
flow_a.fit(
    df,
    epochs=2000,
    batch_size=4096,
    learning_rate=1e-1,
    validation_data=df,
    callbacks=EarlyStopping(patience=60),
)
t_adam = time.perf_counter() - t0

coef_c, coef_a = flow_c.ls_coefficients(), flow_a.ls_coefficients()

```

```

print(f"{'coef':<10}{'classical':>12}{'adam':>12}{'|diff|':>10}")
adam_gaps = []
for node, p in [("x2", "x1"), ("x3", "x1"), ("x3", "x2")]:
    c, aw = coef_cnode[0], coef_anode[0]
    adam_gaps.append(abs(c - aw))
    print(f"{p}->{node:<6}{c:>12.4f}{aw:>12.4f}{adam_gaps[-1]:>10.4f}")
print(f"\nclassical {rep['seconds']:.2f}s vs adam {t_adam:.1f}s")
claim("max |coef_classical - coef_adam|", float(max(adam_gaps)), 1e-3)

coef      classical      adam      |diff|
x1->x2      1.7986      1.7986      0.0001
x1->x3     -1.7796     -1.7800      0.0003
x2->x3     -0.4549     -0.4551      0.0002

classical 1.66s vs adam 15.3s
ok max |coef_classical - coef_adam|: 3.08e-04 <= 1.00e-03

```

2. Ordinal case

For an **ordinal** outcome the classical model is the ordered logit, which `statsmodels.OrderedModel` fits, so here the equivalence is checkable in Python. The data is the paper's mixed triangle (experiments/data/triangle-mixed/linear): x3 has four levels and the two continuous parents x1 and x2; docs/paper-replication.md states the DGP.

```

tm = pd.read_csv(DATA / "triangle-mixed" / "linear" / "obs.csv")

spec_tm = {
    "x1": ContinuousNode(),
    "x2": ContinuousNode(LS("x1")),
    "x3": OrdinalNode(4, LS("x1") + LS("x2")),
}

flow_o = CausalFlowDAG(spec_tm, seed=0)
flow_o.fit_classical(tm)

{'converged': True,
 'final_nll': 1.063860203449753,
 'seconds': 2.4709746849999874,
 'coefficients': {'x2': {'x1': array([2.02988766])},
                  'x3': {'x1': array([0.00963845]), 'x2': array([0.38323205])}},
 'nodes': {'x1': {'converged': True,
                  'stop_reason': 'tolerance',
                  'n_iter': 124,
                  'final_nll': -0.5284964805045219,
                  'grad_norm': 5.95125877766619e-06,
                  'seconds': 0.8075660579999635,
                  'coefficients': {}},
           'x2': {'converged': True,
                  'stop_reason': 'tolerance',
                  'n_iter': 244,
                  'final_nll': 0.3826478325621656,
                  'grad_norm': 3.2687550894614076e-06,
                  'seconds': 1.6484672330000194,
                  'coefficients': {'x1': array([2.02988766])}},
           'x3': {'converged': True,
                  'stop_reason': 'tolerance',
                  'n_iter': 13,

```

```
'final_nll': 1.2097088513921095,
'grad_norm': 9.188159460024257e-07,
'seconds': 0.014941394000004493,
'coefficients': {'x1': array([0.00963845]), 'x2': array([0.38323205])}]})
```

Refitting the same design in statsmodels

`flow.design_matrix(df, node, drop_first=True)` gives the encoding that the flow gives its own shifts. Thus `statsmodels` fits the same design, not a different one.

```
design = flow_o.design_matrix(tm, "x3", drop_first=True)
res = OrderedModel(tm["x3"].astype(int), design, distr="logit").fit(
    method="bfgs", disp=False
)

fitted = flow_o.ls_coefficients()["x3"]
print(f"{'coefficient':<14}{'fit_classical':>14}{'statsmodels':>13}{'|diff|':>9}")
for name in ("x1", "x2"):
    a_, b_ = float(fittedname), res.params[name]
    print(f"{'name':<14}{'a_':>14.4f}{'b_':>13.4f}{'abs(a_ - b_):>9.4f}")
```

coefficient	fit_classical	statsmodels	diff
x1	0.0096	0.0097	0.0000
x2	0.3832	0.3832	0.0000

Standard errors

The helper from Section 1 needs no change. `statsmodels` reports the same quantities, so every number below has an external check.

```
table_o, diag_o = ls_conf_int(flow_o, "x3", tm)
ci_sm = res.conf_int()
print(
    f"{'':<6}{'flow SE':>9}{'sm SE':>8}{'|est|/SE':>10}    "
    f"{'flow 95%':>18}{'statsmodels 95%':>20}"
)
for term in ("x1", "x2"):
    r = table_o.loc[term]
    print(
        f"{'term':<6}{'r['se']':>9.4f}{'res.bse[term]:>8.4f}"
        f"{'abs(r['estimate']) / r['se']':>10.2f}    "
        f"{'[r['lower']:+.3f], {r['upper']:+.3f}]'    "
        f"{'[ci_sm.loc[term, 0]:+.3f], {ci_sm.loc[term, 1]:+.3f}]"
    )

se_gap = max(
    abs(float(table_o.loc[term, "se"]) - float(res.bse[term])) for term in ("x1", "x2")
)
claim("max |SE_flow - SE_statsmodels|", se_gap, 1e-3)
```

	flow SE	sm SE	est /SE	flow 95%	statsmodels 95%
x1	0.1080	0.1080	0.09	[-0.202, +0.221]	[-0.202, +0.221]
x2	0.0735	0.0735	5.21	[+0.239, +0.527]	[+0.239, +0.527]

ok max |SE_flow - SE_statsmodels|: 1.13e-06 <= 1.00e-03

The standard errors agree with `statsmodels`. The cell below checks that the coefficient gap is a small fraction of one standard error.

```

print(f"|grad| at the classical optimum: {diag_o['grad_norm']:.1e}")
for term in ("x1", "x2"):
    observed = abs(float(table_o.loc[term, "estimate"]) - float(res.params[term]))
    print(f"{term:<4} |flow - statsmodels| = {observed:.2e}")
    claim(
        f"{term}: gap against statsmodels / SE",
        observed / float(table_o.loc[term, "se"]),
        1e-3,
    )

|grad| at the classical optimum: 4.3e-03
x1 |flow - statsmodels| = 1.85e-05
ok x1: gap against statsmodels / SE: 1.71e-04 <= 1.00e-03
x2 |flow - statsmodels| = 1.71e-05
ok x2: gap against statsmodels / SE: 2.33e-04 <= 1.00e-03

```

3. Warm start: classical fit, then further training

`fit_classical` leaves the model at the MLE in float32, ready for any normal operation. A continued `fit()` from the classical solution *stays put*. This confirms that the classical solution is the optimum. It also makes the classical fit a warm start for further training.

```

before = {k: v.copy() for k, v in flow_o.ls_coefficients()["x3"].items()}

# continue training from the classical solution with a gentle Adam phase
flow_o.fit(tm, epochs=300, learning_rate=1e-3, batch_size=256)
after = flow_o.ls_coefficients()["x3"]

print("coefficient drift after 300 more Adam epochs from the classical MLE:")
for p in ["x1", "x2"]:
    d = float(np.abs(after[p] - before[p]).max())
    claim(f"{p}: max |drift| / SE", d / float(table_o.loc[p, "se"]), 0.1)

coefficient drift after 300 more Adam epochs from the classical MLE:
ok x1: max |drift| / SE: 2.78e-02 <= 1.00e-01
ok x2: max |drift| / SE: 2.89e-02 <= 1.00e-01

```

TRAM-DAG: one causal model, all three rungs

A TRAM-DAG (Sick & Dürr, 2025) is one normalizing flow wired like the adjacency matrix of a causal DAG. Fit it once on observational data. Then answer all three rungs of Pearl's ladder with the same fitted model:

rung	query	call
L1 association	$p(x)$, sampling	<code>flow.log_prob(df),</code> <code>flow.sample(n)</code>
L2 intervention	$p(x \mid do(x_j=a))$	<code>flow.sample(n, do={...})</code>
L3 counterfactual	what would x_i have been, had x_j been a ?	<code>u = flow.abduct(df), then</code> <code>flow.sample(do={...}, u=u)</code>

This notebook uses the first benchmark of the paper. It is a three-variable process with a bimodal source, and its ground truth is known analytically. Every claim below is therefore a computed check, not a statement. A failed check stops the notebook.

The model theory is in `docs/model.md`. How to read a fitted model is in `docs/interpretation.md`.

The notebook runs on CPU in well under five minutes.

```
import importlib.util
import subprocess
import sys
import time

if importlib.util.find_spec("tramdag") is None: # Colab: install from PyPI
    subprocess.check_call(
        [sys.executable, "-m", "pip", "install", "-q", "tramdag[plots]"]
    )
    # to track active development instead of the latest release, use:
    # pip install "git+https://github.com/tensorchiefs/tramdag.git@main"

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import torch

from tramdag import CausalFlowDAG, ContinuousNode, I, plot_dag
from tramdag.callbacks import EarlyStopping
from tramdag.plots import plot_marginals, plot_training

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
plt.rcParams["figure.dpi"] = 110
t_total = time.perf_counter()
print(f"torch {torch.__version__} device: {DEVICE}")

torch 2.14.0+cu130 device: cpu
```

1. The benchmark process

The process comes from Sánchez-Martín et al. (2022, App. E.1) and from App. C.1 of Sick & Dürr (2025):

$$\begin{aligned}x_1 &\sim \tfrac{1}{2} \mathcal{N}(-2, 1.5) + \tfrac{1}{2} \mathcal{N}(1.5, 1) \quad (\text{bimodal source}) \\x_2 &= -x_1 + \mathcal{N}(0, 1) \\x_3 &= x_1 + 0.25 x_2 + \mathcal{N}(0, 1)\end{aligned}$$

The DAG is $x_1 \rightarrow x_2$, $x_1 \rightarrow x_3$, $x_2 \rightarrow x_3$. The source is bimodal, which is the part that a location-scale model cannot fit; the monotone transform of each node is what fits it (docs/model.md).

```
# The process is written out here, so the notebook needs nothing but tramdag.
# draw_latents and simulate stay separate on purpose. Keeping the noise lets
# section 5 intervene on the same individuals, which is what makes an
# individual counterfactual checkable.
```

```
def draw_latents(n, rng):
    """Draw every noise variable of the process, n rows each."""
    return {
        "x1_mix": rng.uniform(size=n), # which mixture component
        "x1_a": rng.normal(size=n), # the N(-2, 1.5) branch
        "x1_b": rng.normal(size=n), # the N(1.5, 1) branch
        "x2": rng.normal(size=n),
        "x3": rng.normal(size=n),
```

```
}
```

```
def simulate(latents, do=None):
    """Run the process forward. A variable named in `do` is clamped instead."""
    do = do or {}
    n = len(latents["x2"])

    if "x1" in do:
        x1 = np.full(n, float(do["x1"]))
    else:
        x1 = np.where(
            latents["x1_mix"] < 0.5,
            -2.0 + np.sqrt(1.5) * latents["x1_a"],
            1.5 + latents["x1_b"],
        )

    x2 = np.full(n, float(do["x2"])) if "x2" in do else -x1 + latents["x2"]
    x3 = x1 + 0.25 * x2 + latents["x3"]
    return pd.DataFrame({"x1": x1, "x2": x2, "x3": x3})

def sample_dgp(n, seed, do=None):
    """Draw n fresh rows, optionally under an intervention."""
    return simulate(draw_latents(n, np.random.default_rng(seed)), do)
```

```
N = 20_000
df = sample_dgp(N, seed=43)
train, val = df.iloc[: N - 2_000], df.iloc[N - 2_000 :]

fig, ax = plt.subplots(figsize=(5.5, 3))
ax.hist(df["x1"], bins=80, density=True, alpha=0.7)
ax.set_title(f"the source $x_1$ has two modes ({N:,} rows)")
ax.set_xlabel("$x_1$")
ax.set_ylabel("density")
fig.tight_layout()
plt.show()
```

2. The spec is the DAG, and one fit

A spec is one entry per node. Each entry names the terms through which the parents enter. `I(...)` puts the parents in control of the transform parameters, which is the most flexible term. `notebooks/classical_fit_tram_dag.py` writes the same DAG with interpretable terms.

Each node fits on its own Adam loop (`docs/fitting.md`). `validation_data=` scores the validation rows per epoch, and `verbose=` prints progress. The strategy attaches through `callbacks=`. `EarlyStopping` stops each node when its own validation score stops improving, and loads its best weights back. Thus `epochs=200` is a ceiling and not a budget. The `seed=` of `CausalFlowDAG` makes the initial weights reproducible.

`notebooks/training_strategies.py` works through the other recipes.

```
spec = {
    "x1": ContinuousNode(), # the source has no parents
    "x2": ContinuousNode(I("x1")),
    "x3": ContinuousNode(I("x1", "x2")),
```

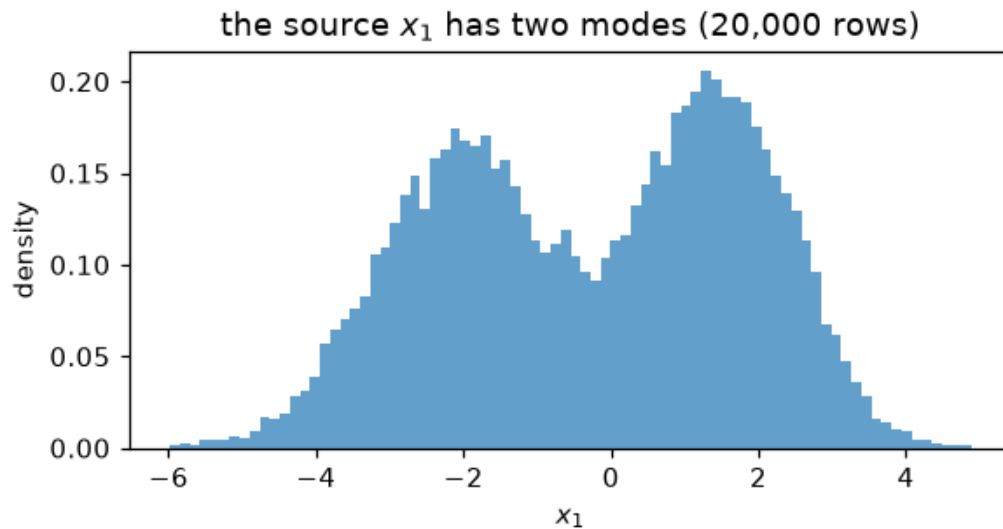


Figure 2: png

```
}
plot_dag(spec)
plt.show()
```

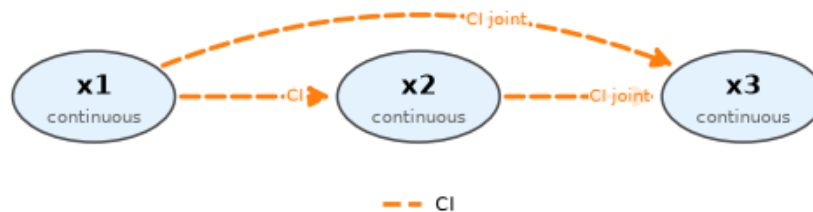


Figure 3: png

```
flow = CausalFlowDAG(spec, device=DEVICE, seed=0)

t0 = time.perf_counter()
flow.fit(
    train,
    epochs=200,
    batch_size=2048,
    validation_data=val,
    verbose=50,
    learning_rate=1e-1,
    callbacks=EarlyStopping(patience=40),
)
t_fit = time.perf_counter() - t0
stopped = {name: len(nd.history["train"]) for name, nd in flow.nodes.items()}
print(f"\nfitted on {DEVICE} in {t_fit:.1f}s")
print(f"each node stopped at epoch: {stopped}")

# The ceiling must not bind. If it does, a node never stopped and the
# numbers below describe an unconverged fit.
```

```

assert max(stopped.values()) < 200, (
    "a node used all 200 epochs, so its early stop never triggered; raise the "
    "ceiling before trusting anything below"
)
x1: epoch 50/200  train 2.0586  val 2.0807

x1: epoch 98/200  train 2.0587  val 2.0804

x2: epoch 50/200  train 1.4182  val 1.4197

x2: epoch 100/200  train 1.4195  val 1.4244

x2: epoch 130/200  train 1.4181  val 1.4216

x3: epoch 50/200  train 1.4185  val 1.4368
x3: epoch 57/200  train 1.4272  val 1.4563

fitted on cpu in 8.7s
each node stopped at epoch: {'x1': 98, 'x2': 130, 'x3': 57}
plot_training(flow, stops=stopped)
plt.show()

```

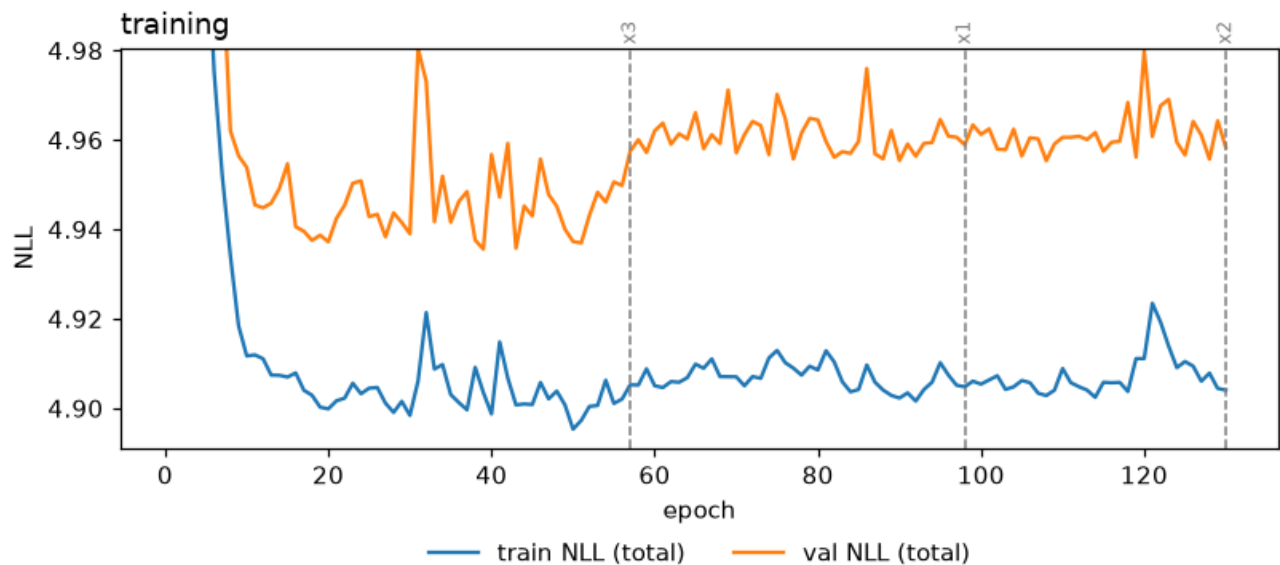


Figure 4: png

3. Rung 1: the observational fit

Samples from the fitted flow must reproduce the observed joint distribution. The marginals are the picture. The correlation matrix is the check, because it also covers the dependence between variables, which a marginal plot does not show.

```
plot_marginals(flow, df, seed=1)
plt.show()
```

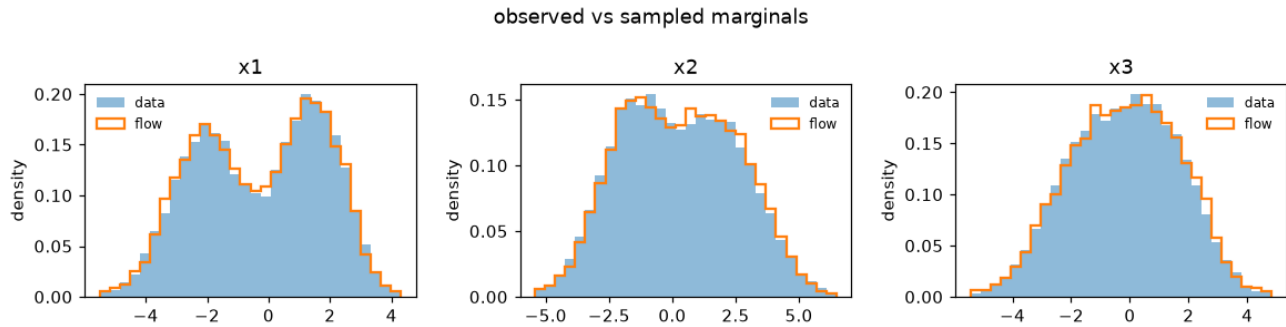


Figure 5: png

```
samp = flow.sample(len(df), seed=1)
corr_gap = float(np.abs(samp.corr().to_numpy() - df.corr().to_numpy()).max())
print(f"largest absolute difference between correlation matrices: {corr_gap:.4f}")
assert corr_gap < 0.05, f"the sampled dependence structure is off by {corr_gap:.4f}"

largest absolute difference between correlation matrices: 0.0061
```

4. Rung 2: interventions

do mutilates the graph. It clamps x_2 , cuts the edge into it, and resamples everything downstream. Under the process, $x_3 \mid do(x_2=a) = x_1 + 0.25a + \mathcal{N}(0, 1)$, so $\mathbb{E}[x_3] = -0.25 + 0.25a$ exactly. That analytic target is what makes an error visible.

```
fig, axes = plt.subplots(1, 3, figsize=(11, 3.2), sharey=True)
errors = []
print("E[x3 | do(x2=a)]    analytic    TRAM-DAG    error")
for i, (ax, a) in enumerate(zip(axes, (-3.0, -1.0, 0.0), strict=True)):
    truth = sample_dgp(len(df), seed=543, do={"x2": a})
    fitted = flow.sample(len(df), do={"x2": a}, seed=2 + i)
    bins = np.linspace(truth["x3"].quantile(0.001), truth["x3"].quantile(0.999), 70)
    ax.hist(truth["x3"], bins=bins, density=True, alpha=0.5, label="process")
    ax.hist(
        fitted["x3"], bins=bins, density=True, histtype="step", lw=1.6, label="TRAM-DAG"
    )
    ax.set_title(f"$p(x_3 \mid do(x_2={a:g}))$")
    ax.set_xlabel("$x_3$")

    analytic = -0.25 + 0.25 * a
    got = float(fitted["x3"].mean())
    errors.append(abs(got - analytic))
    print(
        f"    a = {a:+.0f}:           {analytic:+.4f}    {got:+.4f}    {errors[-1]:.4f}"
    )
axes[0].legend()
fig.tight_layout()
plt.show()
```

```
# The bound leaves room for another machine and another draw.
assert max(errors) < 0.15, f"interventional mean off by {max(errors):.4f}"
```

$E[x_3 \mid do(x_2=a)]$	analytic	TRAM-DAG	error
$a = -3:$	-1.0000	-1.0186	0.0186
$a = -1:$	-0.5000	-0.5321	0.0321
$a = +0:$	-0.2500	-0.2927	0.0427

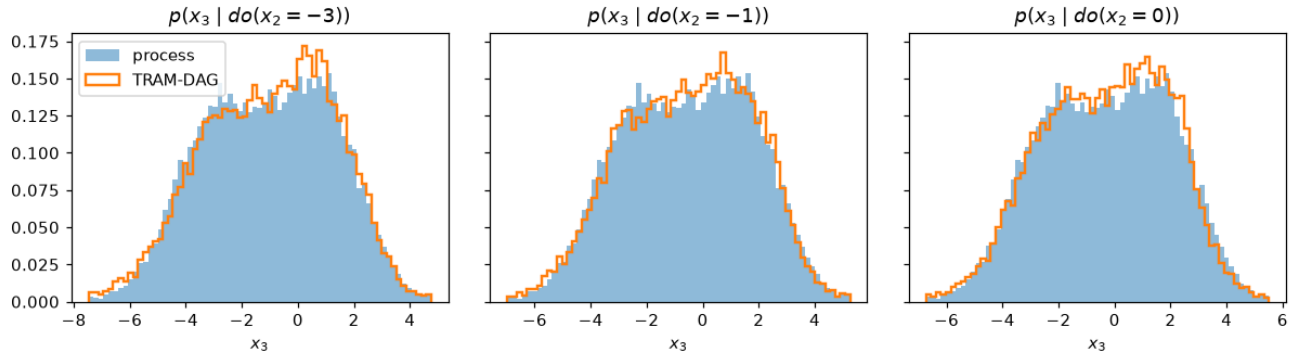


Figure 6: png

5. Rung 3: individual counterfactuals

Take 1,000 held-out individuals. Abduction inverts the flow and recovers the latent noise of each one. That noise is everything about the individual that the model does not attribute to the parents. Pushing it back through the flow must return the observed row, which is the first check.

Action and prediction then rerun each individual under $do(x_1 = 0)$ with the same noise. The process keeps its own noise, so the true individual counterfactual is known and the flow can be scored per individual. Real data never permits this check, because only one of the two outcomes exists.

```

lat = draw_latents(1_000, np.random.default_rng(7))
factual = simulate(lat) # the same noise on both sides,
cf_true = simulate(lat, do={"x1": 0.0}) # so these are true counterfactuals

u = flow.abduct(factual)
recon_err = float(np.abs(flow.sample(u=u).to_numpy() - factual.to_numpy()).max())
print(f"abduction then reconstruction: max |error| = {recon_err:.2e}")
# This inverts a monotone transform in float32, so the bound is a precision
# bound rather than a statistical one.
assert recon_err < 1e-4, f"abduction did not round-trip: {recon_err:.2e}"

cf_flow = flow.sample(do={"x1": 0.0}, u=u)
fig, axes = plt.subplots(1, 2, figsize=(9, 3.6))
for ax, c in zip(axes, ["x2", "x3"], strict=True):
    r = float(np.corrcoef(cf_true[c], cf_flow[c])[0, 1])
    ax.scatter(cf_true[c], cf_flow[c], s=4, alpha=0.4)
    lims = [cf_true[c].min(), cf_true[c].max()]
    ax.plot(lims, lims, "k--", lw=1)
    ax.set_title(f"${c[0]}_{c[1]}$ under $do(x_1{{=}}0)$ (r = {r:.4f})")
    ax.set_xlabel("true, shared noise")
    ax.set_ylabel("TRAM-DAG")

```

```
assert r > 0.99, f"counterfactual {c} correlates only {r:.4f} with the truth"
fig.suptitle("L3: individual counterfactuals, scored one unit at a time")
fig.tight_layout()
plt.show()

abduction then reconstruction: max |error| = 5.50e-06
```

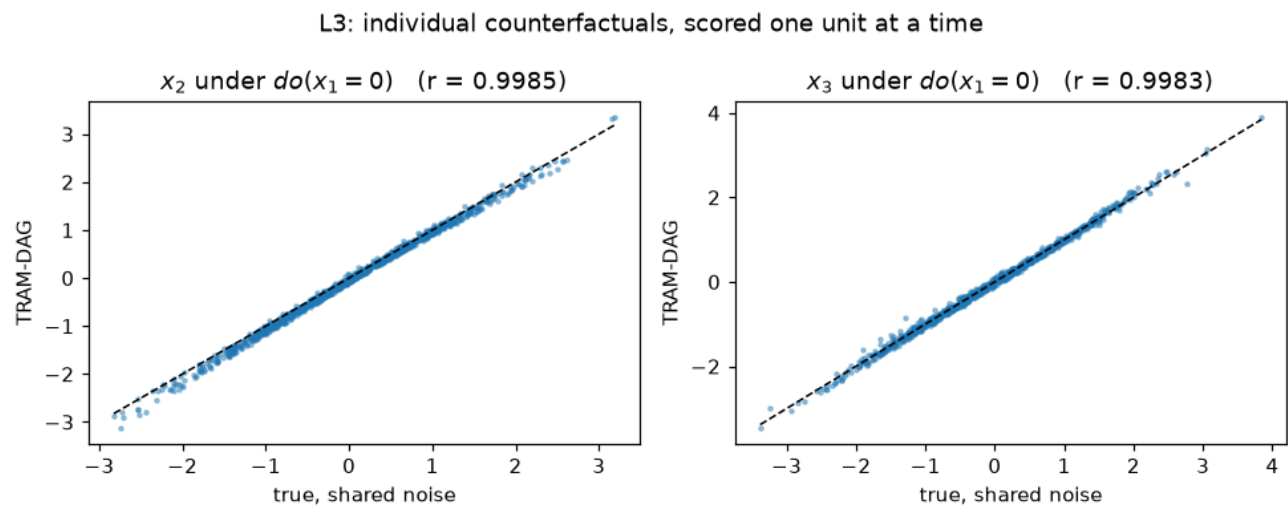


Figure 7: png

What this notebook checked

rung	call	checked against
L1	flow.sample(n)	the correlation matrix of the data
L2	flow.sample(n, do=...)	the analytic mean $-0.25 + 0.25a$
L3	flow.abduct(df), then flow.sample(do=..., u=u)	the true per-unit counterfactual

- Where to go next:
- docs/model.md, the model and its notation.
 - docs/interpretation.md, reading a fitted model, including ordinal nodes and log-odds ratios.
 - notebooks/training_strategies.py, the fitting API and every shipped strategy.
 - notebooks/classical_fit_tram_dag.py, the agreement with statsmodels and R.
 - notebooks/varying_coefficients.py, effects that vary with a covariate.
 - repo · paper

```
elapsed = time.perf_counter() - t_total
print(f"whole notebook: {elapsed:.0f}s")
assert elapsed < 300, f"the notebook took {elapsed:.0f}s, over its five-minute claim"

whole notebook: 11s
```

Training strategies: driving fit from the API side

fit runs one minibatch Adam loop per node and keeps the final weights. It owns three things only: the loop, the per-epoch validation score, and progress printing. Everything else is the caller's, through `optimizer=` and `callbacks=`.

This notebook runs every shipped strategy on one small workload, so the comparison is like for like. It prints what each strategy leaves in `flow.history`. At the end, a scoreboard compares the runtime of the recipes on this workload. The table in `docs/fitting.md` tells which strategy to pick.

```
import time
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import torch

from tramdag import CausalFlowDAG, ContinuousNode, I, Node
from tramdag.callbacks import Callback, EarlyStopping
from tramdag.plots import plot_training

plt.rcParams["figure.dpi"] = 110

# repo-relative data, whether the notebook runs from the repo root or notebooks/
REPO = next(
    p for p in [Path.cwd(), *Path.cwd().parents] if (p / "pyproject.toml").exists()
)

# The tracked 1000-row VACA sample, drawn from the same process as the Colab demo.
df = pd.read_csv(REPO / "notebooks" / "data" / "vaca.csv")
train, val = df.iloc[:900], df.iloc[900:]

SPEC = {
    "x1": ContinuousNode(),
    "x2": ContinuousNode(I("x1")),
    "x3": ContinuousNode(I("x1", "x2")),
}
CEILING = 600 # epoch ceiling for every recipe below
scoreboard = []

def build():
    """Give a fresh unfitted flow, seeded so every recipe starts identically."""
    torch.manual_seed(0)
    return CausalFlowDAG(SPEC)

def record(name, flow, seconds):
    """Add one finished recipe to the scoreboard and print its line."""
    nll = sum(flow.nll(val).values())
    node_epochs = sum(len(nd.history["train"]) for nd in flow.nodes.values())
    scoreboard.append(
        {
            "strategy": name,
```

```

        "val_nll": nll,
        "node_epochs": node_epochs,
        "seconds": seconds,
    }
)
print(
    f"{name:22s} val NLL {nll:.4f}    {node_epochs:4d} node-epochs    {seconds:5.1f}s"
)
return nll

print(f"{len(train)} train rows, {len(val)} validation rows")
900 train rows, 100 validation rows

```

1. What fit records without any callback

Pass `validation_data=` and fit scores each node's validation NLL once per epoch. `EarlyStopping` reads it from `node.history`. Four keys appear in `flow.history`:

- `train`, one dict of per-node negative log-likelihood per epoch,
- `val`, the same for the validation rows, only when validation is on,
- `val_epoch`, the train epoch each `val` entry belongs to,
- `lr`, each node's rates `{group index: lr}` after each epoch, recorded after the callbacks ran, so a schedule's decision for that epoch is what gets stored.

Each node keeps its own history in `flow.nodes[name].history`, one entry per epoch. `flow.history` is a view over them.

```

flow = build()
flow.fit(train, epochs=20, batch_size=256, validation_data=val)

print("history keys:", sorted(flow.history))
print("epochs recorded:", len(flow.history["train"]))
print("per-node train NLL, last epoch:")
for node, value in flow.history["train"].items():
    print(f"    {node}: {value:.4f}")
print("learning rate, last epoch:", flow.history["lr"])

assert len(flow.history["train"]) == len(flow.history["val"]) == 20
assert set(flow.history["train"]) == set(SPEC)

history keys: ['lr', 'train', 'val', 'val_epoch']
epochs recorded: 20
per-node train NLL, last epoch:
  x1: 2.6743
  x2: 1.4055
  x3: 1.4272
learning rate, last epoch: {'x1': {0: 0.01}, 'x2': {0: 0.01}, 'x3': {0: 0.01}}

```

Without `validation_data=` there is no `val` key, and `EarlyStopping` refuses rather than guesses.

```

bare = build()
bare.fit(train, epochs=5, batch_size=256)
print("history keys without validation:", sorted(bare.history))
assert "val" not in bare.history

try:
    build().fit(train, epochs=5, batch_size=256, callbacks=EarlyStopping())

```

```

except RuntimeError as err:
    print("\nEarlyStopping without validation refuses:\n    ", err)
else:
    raise AssertionError("EarlyStopping should refuse without validation data")
history keys without validation: ['lr', 'train']

EarlyStopping without validation refuses:
    EarlyStopping reads node.history['val']; pass validation_data= or validation_split= to fit()

```

2. Plain Adam

One rate, one loop, and the weights at the last epoch. This is the baseline every other recipe is measured against. An all-LS model trained this way to convergence reaches the classical maximum-likelihood estimate, which `classical_fit_tram_dag.py` checks against `statsmodels` and `R`.

```

flow = build()
t0 = time.perf_counter()
flow.fit(train, epochs=CEILING, batch_size=256, learning_rate=1e-2, validation_data=val)
nll_plain = record("plain Adam", flow, time.perf_counter() - t0)
history_plain = [sum(d.values()) for d in flow.history["val"]]

plain Adam          val NLL 4.9987    1800 node-epochs    10.1s

```

3. Two phases

Calling `fit` again continues training. Each call builds a new optimizer, so a second call at a lower rate is a coarse-then-fine schedule with no callback at all. History accumulates across the calls.

```

flow = build()
t0 = time.perf_counter()
flow.fit(train, epochs=200, batch_size=256, learning_rate=1e-2, validation_data=val)
after_coarse = sum(flow.nll(val).values())
flow.fit(train, epochs=100, batch_size=256, learning_rate=1e-3, validation_data=val)
nll_two_phase = record("two phases", flow, time.perf_counter() - t0)

print(f"    after the coarse phase: {after_coarse:.4f}")
print(f"    after the fine phase:    {nll_two_phase:.4f}")
assert len(flow.history["train"]) == 300, "the second fit did not continue the history"

two phases          val NLL 4.9707    900 node-epochs    5.6s
    after the coarse phase: 4.9783
    after the fine phase:   4.9707

```

4. EarlyStopping: keep the best weights

The loop keeps the final weights. An all-LS model wants the final weights, because they are the MLE (`docs/fitting.md`). A flexible model does not always want them. It can reach a better validation score part way through and then drift. `EarlyStopping` snapshots each node's best epoch and loads it back at the end of that node's fit.

The check below confirms it. After the fit, each node's validation score equals the *minimum* over its recorded epochs, not the last one.

```

flow = build()
t0 = time.perf_counter()
stopper = EarlyStopping() # restore_best is on by default, no patience

```

```

flow.fit(
    train,
    epochs=CEILING,
    batch_size=256,
    learning_rate=1e-2,
    validation_data=val,
    callbacks=stopper,
)
nll_best = record("EarlyStopping", flow, time.perf_counter() - t0)

restored = flow.nll(val)
for name, nd in flow.nodes.items():
    recorded = nd.history["val"]
    best = int(np.argmin(recorded)) + 1
    print(
        f"    {name}: best epoch {best} of {len(recorded)}, "
        f"minimum {min(recorded):.4f}, last {recorded[-1]:.4f}"
    )
    assert abs(restored[name] - min(recorded)) < 1e-4, (
        f"the restored weights of {name} do not match its best recorded epoch"
    )

EarlyStopping          val NLL 4.9465    1800 node-epochs    10.8s
x1: best epoch 483 of 600, minimum 2.0991, last 2.1029
x2: best epoch 174 of 600, minimum 1.4721, last 1.5049
x3: best epoch 327 of 600, minimum 1.3754, last 1.3909

```

5. EarlyStopping(patience=): stop each node on its own

Without patience each node spends its whole budget and only the restoration happens. With it, a node also stops once its best epoch is that many epochs old. Nodes converge at different speeds, so they stop at different epochs.

One shared instance resets at each node's fit begin, so its attributes describe the last node only. A function of the node name, `callbacks=lambda name: ...`, gives each node its own instance.

```

flow = build()
t0 = time.perf_counter()
stoppers = {}
flow.fit(
    train,
    epochs=CEILING,
    batch_size=256,
    learning_rate=1e-2,
    validation_data=val,
    callbacks=lambda name: stoppers.setdefault(name, EarlyStopping(patience=25)),
)
record("EarlyStopping(25)", flow, time.perf_counter() - t0)

stopped = {name: len(nd.history["train"]) for name, nd in flow.nodes.items()}
for name, spent in stopped.items():
    best = stoppers[name].best_epoch
    print(f"    {name}: stopped after {spent} of {CEILING} epochs, best was {best}")
    assert spent < CEILING, f"patience never triggered for {name}"
    assert spent - best >= 25

EarlyStopping(25)      val NLL 4.9775    343 node-epochs    2.0s

```

```

x1: stopped after 218 of 600 epochs, best was 193
x2: stopped after 82 of 600 epochs, best was 57
x3: stopped after 43 of 600 epochs, best was 18

```

flow.history repeats the last entry of a node that stopped earlier, so the summed curve stays defined. plot_training marks the stop epochs.

```

plot_training(flow, stops=stopped)
plt.show()

```

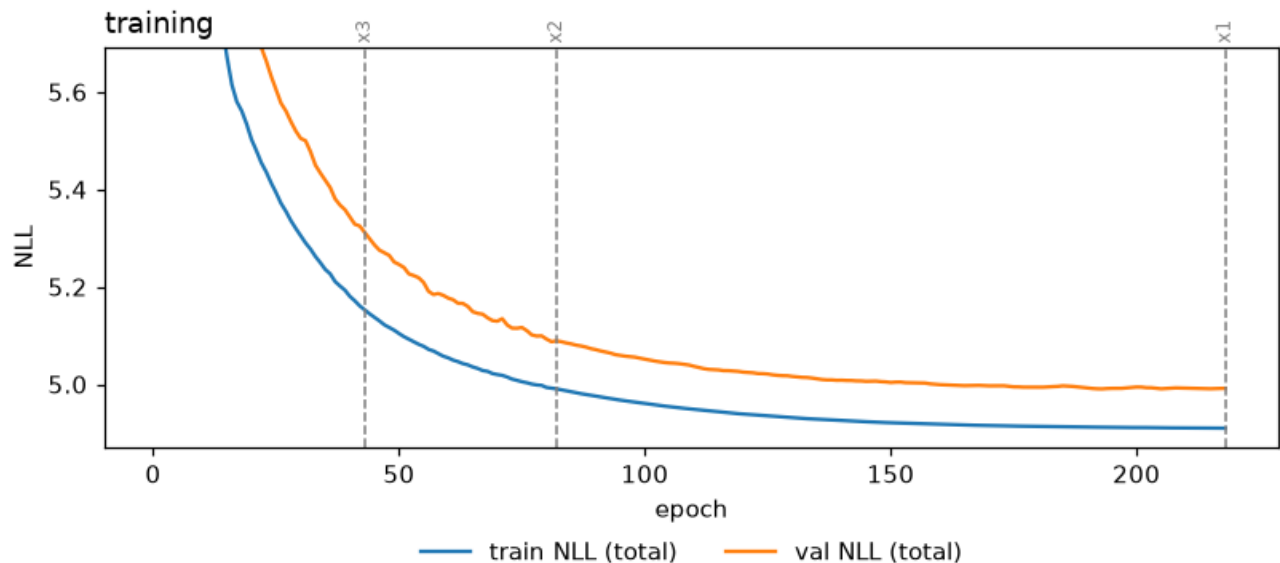


Figure 8: png

6. Writing your own

The callback contract is in docs/fitting.md: every hook receives the node, a bare callable in a list is an on_epoch_end hook, and a Callback subclass gets the other two hooks. Plateau below builds torch's ReduceLR0nPlateau on each node's optimizer and steps it on the validation score fit has already put in node.history["val"].

```

class Plateau(Callback):
    """One torch ReduceLR0nPlateau per node, driven by fit's own score."""

    def __init__(self, factor=0.3, patience=10):
        self.factor, self.patience = factor, patience
        self.scheduler = None

    def on_fit_begin(self, node, optimizer):
        """Build the scheduler here: the node's optimizer exists only now."""
        self.scheduler = torch.optim.lr_scheduler.ReduceLR0nPlateau(
            optimizer, factor=self.factor, patience=self.patience
        )

    def on_epoch_end(self, self, node, epoch, optimizer):
        """Step the scheduler, and stop once the rate has bottomed out."""
        self.scheduler.step(node.history["val"])
        return optimizer.param_groups[0] < 1e-5

```

```

flow = build()
t0 = time.perf_counter()
flow.fit(
    train,
    epochs=CEILING,
    batch_size=256,
    learning_rate=1e-2,
    validation_data=val,
    callbacks=Plateau(),
)
record("Plateau", flow, time.perf_counter() - t0)

for name, rates in flow.history["lr"].items():
    print(f"    {name}: rate went 1.0e-02 -> {rates[0]:.1e}")
    assert rates[0] < 1e-2, f"the scheduler never decayed the rate of {name}"

```

Plateau val NLL 4.9907 500 node-epochs 2.7s
x1: rate went 1.0e-02 -> 7.3e-06
x2: rate went 1.0e-02 -> 7.3e-06
x3: rate went 1.0e-02 -> 7.3e-06

A one-line callable is often enough. This one records each node's validation NLL after every epoch. Put it in a list: on the flow a callable alone is a function of the node name.

```

trace = []
build().fit(
    train,
    epochs=5,
    batch_size=256,
    callbacks=[lambda node, epoch, opt: trace.append((node.name, node.nll(val)))],
)
print(f"{len(trace)} entries, the first: {trace[0]}")
assert len(trace) == 5 * len(SPEC)

```

15 entries, the first: ('x1', 3.561253309249878)

optimizer= takes a factory f(node). This one gives the networks weight decay in their own parameter group. flow.history["lr"] records the rate of each group, so it shows both groups of every node.

```

def adam_with_net_decay(node):
    """Adam with weight decay on the network parameters only."""
    named = list(node.named_parameters())
    nets = [p for name, p in named if ".net." in name]
    rest = [p for name, p in named if ".net." not in name]
    groups = [{"params": rest}, {"params": nets, "weight_decay": 1e-4}]
    return torch.optim.Adam(groups, lr=1e-2)

```

```

flow = build()
flow.fit(train, epochs=20, batch_size=256, optimizer=adam_with_net_decay)
print("rates per group, last epoch:", flow.history["lr"])
assert all(set(rates) == {0, 1} for rates in flow.history["lr"].values())

```

rates per group, last epoch: {'x1': {0: 0.01, 1: 0.01}, 'x2': {0: 0.01, 1: 0.01}, 'x3': {0: 0.01, 1: 0.01}}

7. One node alone

A node is a TRAM regression of one variable on its parents and fits without the DAG. It takes its name, its node spec and a parent schema, {parent: "continuous" | n_levels}. The frame needs only the node's own column and its parents' columns.

```
x3 = Node("x3", SPEC["x3"], {"x1": "continuous", "x2": "continuous"})
x3.fit(
    train[["x1", "x2", "x3"]],
    epochs=CEILING,
    validation_data=val[["x1", "x2", "x3"]],
    callbacks=EarlyStopping(patience=25),
)
print(f"x3 alone: {len(x3.history['train'])} epochs, val NLL {x3.nll(val):.4f}")
assert abs(x3.nll(val) - min(x3.history["val"])) < 1e-4 # the restored best
x3 alone: 178 epochs, val NLL 1.3967
```

8. The scoreboard: a runtime comparison

The board has one workload, one seed, one machine and every recipe. Each row gives the wall-clock seconds and the node-epochs (summed over the nodes). It also gives the validation NLL and the gap to the best NLL on the board. A recipe that stops itself wins on seconds only if its NLL also stays near the best. The absolute seconds apply to this machine only.

```
board = pd.DataFrame(scoreboard).set_index("strategy")
board["nll_gap"] = board["val_nll"] - board["val_nll"].min()
board["s_per_node_epoch"] = board["seconds"] / board["node_epochs"]
print(board.to_string(float_format=lambda v: f"{v:.4f}"))

fig, ax = plt.subplots(figsize=(7, 3.4))
ax.scatter(board["seconds"], board["nll_gap"])
for name, row in board.iterrows():
    ax.annotate(
        name,
        (row["seconds"], row["nll_gap"]),
        xytext=(4, 4),
        textcoords="offset points",
        fontsize=8,
    )
ax.margins(x=0.15)
ax.set_xlabel("wall-clock seconds")
ax.set_ylabel("validation NLL above the best recipe")
ax.set_title("cost against quality, one workload")
fig.tight_layout()
plt.show()

# Keeping the best weights cannot score worse than keeping the last ones.
assert nll_best <= nll_plain + 1e-6, (
    f"EarlyStopping scored {nll_best:.4f} against plain Adam's {nll_plain:.4f}"
)
```

	val_nll	node_epochs	seconds	nll_gap	s_per_node_epoch
strategy					
plain Adam	4.9987	1800	10.0838	0.0521	0.0056
two phases	4.9707	900	5.6375	0.0241	0.0063
EarlyStopping	4.9465	1800	10.7545	0.0000	0.0060
EarlyStopping(25)	4.9775	343	1.9506	0.0309	0.0057

Plateau 4.9907 500 2.7216 0.0442 0.0054

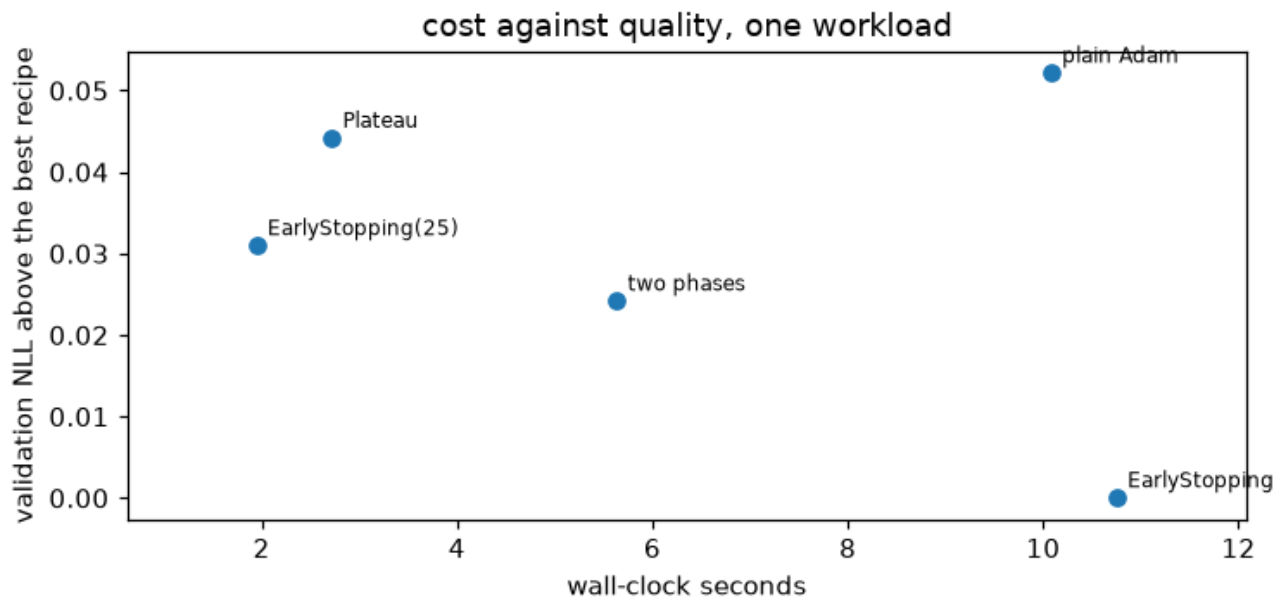


Figure 9: png

The restored EarlyStopping score lies below the minimum of the plain Adam curve. Each node restores its own best epoch, but the summed curve has one epoch for all nodes.

```
assert nll_best <= min(history_plain) + 1e-6
fig, ax = plt.subplots(figsize=(7, 3.4))
ax.plot(np.arange(1, len(history_plain) + 1), history_plain, label="plain Adam")
ax.axhline(nll_best, ls="--", lw=1, color="C1", label="EarlyStopping, restored")
ax.set_xlabel("epoch")
ax.set_ylabel("validation NLL (total)")
ax.set_ylim(min(nll_best, min(history_plain)) - 0.02, min(history_plain) + 0.4)
ax.legend()
ax.set_title("keeping the best epoch against keeping the last")
fig.tight_layout()
plt.show()
```

Which one, when

The decision table lives in docs/fitting.md.

Heterogeneous treatment effects: the VC term

Most causal questions are not “does the treatment help?” but “**whom** does it help, and by how much?”. A TRAM-DAG answers that with a **varying-coefficient** term, which gives one edge of the DAG a treatment effect that depends on covariates:

$$\text{shift} += \beta(\text{modifiers}) \cdot x_t, \quad \beta(x) = \beta_0 + b_{\Theta}(x)$$

docs/varying-coefficients.md states the model, its penalty and its centering; this notebook runs it.

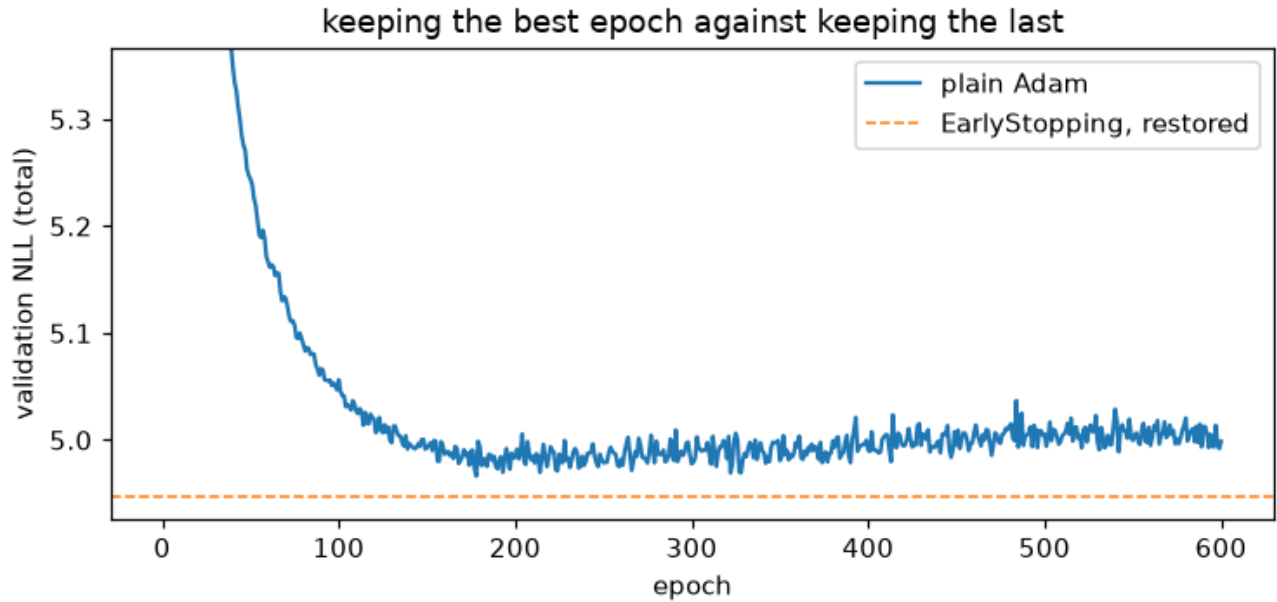


Figure 10: png

This notebook builds a DGP with a known effect function, fits the model, and scores the recovered $\beta(x)$ against the truth. It then shows the two things the term exists for that a plain flexible shift does not give: a read-out you can plot, and a **propensity-centered** variant that survives confounding.

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from tramdag import CS, LS, SI, VC, CausalFlowDAG, ContinuousNode, OrdinalNode
from tramdag.callbacks import EarlyStopping
from tramdag.plots import plot_varying_coef

plt.rcParams["figure.dpi"] = 110
```

1. A DGP with a known effect function

Three covariates, a **confounded** treatment (X_1 and X_2 push assignment), and an outcome whose conditional is exactly a logistic-latent flow:

$$h(y) + g(x) + \beta(x)t = U, \quad U \sim \text{Logistic}(0, 1)$$

with $h(y) = 2y$, a nonlinear prognostic part $g(x) = \frac{1}{2}X_1^2 + X_2 - \frac{1}{2}X_3$, and the effect function

$$\beta(x) = -1 + 0.8X_2 - 0.6X_3.$$

X_2 is deliberately *both* a confounder and an effect modifier. In this case a naive estimate fails hardest.

$B_0, B_2, B_3 = -1.0, 0.8, -0.6$

```
def true_beta(df):
    """The effect function the model has to recover, on the latent scale."""
    return B0 + B2 * df["X2"].to_numpy() + B3 * df["X3"].to_numpy()

def simulate(n, seed):
    rng = np.random.default_rng(seed)
    x1, x2, x3 = (rng.normal(size=n) for _ in range(3))
    t = (rng.logistic(size=n) > -(0.4 * x1 + 0.4 * x2)).astype(float)
    g = 0.5 * x1**2 + x2 - 0.5 * x3
    beta = B0 + B2 * x2 + B3 * x3 # the same expression true_beta reports
    y = (rng.logistic(size=n) - g - beta * t) / 2.0
    return pd.DataFrame({"X1": x1, "X2": x2, "X3": x3, "T": t, "Y": y})

train, val, test = simulate(4500, 1), simulate(500, 2), simulate(2000, 3)
print(f"treated fraction: {train['T'].mean():.2f}")
print(
    f"true effect ranges from {true_beta(test).min():+.2f} to {true_beta(test).max():+.2f}"
)

treated fraction: 0.48
true effect ranges from -4.41 to +2.29
```

2. Which covariates modify the effect? (effect_modifier_scan)

Before you choose the modifiers, ask the data. Fit the **simple all-LS model**, which takes seconds and is deterministic. Then look at the per-row scores of the treatment coefficient. With a constant effect, these scores show no pattern along any covariate. The scan measures that pattern per covariate and ranks the covariates.

```
def cheap_all_ls():
    return {
        "X1": ContinuousNode(),
        "X2": ContinuousNode(),
        "X3": ContinuousNode(),
        "T": OrdinalNode(2, LS("X1") + LS("X2")),
        "Y": ContinuousNode(LS("X1") + LS("X2") + LS("X3") + LS("T")),
    }
```

```
screen = CausalFlowDAG(cheap_all_ls(), seed=0)
screen.fit_classical(train)
scan = screen.effect_modifier_scan(train, "Y", t="T")
print(scan)
assert scan["flag"].all()
```

	stat	p_value	crit_5pct	flag
X2	4.164390	1.729171e-15	1.3581	True
X1	4.141048	2.548149e-15	1.3581	True
X3	3.161705	4.152248e-09	1.3581	True

X2 and X3 are the true modifiers, and both flag. X1 flags too, but it does **not** enter $\beta(x)$. Here g contains $\frac{1}{2}X_1^2$, which the simple linear model cannot represent. docs/scores.md explains why a misspecified prognostic part flags too.

To show this, generate the data again with a *linear* prognostic X_1 and change nothing else. X1 then does not flag.

```
def simulate_linear_x1(n, seed):
    rng = np.random.default_rng(seed)
    x1, x2, x3 = (rng.normal(size=n) for _ in range(3))
    t = (rng.logistic(size=n) > -(0.4 * x1 + 0.4 * x2)).astype(float)
    g = 0.5 * x1 + x2 - 0.5 * x3 # linear, so the simple model has the right form
    y = (rng.logistic(size=n) - g - (B0 + B2 * x2 + B3 * x3) * t) / 2.0
    return pd.DataFrame({"X1": x1, "X2": x2, "X3": x3, "T": t, "Y": y})
```

```
well_specified = simulate_linear_x1(4500, 1)
screen2 = CausalFlowDAG(cheap_all_ls(), seed=0)
screen2.fit_classical(well_specified)
scan2 = screen2.effect_modifier_scan(well_specified, "Y", t="T")
print(scan2)
assert scan2["flag"].all()
assert not scan2["flag"]
```

	stat	p_value	crit_5pct	flag
X2	4.609048	7.068293e-19	1.3581	True
X3	3.474039	6.577742e-11	1.3581	True
X1	0.553723	9.190288e-01	1.3581	False

The figure draws the scaled running sum B_j of the scan for each candidate (docs/scores.md explains it). The left panel uses the quadratic X1, the right panel the linear X1. stat is the largest distance of the path from zero, the dot on each line. If a path leaves the grey band, the scan flags its covariate.

```
CANDIDATES = {"X1": "#2a78d6", "X2": "#eb6834", "X3": "#1baf7a"}
```

```
def plot_cusum(ax, flow, df, title):
    """Draw each candidate's scaled running sum of the T scores, as the scan forms it."""
    psi = flow.scores(df, "Y")["T[1]"].to_numpy()
    n = len(psi)
    scan = flow.effect_modifier_scan(df, "Y", t="T")
    crit = scan["crit_5pct"].iloc[0]
    ax.axhspan(
        -crit, crit, color="0.93", lw=0, label=f"$|B_j| < {crit:.3f}$: no flag at 5%"
    )
    for c in (-crit, crit):
        ax.axhline(c, color="0.55", lw=1, ls="--")
    ax.axhline(0, color="0.75", lw=0.8)
    frac = np.arange(1, n + 1) / n
    for name, color in CANDIDATES.items():
        b = np.cumsum(psi[np.argsort(df[name].to_numpy(), kind="stable")])
        b /= psi.std() * np.sqrt(n)
        j = np.abs(b).argmax()
        ax.plot(frac, b, color=color, lw=2, label=name)
        ax.plot(frac[j], b[j], "o", ms=8, color=color, mec="white", mew=2)
        ax.annotate(
            f"{name}: stat {scan.loc[name, 'stat']:.2f}",
            (frac[j], b[j]),
            xytext=(0, 10 if b[j] > 0 else -16),
            textcoords="offset points",
            ha="center",
            color="0.2",
        )
    )
```

```

ax.set(
    title=title, xlabel="fraction of rows, sorted by the candidate", ylim=(-6, 5)
)

fig, axes = plt.subplots(1, 2, figsize=(11, 4), sharey=True)
plot_cusum(axes[0], screen, train, "X1 quadratic in the outcome: X1 flags too")
plot_cusum(
    axes[1], screen2, well_specified, "X1 linear in the outcome: only X2, X3 flag"
)
axes[0].set_ylabel("scaled running sum  $B_j$  of the T scores")
fig.legend(
    *axes[1].get_legend_handles_labels(), loc="lower center", ncol=4, frameon=False
)
fig.tight_layout(rect=(0, 0.07, 1, 1))
plt.show()

```

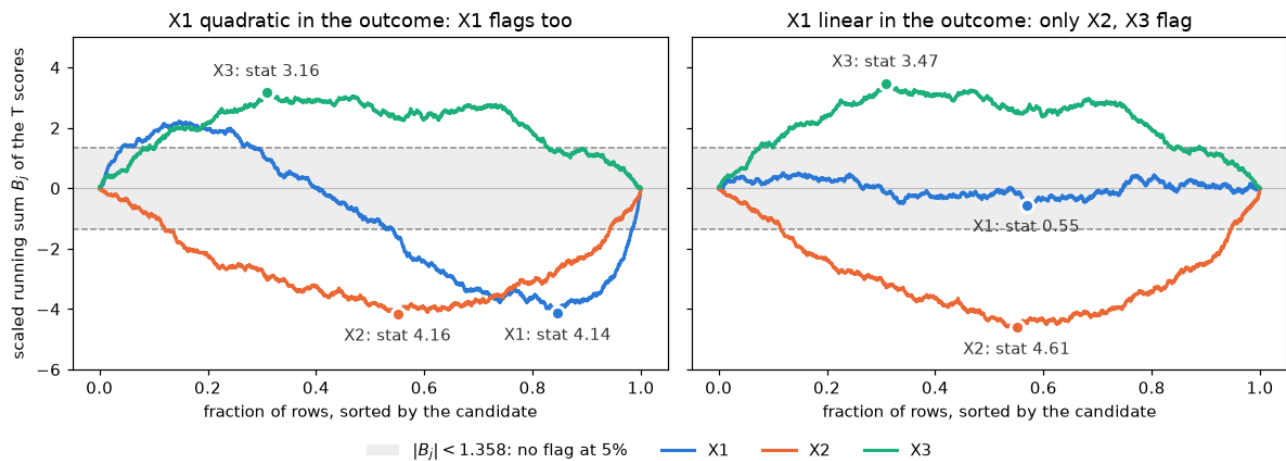


Figure 11: png

3. The spec: prognostic part and effect head are separate

CS("X1", "X2", "X3") absorbs the prognostic signal, as flexibly as necessary. VC("X2", "X3", t="T") carries the effect. X2 and X3 appear twice on purpose: prognostically through the shift, and as modifiers through the head. Only the treatment named by t= owns its edge.

```

spec = {
    "X1": ContinuousNode(),
    "X2": ContinuousNode(),
    "X3": ContinuousNode(),
    "T": OrdinalNode(2, LS("X1") + LS("X2")),
    "Y": ContinuousNode(CS("X1", "X2", "X3") + VC("X2", "X3", t="T")),
}
flow = CausalFlowDAG(spec, seed=0)
flow.fit(
    train,
    epochs=300,
    learning_rate=1e-2,
    batch_size=512,
    validation_data=val,
    seed=0,
)

```

```

    callbacks=EarlyStopping(), # keep the best-validation weights
)
print(flow.to_matrix())

```

	X1	X2	X3	T	Y
X1				LS	CS['X1', 'X2', 'X3']
X2				LS	CS['X1', 'X2', 'X3']+VCm
X3					CS['X1', 'X2', 'X3']+VCm
T					VC
Y					

In the meta-adjacency view (docs/interpretation.md) VC marks the treatment edge and VCm the modifiers, so X2 visibly enters Y twice.

4. Reading the effect out

varying_coef gives $\beta(x)$ per row. It is **deterministic** and does not look at the outcome. It is a property of the fitted model, not of the rows' Y values.

```

beta_hat = flow.varying_coef(test, "Y")
beta_true = true_beta(test)
corr = float(np.corrcoef(beta_hat, beta_true)[0, 1])
beta0 = flow.nodes["Y"].shifts["T"].beta0.item()

print(f"corr(beta_hat, beta_true) = {corr:.3f}")
print(f"beta0 = {beta0:+.3f} (true constant part {B0:+.1f})")
print(f"mean |error| = {np.abs(beta_hat - beta_true).mean():.3f}")
assert corr > 0.95, f"the fitted beta(x) correlates only {corr:.3f} with the truth"

```

```

fig, axes = plt.subplots(1, 2, figsize=(10, 3.8))
axes[0].scatter(beta_true, beta_hat, s=6, alpha=0.35)
lims = [min(beta_true.min(), beta_hat.min()), max(beta_true.max(), beta_hat.max())]
axes[0].plot(lims, lims, "k--", lw=1)
axes[0].set_xlabel(r"true $\beta(x)$")
axes[0].set_ylabel(r"fitted $\hat{\beta}(x)$")
axes[0].set_title(f"recovery, corr = {corr:.3f}")

```

```

plot_varying_coef(flow, test, "Y", by="X2", ax=axes[1])
x3_med = test["X3"].median()
grid = np.linspace(test["X2"].min(), test["X2"].max(), 2)
axes[1].plot(grid, B0 + B2 * grid + B3 * x3_med, "k--", lw=1)
along_x2 = test.assign(X3=x3_med)
gap_x2 = np.abs(flow.varying_coef(along_x2, "Y") - true_beta(along_x2)).max()
print(f"X3 at its median: max |beta_hat - beta_true| along X2 = {gap_x2:.3f}")
assert gap_x2 < 0.75, f"the fitted beta(X2) leaves the true line by {gap_x2:.3f}"
axes[1].legend(["fitted", r"$\beta = 0$", "true"])
axes[1].set_title(r"$\beta$ along $X_2$, $X_3$ at its median")
fig.tight_layout()
plt.show()

```

```

corr(beta_hat, beta_true) = 0.995
beta0 = -0.923 (true constant part -1.0)
mean |error| = 0.110
X3 at its median: max |beta_hat - beta_true| along X2 = 0.375

```

The right panel holds X3 at its median, because the effect also depends on X3. Along X2, the fitted line follows the true line.

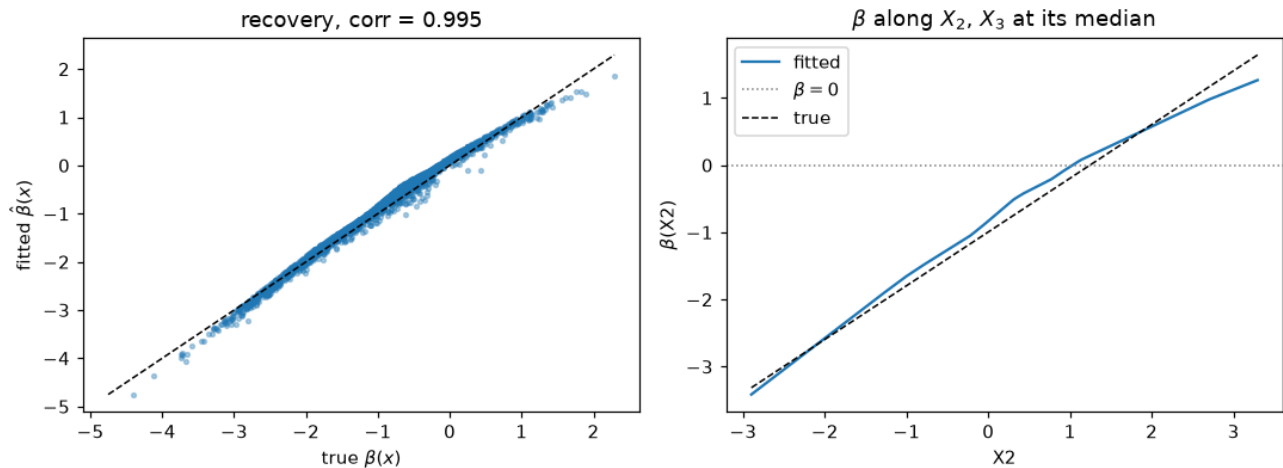


Figure 12: png

5. The read-out is an identity, not a summary

For a binary treatment, $\beta(x)$ equals the difference of the abducted latents between the two arms, with the outcome held fixed. The implementation must satisfy this definition to float precision:

```
u1 = flow.abduct(test.assign(T=1.0), seed=0)["Y"].to_numpy()
u0 = flow.abduct(test.assign(T=0.0), seed=0)["Y"].to_numpy()
gap = np.abs(beta_hat - (u1 - u0)).max()
print(f"max |beta(x) - (u(T=1) - u(T=0))| = {gap:.2e}")
assert gap < 1e-5, f"the read-out differs from the latent difference by {gap:.2e}"
max |beta(x) - (u(T=1) - u(T=0))| = 4.77e-07
```

6. Confounding: why propensity= exists

Everything above had a prognostic part flexible enough to absorb g . Real models are misspecified, and when the misfit correlates with **treatment assignment**, the effect head absorbs the confounding instead of the effect.

The configuration where this bites hardest is deliberately simple: one covariate, a strong propensity $e(x) = \sigma(2x)$, a constant true effect $\tau = -1$, and a quadratic prognostic part fitted with a linear term. propensity="ps" replaces t by $t - \hat{e}(x)$ with out-of-fold propensities from the training-frame column ps. docs/varying-coefficients.md says why they must be out of fold. Stage 1, the propensities, is yours. Here, five classical fits of the treatment spec give them.

TAU = -1.0

```
def confounded(n, seed):
    rng = np.random.default_rng(seed)
    x = rng.normal(size=n)
    t = (rng.logistic(size=n) > -2.0 * x).astype(float) # strong confounding
    y = (rng.logistic(size=n) - 1.2 * x**2 - TAU * t) / 2.0 # quadratic prognostic
    return pd.DataFrame({"X": x, "T": t, "Y": y})
```

```
c_train, c_val, c_test = confounded(5400, 0), confounded(600, 7), confounded(3000, 1000)
# stage 1 of the centered design, the caller's job: cross-fitted P(T=1|X), each
```

```

# fold predicted by a treatment model that never saw it (as double machine
# learning requires)
fold_id = np.random.default_rng(0).permutation(len(c_train)) % 5
e_oof = np.empty(len(c_train))
for j in range(5):
    t_spec = {
        "X": ContinuousNode(SI(transform="affine")),
        "T": OrdinalNode(2, LS("X")),
    }
    proxy = CausalFlowDAG(t_spec, seed=0)
    proxy.fit_classical(c_train.iloc[fold_id != j])
    e_oof[fold_id == j] = proxy.pmf(c_train.iloc[fold_id == j], "T")[:, 1]

mae = {}
for propensity in (None, "ps"):
    spec_c = {
        "X": ContinuousNode(SI(transform="affine")),
        "T": OrdinalNode(2, LS("X")),
        # linear prognostic term, though the truth is quadratic
        "Y": ContinuousNode(LS("X") + VC("X", propensity=propensity, t="T")),
    }
    fc = CausalFlowDAG(spec_c, seed=0)
    fc.fit(
        # the out-of-fold propensities ride the frame as the column propensity= names
        c_train.assign(ps=e_oof) if propensity else c_train,
        epochs=250,
        learning_rate=1e-2,
        batch_size=512,
        validation_data=c_val,
        seed=0,
        callbacks=EarlyStopping(), # keep the best-validation weights
    )
    b = fc.varying_coef(c_test, "Y")
    centered = propensity is not None
    mae[centered] = float(np.abs(b - TAU).mean())
    print(
        f"centered={centered!s:5s} mean |beta - tau| = {mae[centered]:.3f}"
        f" mean beta = {b.mean():+.3f} (true tau {TAU:+.1f})"
    )
ratio = mae[False] / mae[True]
print(f"centering cuts the bias by a factor of {ratio:.1f}")
assert ratio >= 2, f"centering should at least halve the bias: {ratio:.2f}"

centered=False mean |beta - tau| = 1.294 mean beta = +0.007 (true tau -1.0)

centered=True mean |beta - tau| = 0.267 mean beta = -0.733 (true tau -1.0)
centering cuts the bias by a factor of 4.8

```

Without centering the confounding swallows most of the effect; with centering the estimate returns near the truth. The printed factor is the measured bias reduction, and tests/test_vc_centered.py requires at least two.

What to take away

you want	use	read it with
one interpretable effect	<code>LS("T")</code>	<code>ls_coefficients()</code>
an effect that varies with covariates	<code>VC(*modifiers, t="T")</code>	<code>varying_coef(df, node)</code>
the same under confounding + a misspecified prognostic part	<code>VC(..., propensity="ps")</code>	the same read-out
a shortlist of modifiers before you commit	<code>effect_modifier_scan</code> on a simple all-LS fit	its flag column, read as screening

Why not a joint CS over treatment and modifiers, and what the penalty and the centering mean: docs/varying-coefficients.md. The scan: docs/scores.md.

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